CS3240

Evaluation

How Do We Know Our Design Works?

Two methods: Heuristic Evaluation[L12] & Usability Testing[L13]

S2AY2025-26

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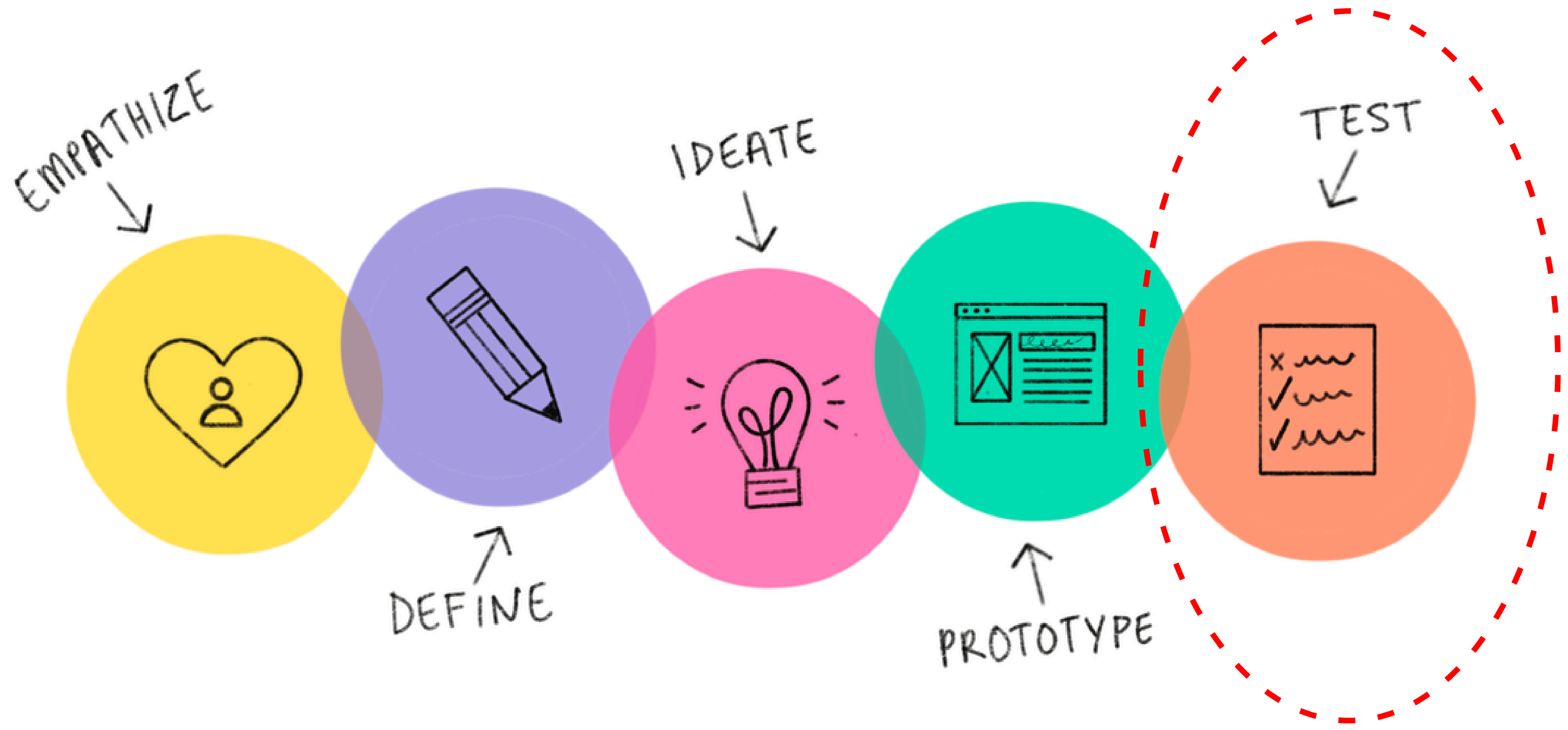
Evaluation



Objectives of L12 and L13 :

- Choose the right **evaluation method**
- **Apply heuristics** to critique interfaces
- Design a basic **usability test**

Sub-topics for this lecture

Why, What, Who, When & Where of Evaluation

Heuristic Evaluation

What is meant by Heuristics
Nielsen Heuristics
Heuristic evaluation process

Heuristics for evaluating AI Features
Appendix: Chatbot Heuristics

Why, When to and What to Evaluate ?

Why Evaluate?

- Avoid costly mistakes
- Improve designs early
- Discover new ideas

When to Evaluate ?

- **Formative -** During design
- **Summative** - After design

What to Evaluate

- Task flows and navigation
- UI clarity and feedback
- Overall user experience

Who and Where to Evaluate ?

Who Evaluates?

- Peers
- Experts
- Users

Where to Evaluate ?

- Lab / Living Lab
- Natural settings /Field
- Remote

Example

For evaluation of **a social media website**

- Do the clickable buttons have labels that conform to industry conventions?
- Does it provide confirmation and auto-save when users attempt to exit chats?
- Does it provide clear error messages e.g. it gives a message why a media file could not upload and also provides options for resolving the issue?
- Is there a search feature for finding specific information e.g. finding information within the Help page?

- Users can be asked to evaluate
- Anyone with UI design knowledge and experience could be asked to evaluate

Example

For evaluation of **an interactive prototype of a personal music service**

- How well users can select tracks from potentially thousands of tunes ?
- Can they easily add and store new music?
- Do they like it?
- What problems they experience with the functions?

- User tests can be designed to observe and colleact data about what users do , when they do it, and for how long.
- Users can be asked to carry out the tasks in their natural settings or they can be invited to a ‘lab’ to test the interface for the designed tasks.
- Users are observed for how they carry out various tasks using the interface controls and navigation.

How to evaluate design?
UI & Interaction Design Evaluation (Focus of this course)

Two Core Methods

Heuristic Evaluation
- Experts review
- Fast & early
- Find known issues

Usability Testing
- Users perform tasks
- Real behavior
- Find real issues

Beyond UI: Broader UX Evaluation Methods
This course focuses on two UI & interaction evaluation methods.
UX evaluation includes a wider range of methods:
- User interviews
- Surveys & questionnaires (e.g., SUS, etc)
- Field studies / observations
- A/B testing
- Analytics (usage data, drop-offs)
- Diary studies

What is Heuristic Evaluation?

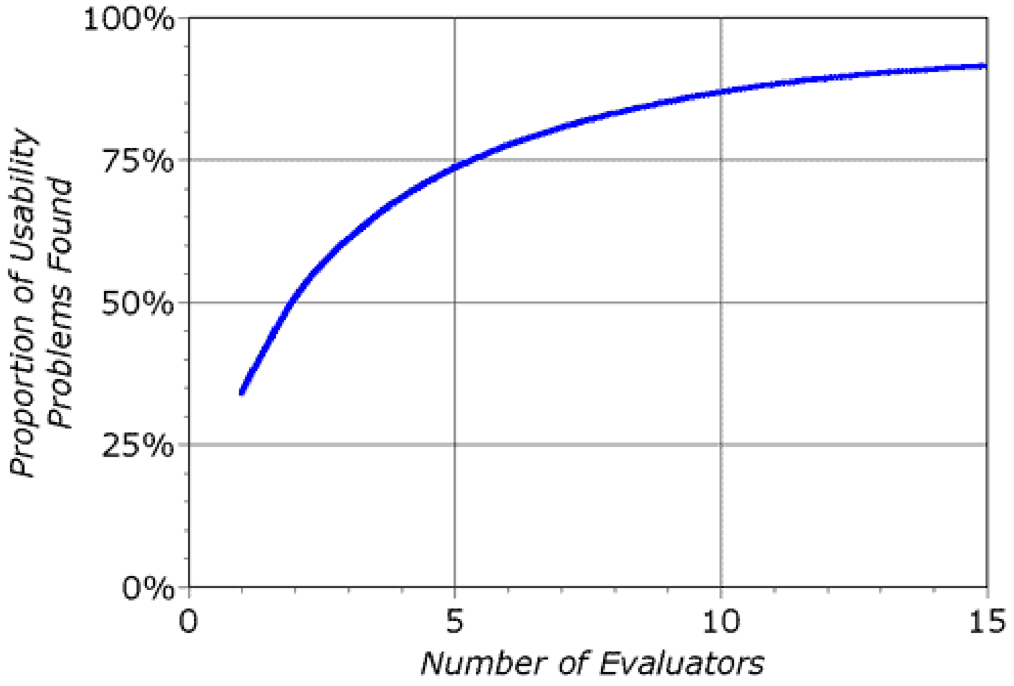
- **Experts** review designs using Heuristics (rules of thumb)
- Identify violations
- Rate severity

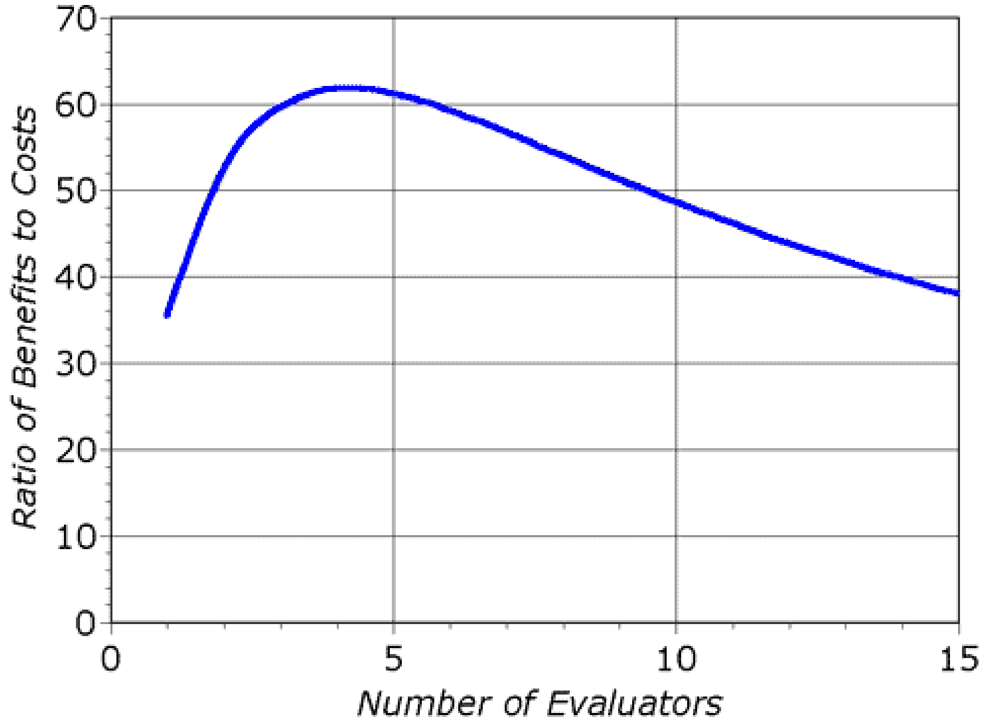
Examples

What do you think is the problem and what can be improved ?

Context	Task/ Screen	Output
In an online hotel booking site	Cancel booking	On canceling a booking an error is shown: Invalid Transaction Code: -54x3
A file upload tool	Upload file	Click on “Upload” but nothing happens. *(You're not sure if it's uploading.)*
ATM machine	Landing screen	Screen displays shows: Please insert monetary instrument.
Online Form filling	Form filling screen	No ‘Back’ or ‘Cancel’ button during online form filling.

How Many Experts Are Needed?





Diminishing returns, but not a fixed rule

Source: nngroup.com

Key Insights

- Each evaluator finds different issues
- Overlap increases after a few evaluators
- But coverage improves with diversity

Practical Guidelines

- Start with 3–5 evaluators
- Add more if:
  - System is complex
  - Multiple user types
  - Safety-critical domain
  - AI / adaptive systems
- Not just number but also type of evaluators
  - UX expert
  - Domain expert
  - Accessibility expert
  - AI/system expert

10 Nielsen Heuristics

Visibility
Show system status and provide immediate feedback

Mapping
Use familiar metaphors and language

Control and freedom
Provide good defaults and undo

Consistency and standards
Use the same interface and language throughout

Error Prevention
Help users avoid making mistakes

Recognition
Make information easy to discover

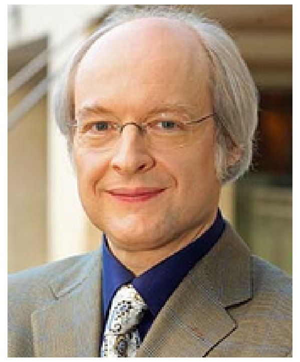
Flexibility
Allow for tasks that cater to both new and advanced users

Minimalism
Provide only necessary information in an elegant way

Error Recovery
Help users recognize, diagnose, and recover from errors

Help and documentation
Use proactive and in-place hints to guide users

Jacob Nielsen
http://www.nngroup.com



https://www.nngroup.com/articles/ten-usability-heuristics/

https://www.nngroup.com/articles/usability-heuristics-applied-video-games/

https://www.nngroup.com/articles/usability-heuristics-virtual-reality/

H-1 Visibility

Show system status and provide immediate feedback

User expects to know what's the outcome of their action or interaction.

Keep users informed through **appropriate feedback** .

Poor Visibility
Click "Pay" → nothing happens
No loading, no message

Good Visibility
"Processing payment..."

In practice, users should always know:

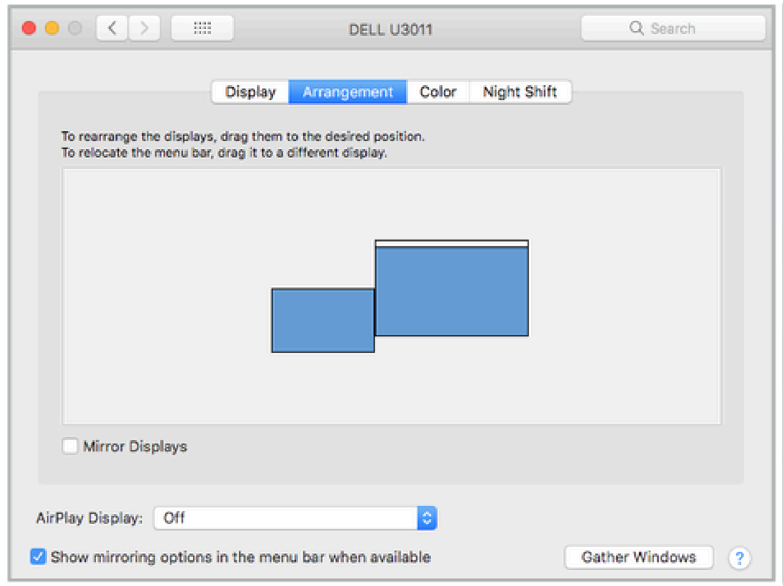
- What is happening right now
  - Uploading 3 files... (60%)
  - Driver arriving in 4 mins
- What just happened
  - Payment successful
  - Item added to cart
- What will happen next
  - Order will arrive by 12:30 PM
  - You will be charged $9.80
- What they can do now
  - Cancel order
  - Retry payment

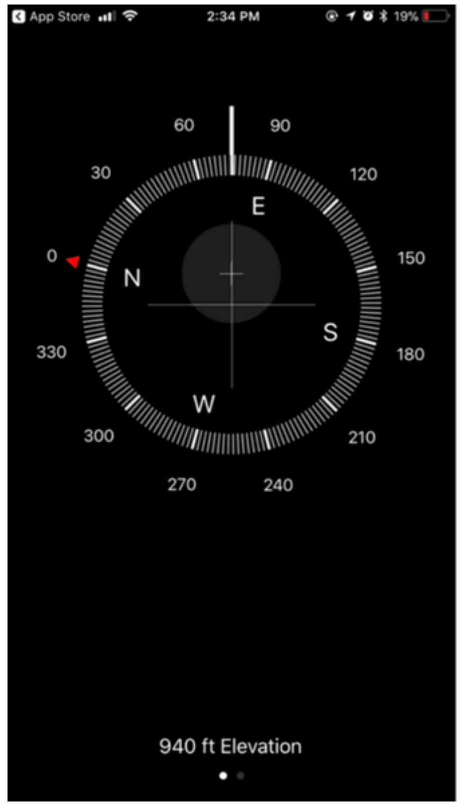
H-2 Mapping

Use familiar metaphors and language

The language, the mappings, the images perceived by users should **match the mental model** the users have about the system.

- Avoid technical terms; use words and phrases that are familiar to the user.
- Use familiar icons. Unfamiliar icons can make users search for meaning.
- Follow real-world conventions.
- Create **visual affordances**—visual cues that give users hints on how they can interact with certain objects.





context: Cancel booking

On canceling a booking an error is shown:
Invalid Transaction Code: -54x3

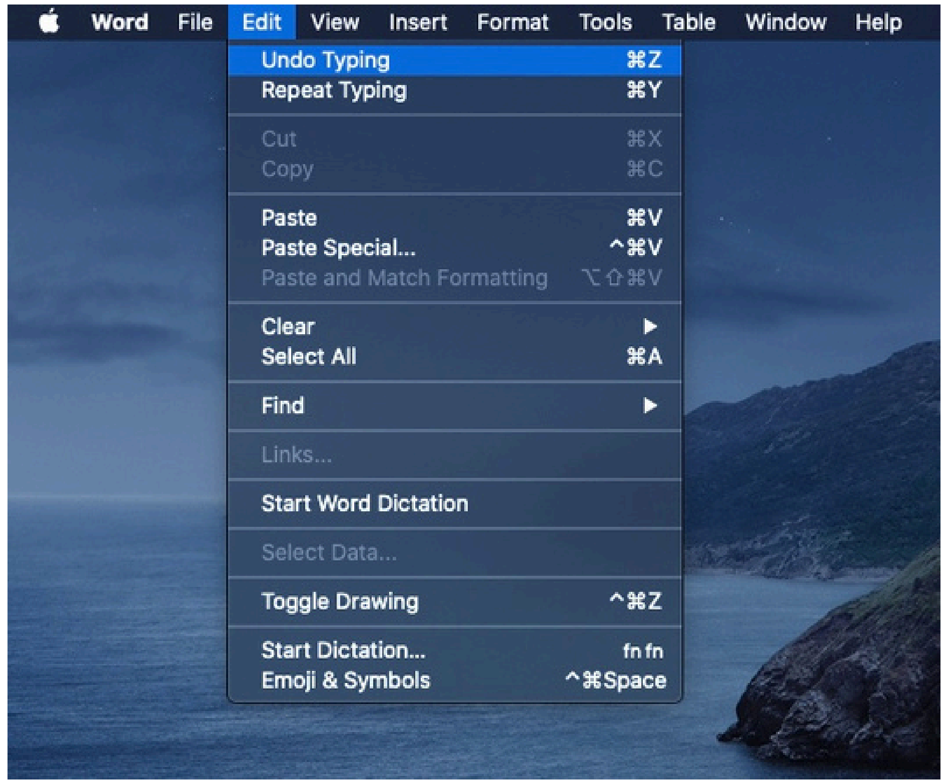
Source: https://www.nngroup.com/articles/match-system-real-world/

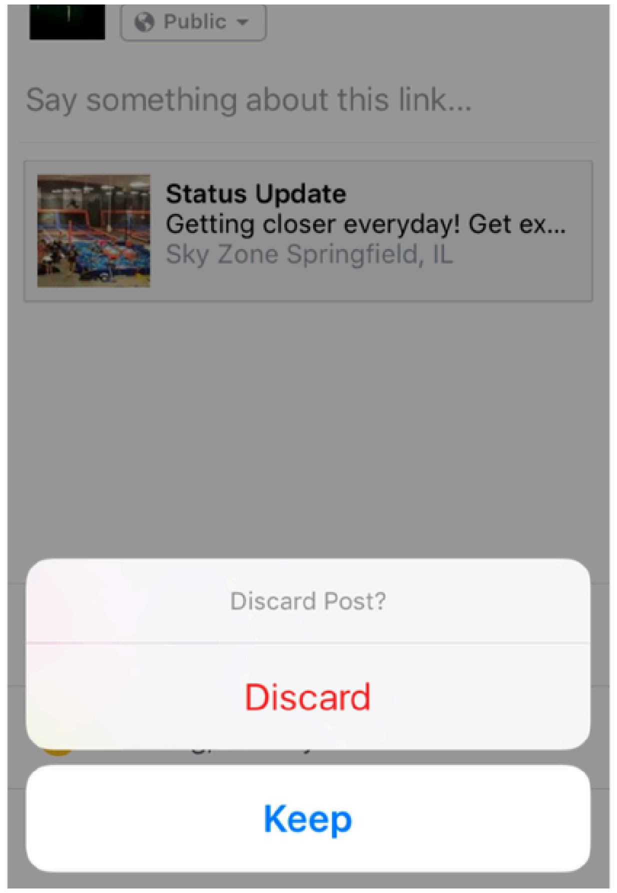
H-3 Control and Freedom

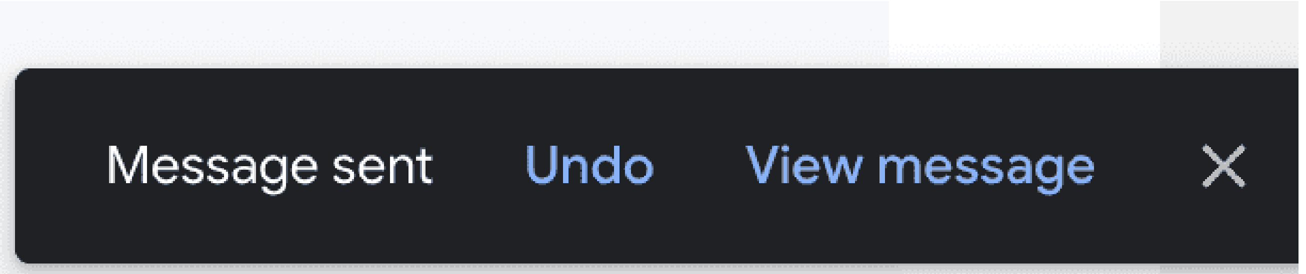
Provide good defaults and undo

Users should be able to undo, cancel, and recover easily

- Provide Undo / Redo for key actions
- Always include Back / Cancel
- Avoid trapping users in flows
- Allow safe exploration





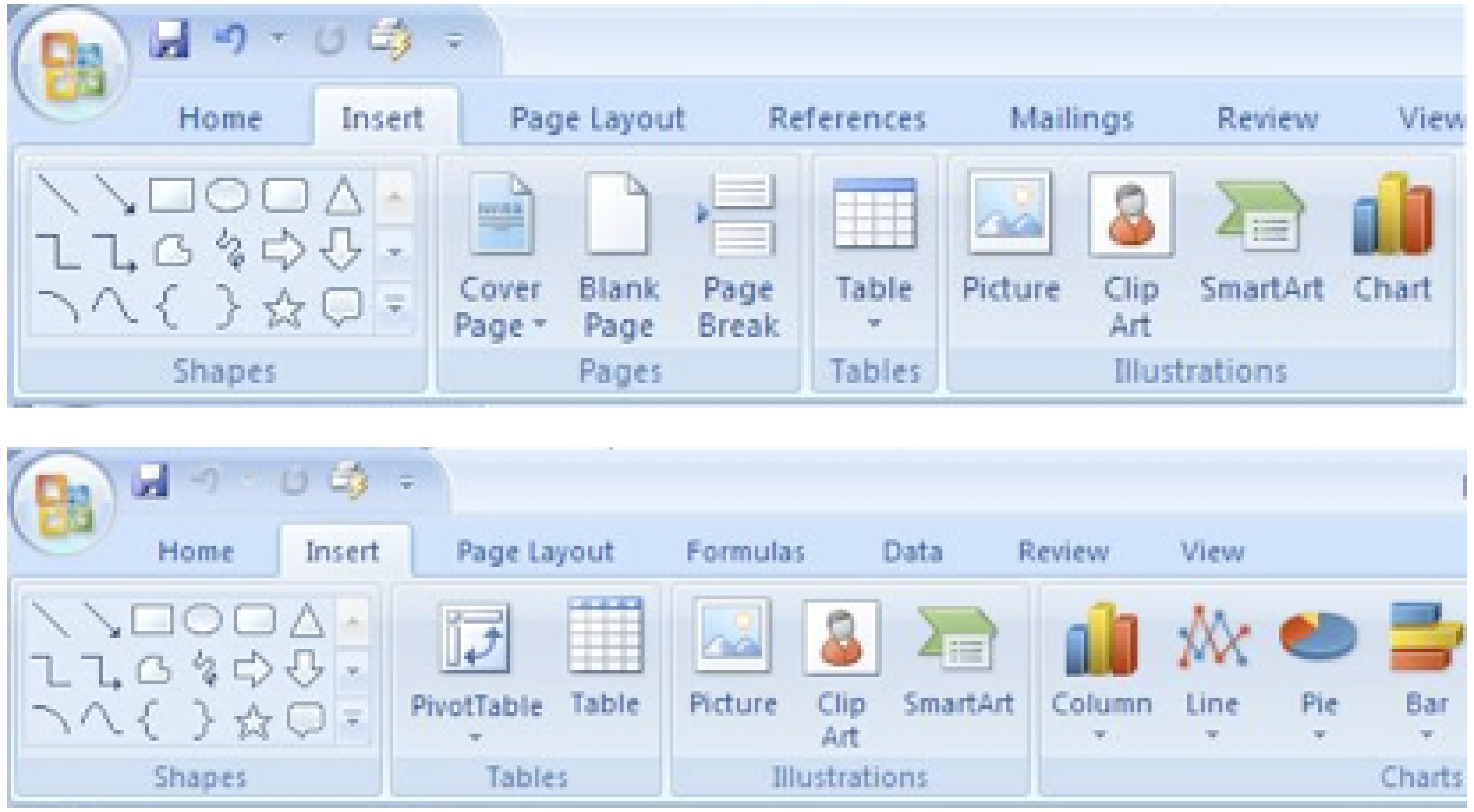


H-4 Consistency and Standards

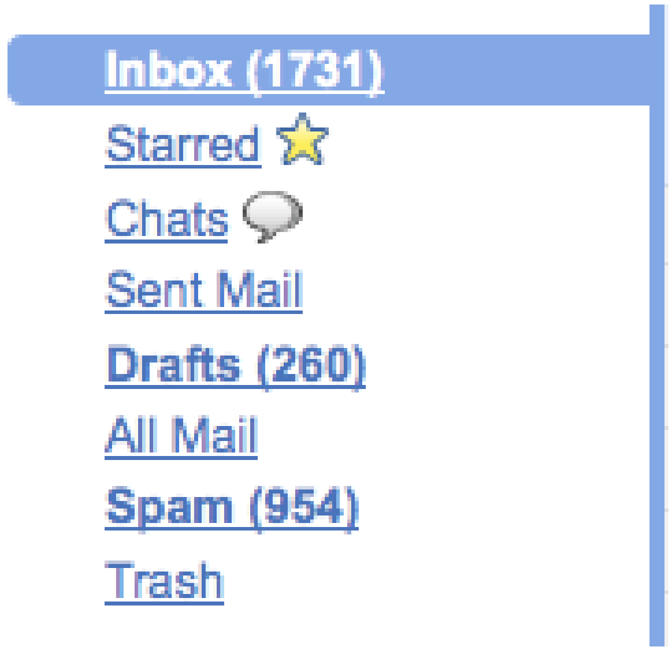
Use the same interface/interactions and language throughout

Use same patterns, labels, and behaviors across the system

- Use consistent labels (Cart vs Basket)
- Keep UI patterns consistent
- Follow platform conventions
- Maintain visual consistency (color, layout)



Example : consistency within a product or a family of products



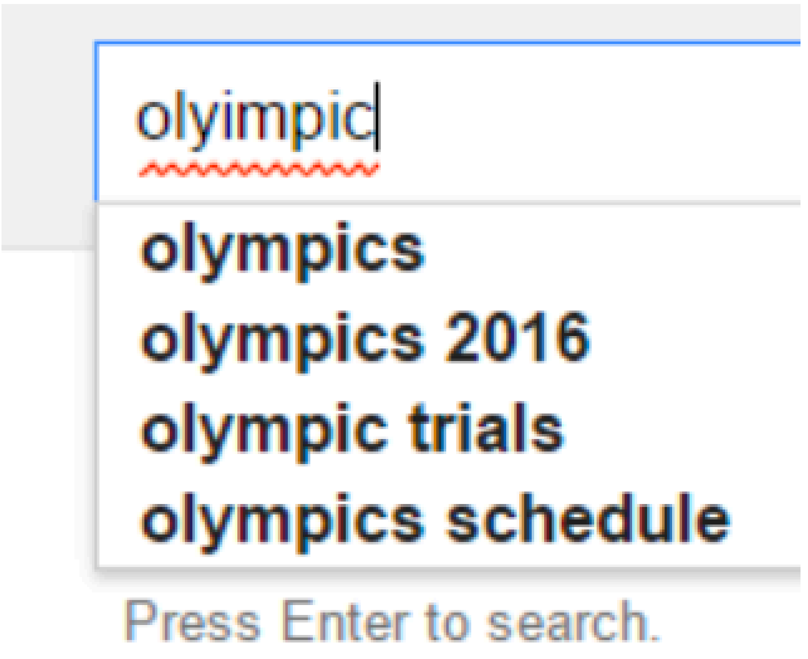
Example: Consistency by following established conventions

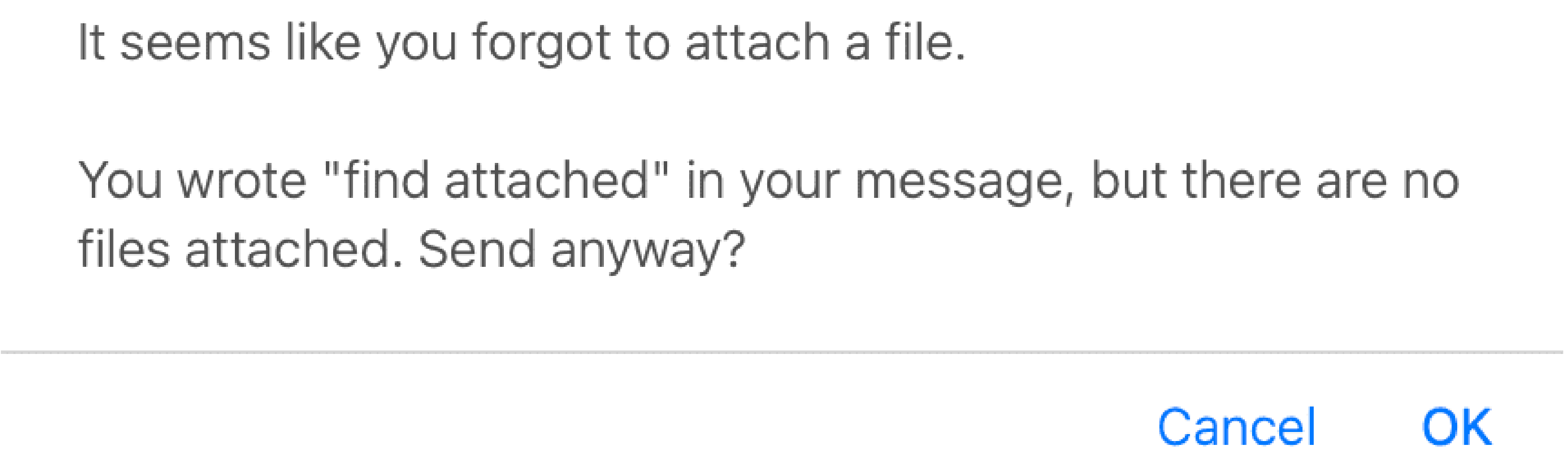
H-5 Error Prevention

Help users avoid making mistakes

Prevent errors before they happen

- Disable invalid actions (e.g., empty cart checkout)
- Provide inline validation (Cannot book ride without destination)
- Confirm destructive actions
- Use constraints & defaults





H-6 Recognition

Make information easy to discover

Make users recognize options instead of remembering

Recognition vs Remebering

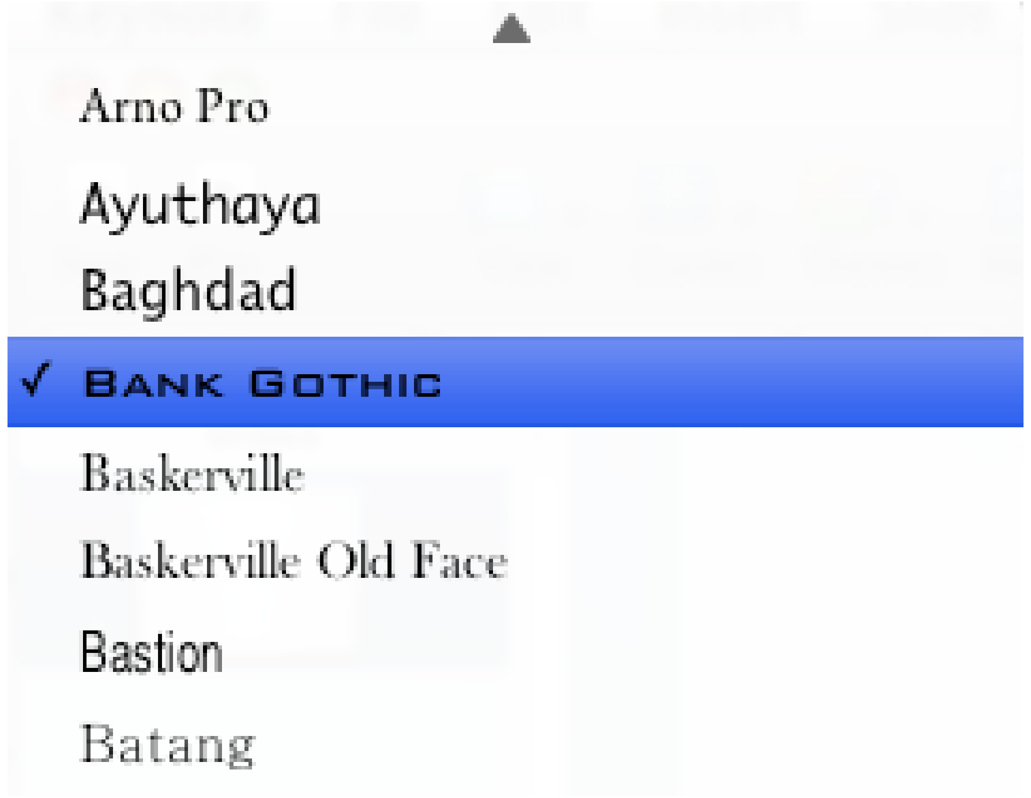
Recognition= seeing
It is our ability to “recognize” a piece of information as being familiar,

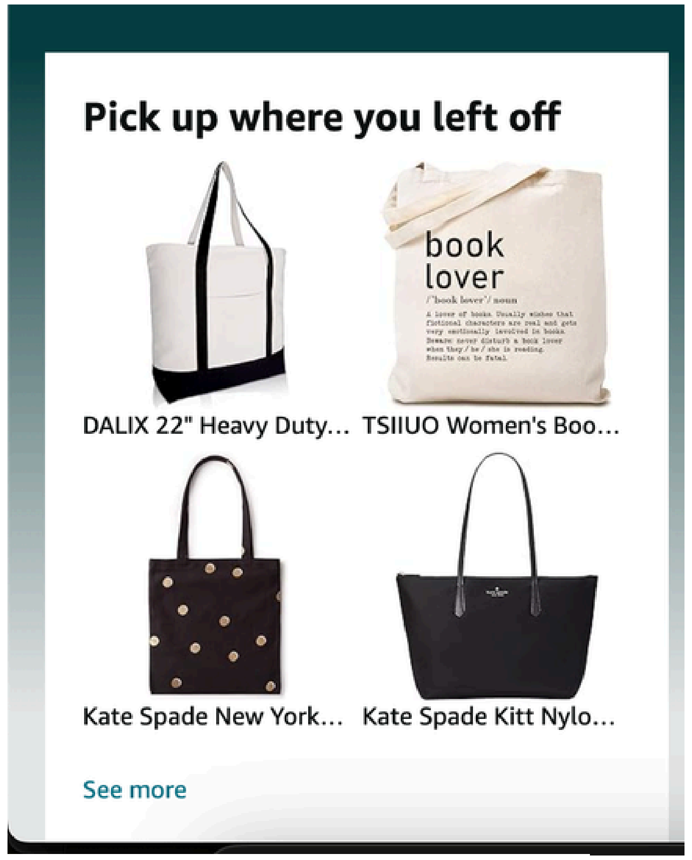
Remembering: the retrieval of related details from memory.
Humans often use a combination of **recognition and remembering** to retrieve information from memory.

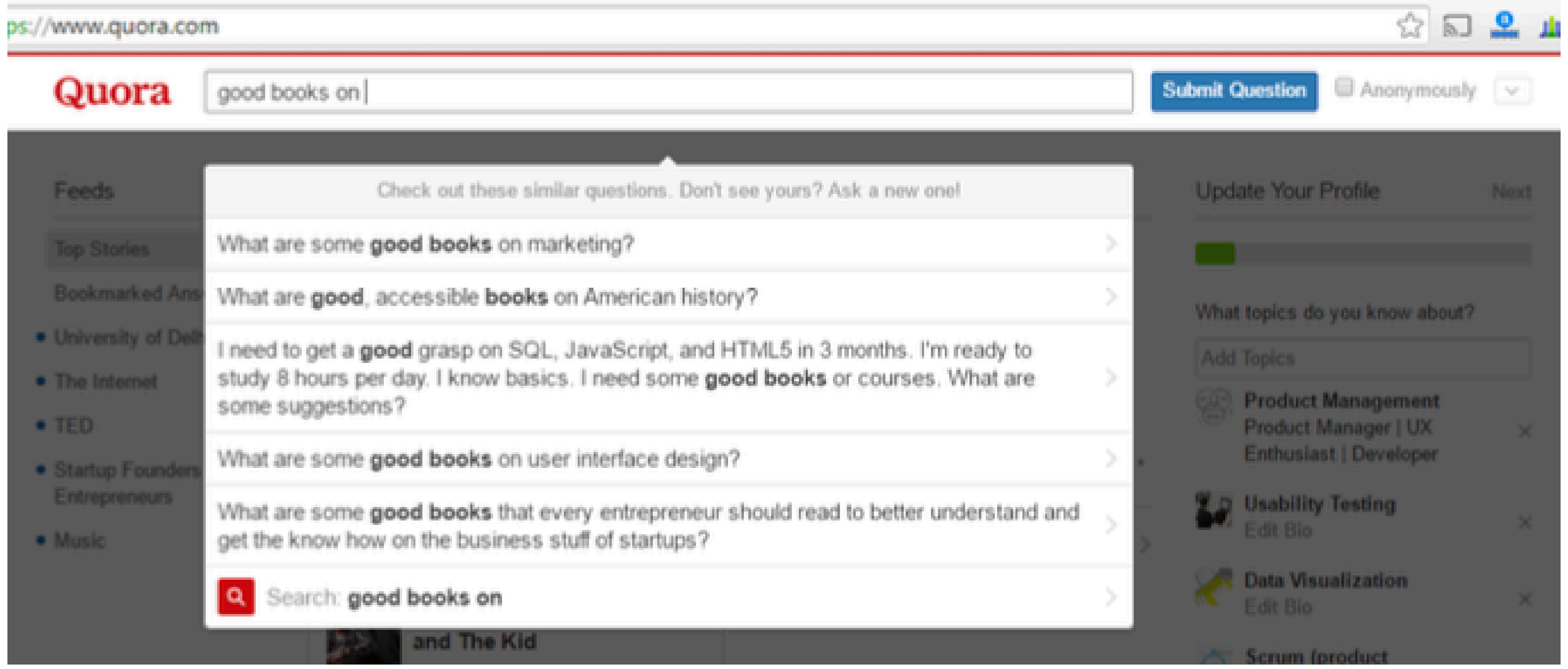
“ The **difference between recognition and remembering is the number of cues** that help memory retrieval
More cues - easy retrieval from memory”

H-6 Recognition

Make information easy to discover







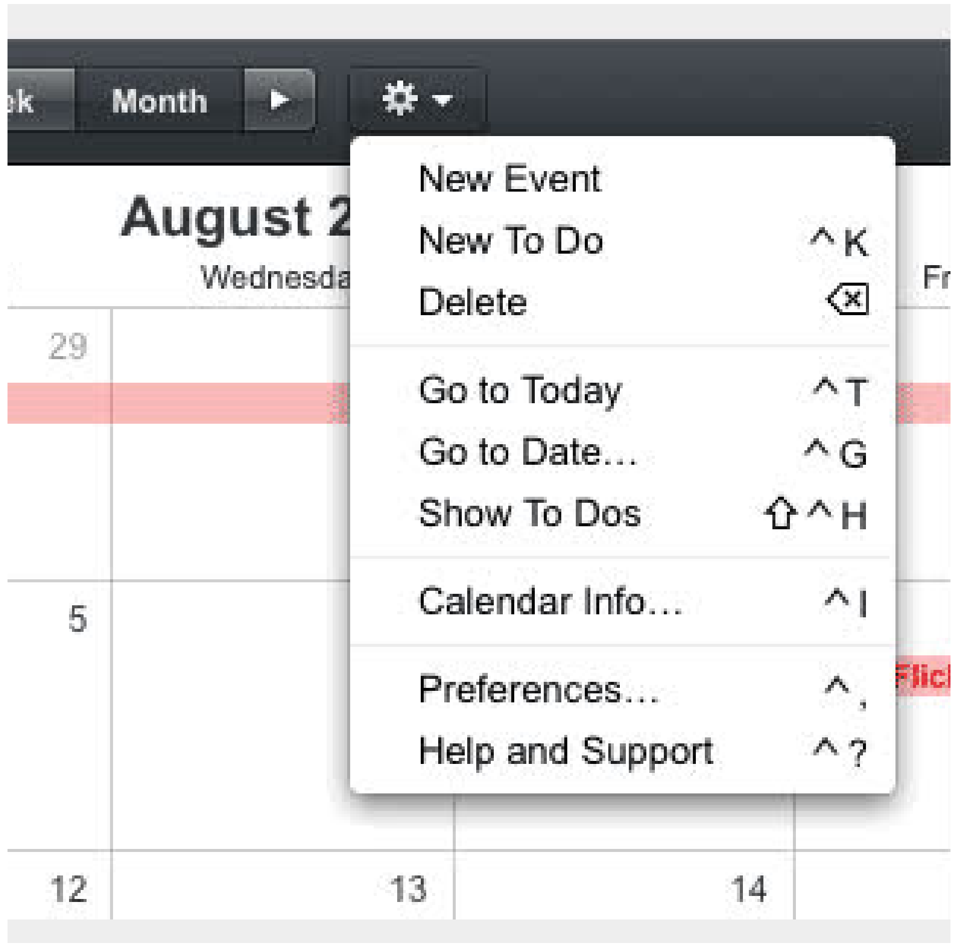
Suggestions help user recognising over recalling the query

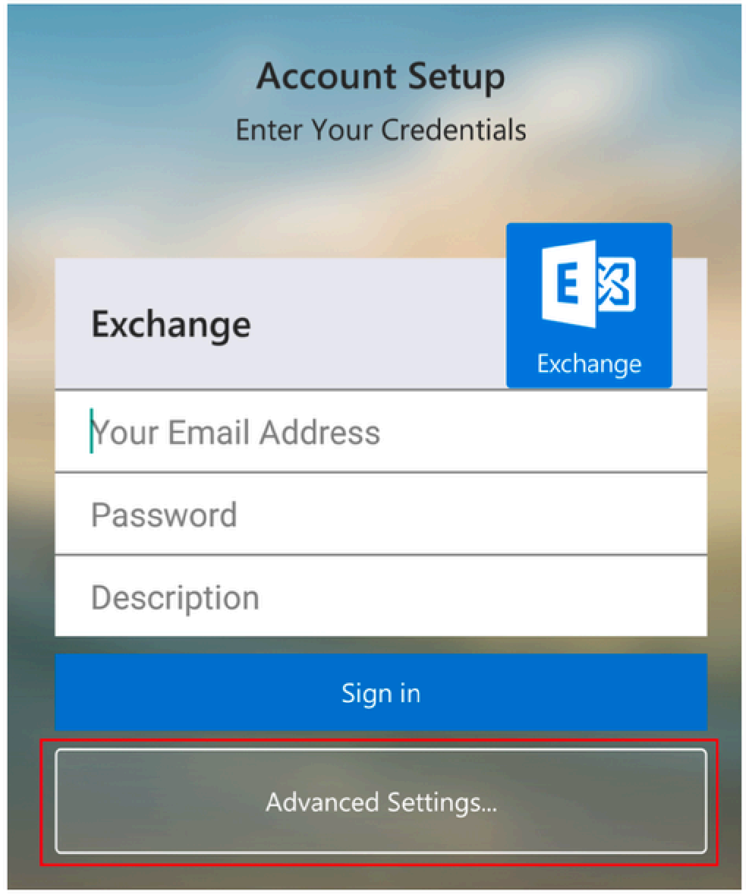
H-7 Flexibility

Allow for tasks that cater to both new and advanced users

Support both beginners and advanced users

- Provide shortcuts (power users)
- Offer defaults (beginners)
- Allow personalization
- Enable faster repeated actions



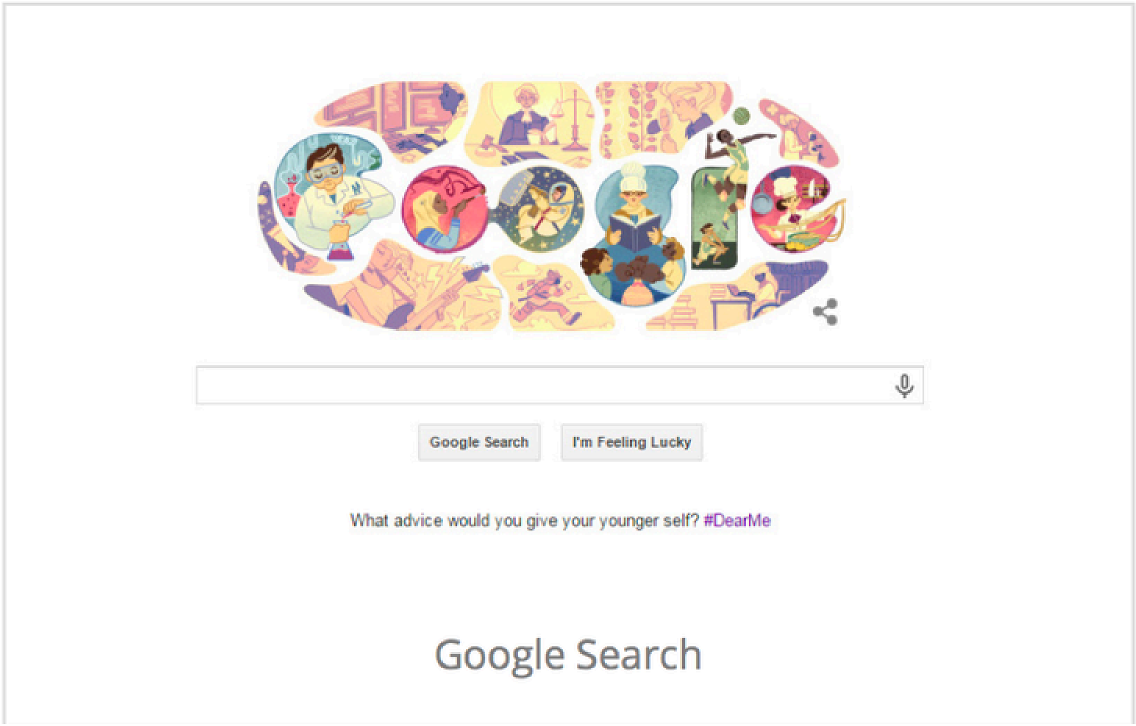


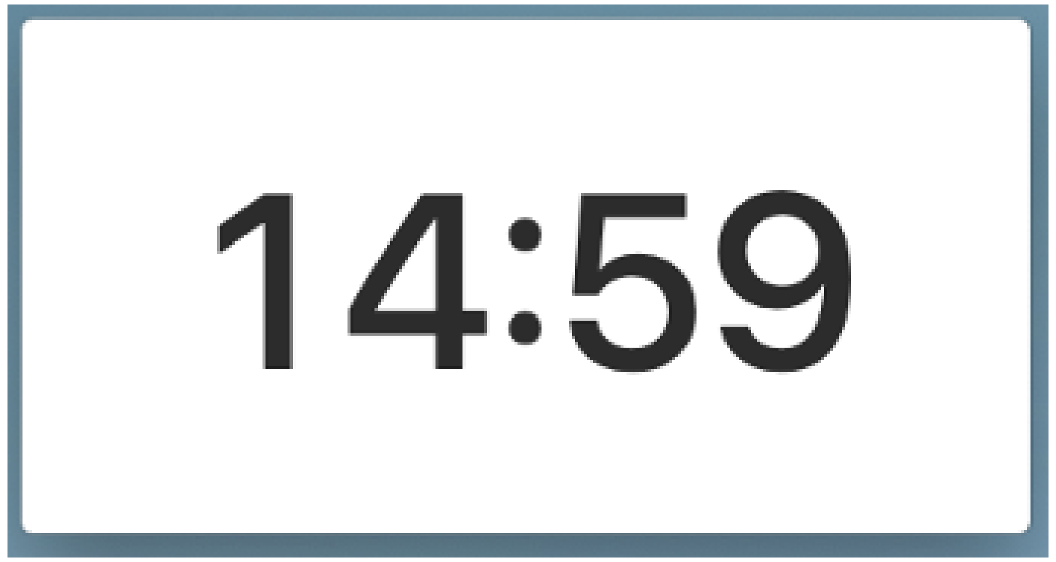
H-8 Minimalism

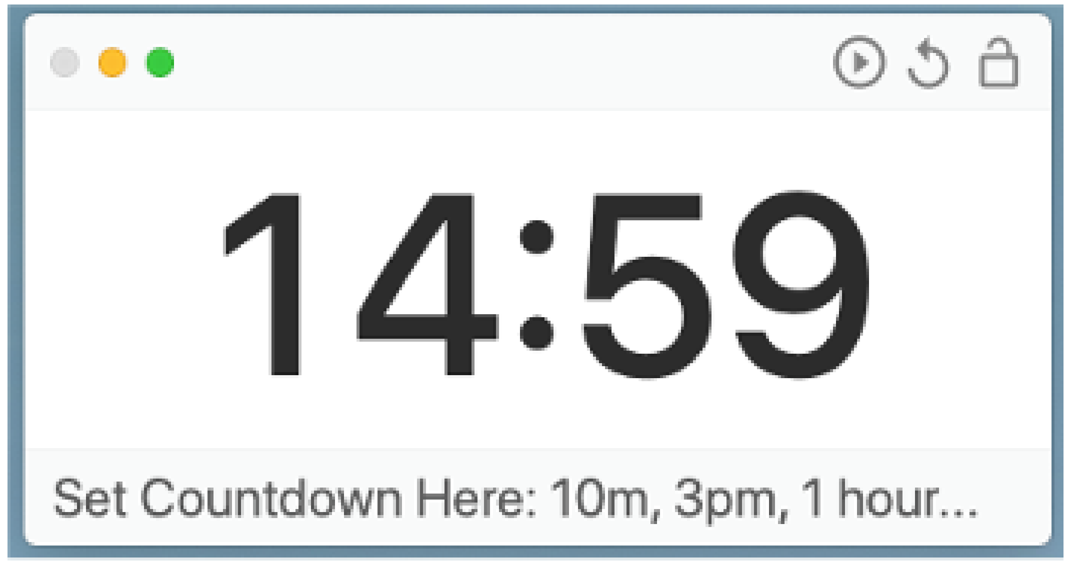
Provide only necessary information in an elegant way

Show only what is necessary right now

- Remove unnecessary elements
- Prioritize key information
- Use progressive disclosure
- Reduce visual clutter





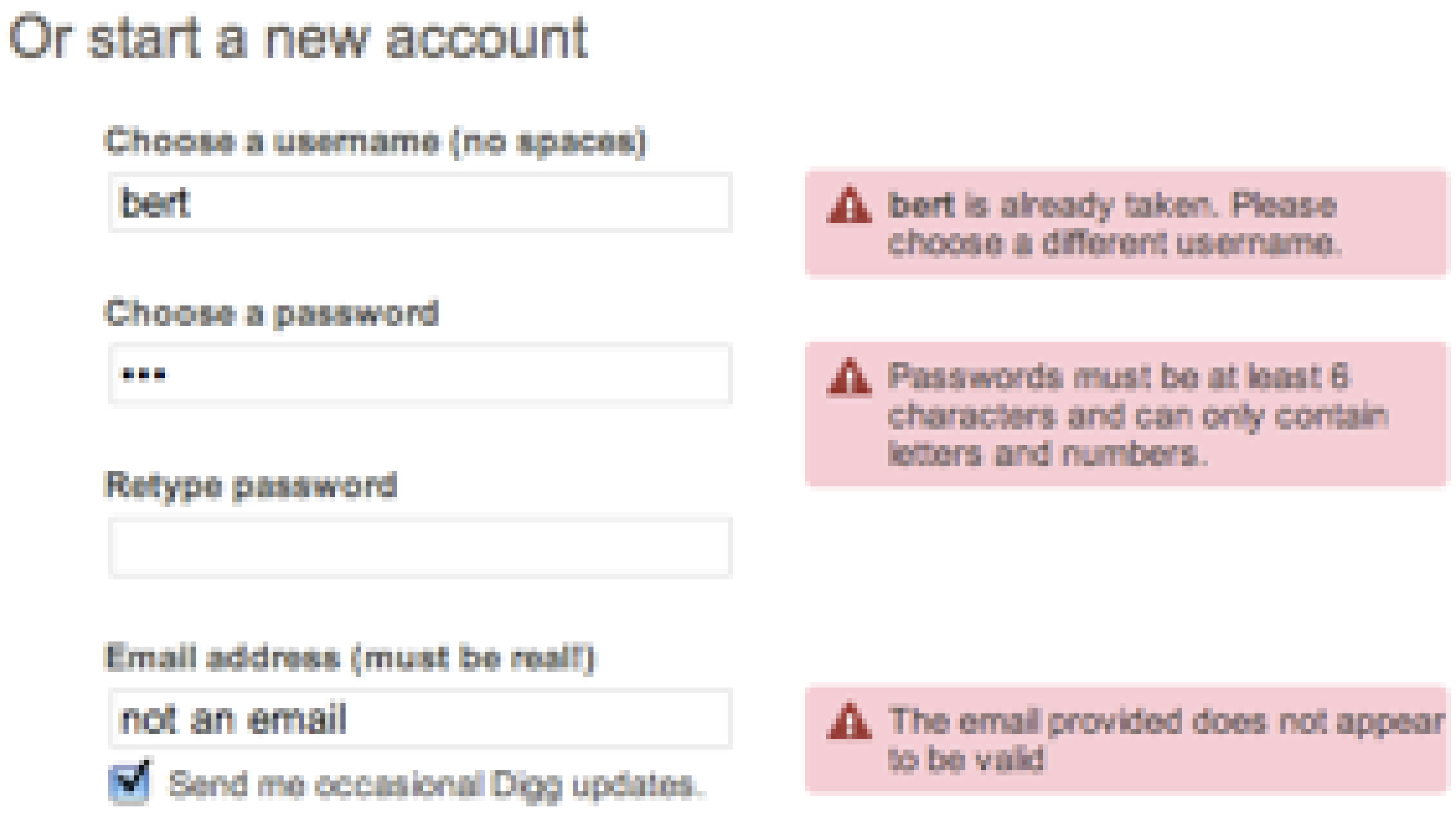


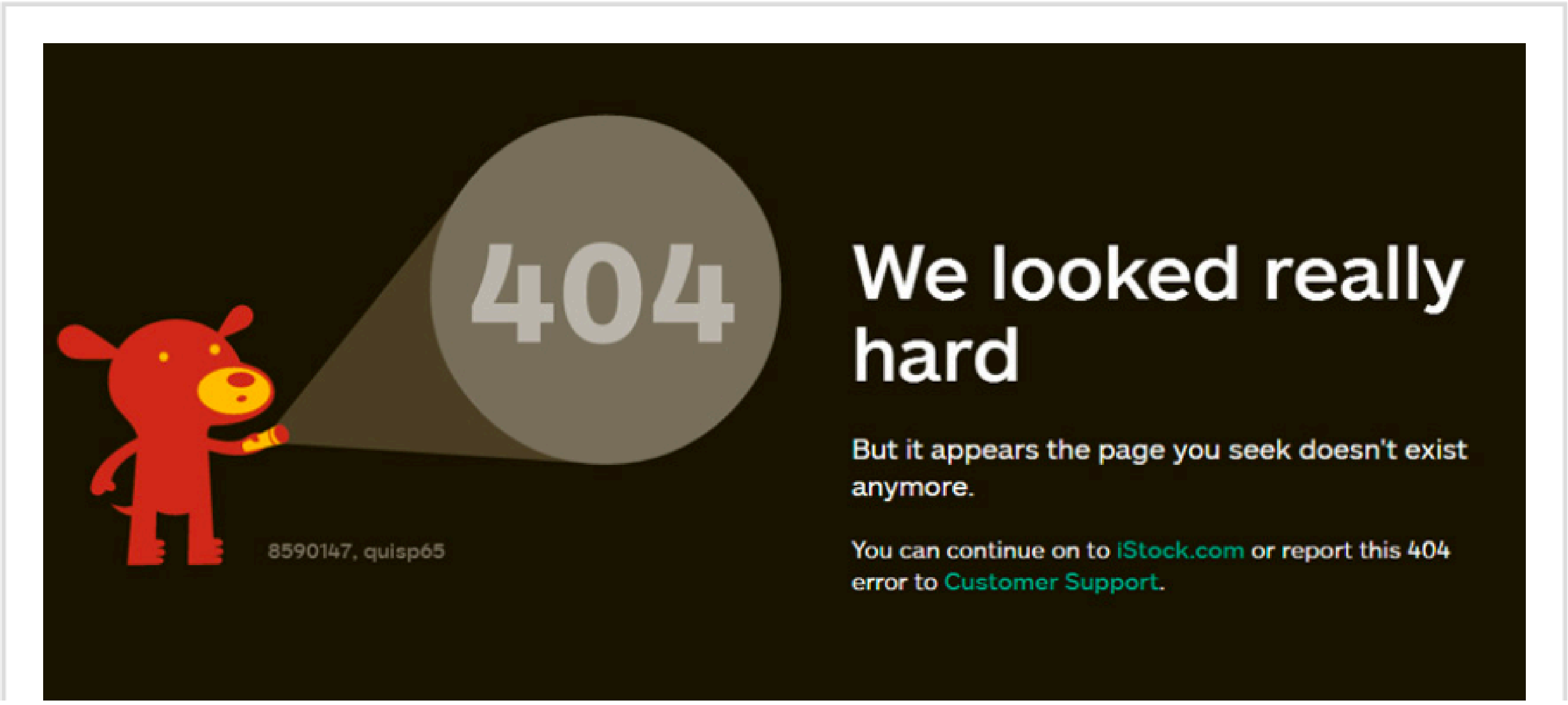
Provide progressive levels of detail.

H-9 Error Recovery

Help Users recognise, diagnose, and recover from errors

Help users understand and fix errors quickly
- Use clear, human-readable messages
- Explain what went wrong
- Suggest how to fix
- Provide recovery actions





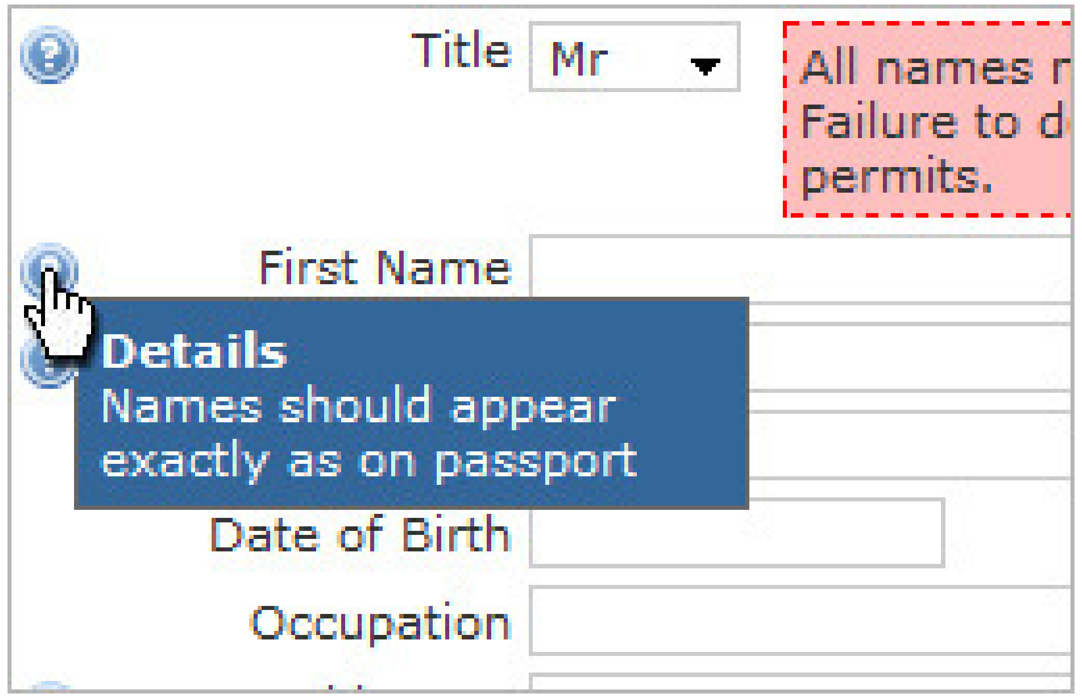
H-10 Help and Documentation

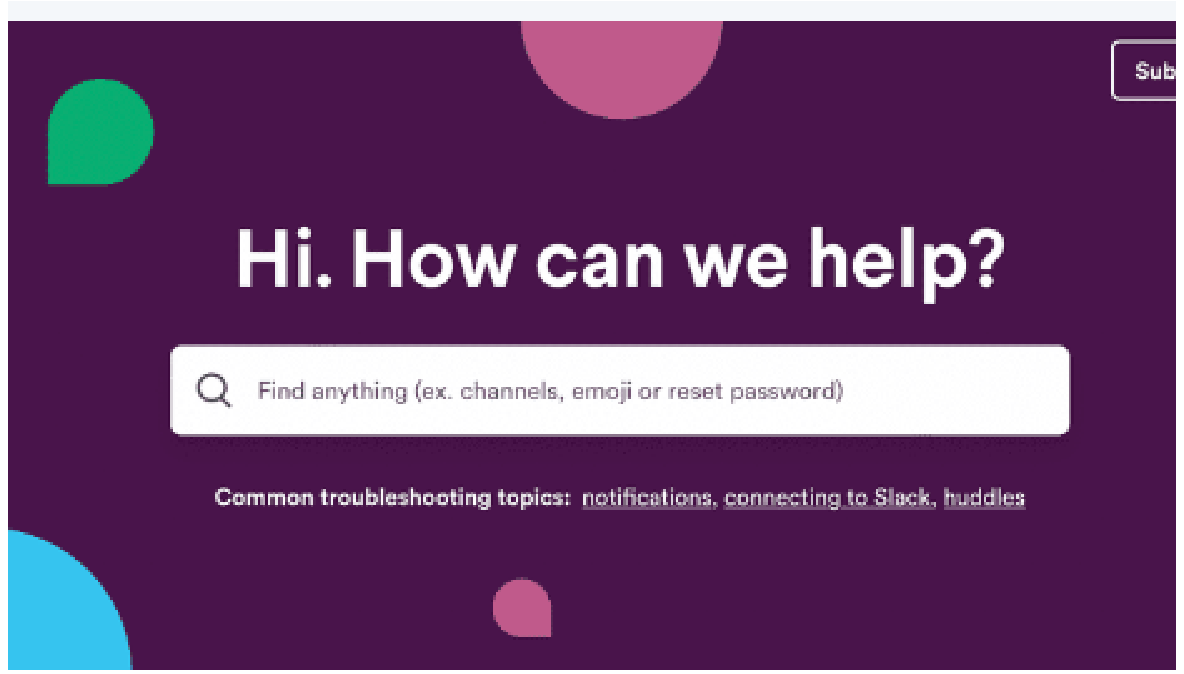
Use pro-active and in-place hints to guide users

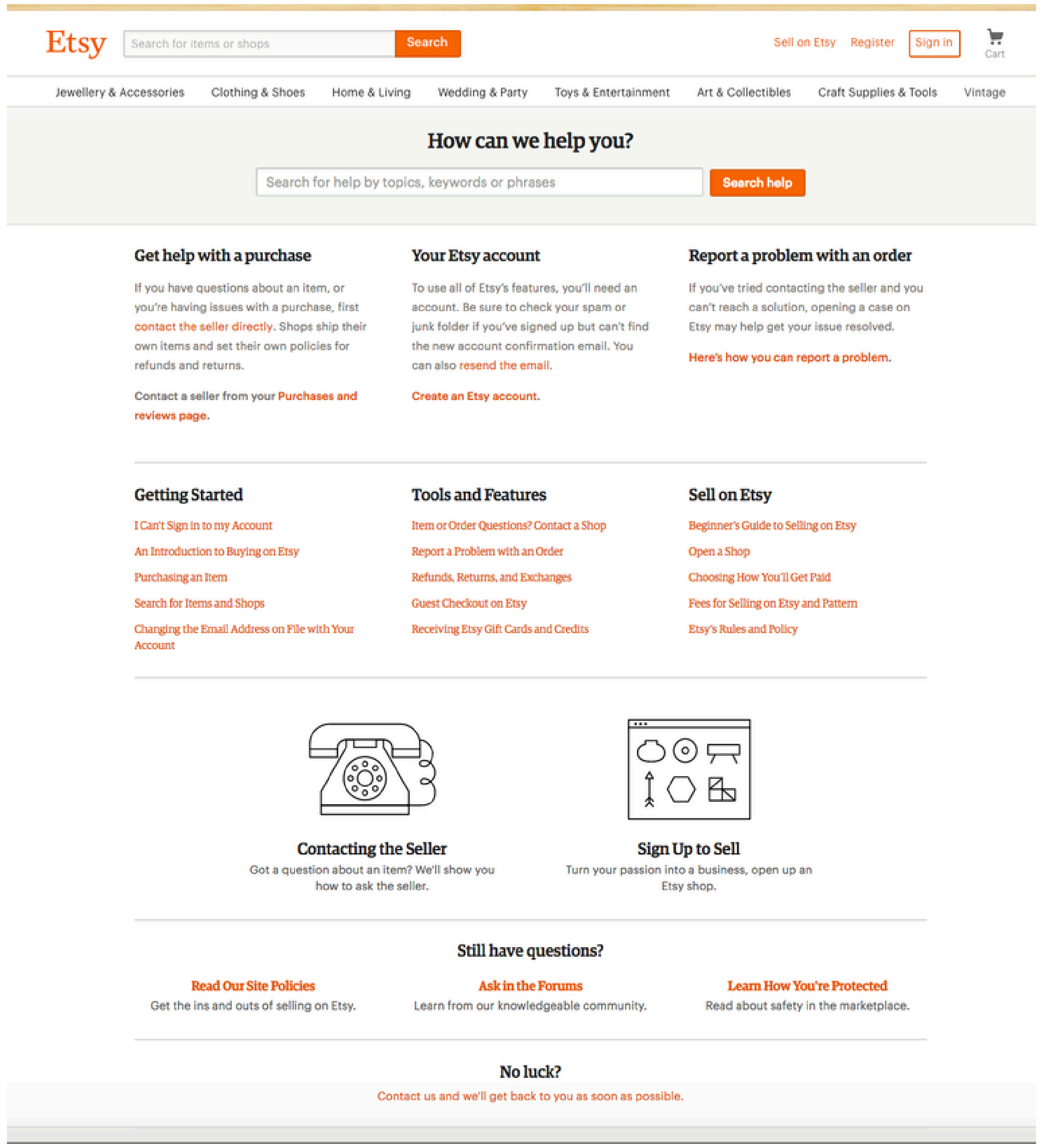
Provide help when and where needed

- Use inline hints (tooltips, placeholders)
- Provide contextual help
- Enable search in help
- Avoid forcing users to leave flow

(Users should be able to get help without interrupting their task)







Summary – Nielsen Heuristics

Visibility
Show system status and provide immediate feedback

Mapping
Use familiar metaphors and language

Control and freedom
Provide good defaults and undo

Consistency and standards
Use the same interface and language throughout

Error Prevention
Help users avoid making mistakes

Recognition
Make information easy to discover

Flexibility
Allow for tasks that cater to both new and advanced users

Minimalism
Provide only necessary information in an elegant way

Error Recovery
Help users recognize, diagnose, and recover from errors

Help and documentation
Use proactive and in-place hints to guide users

The 10 heuristics are good practices that can be applied to interface design to evaluate its usability.

These are based on the observation of real users and the problems they face when using any type of interface.

They have been used for many years.

They continue to be a valid instrument to detect usability problems today.

Heauristic Evaluation process

Choose heuristics → Experts review → Record issues → Rate severity

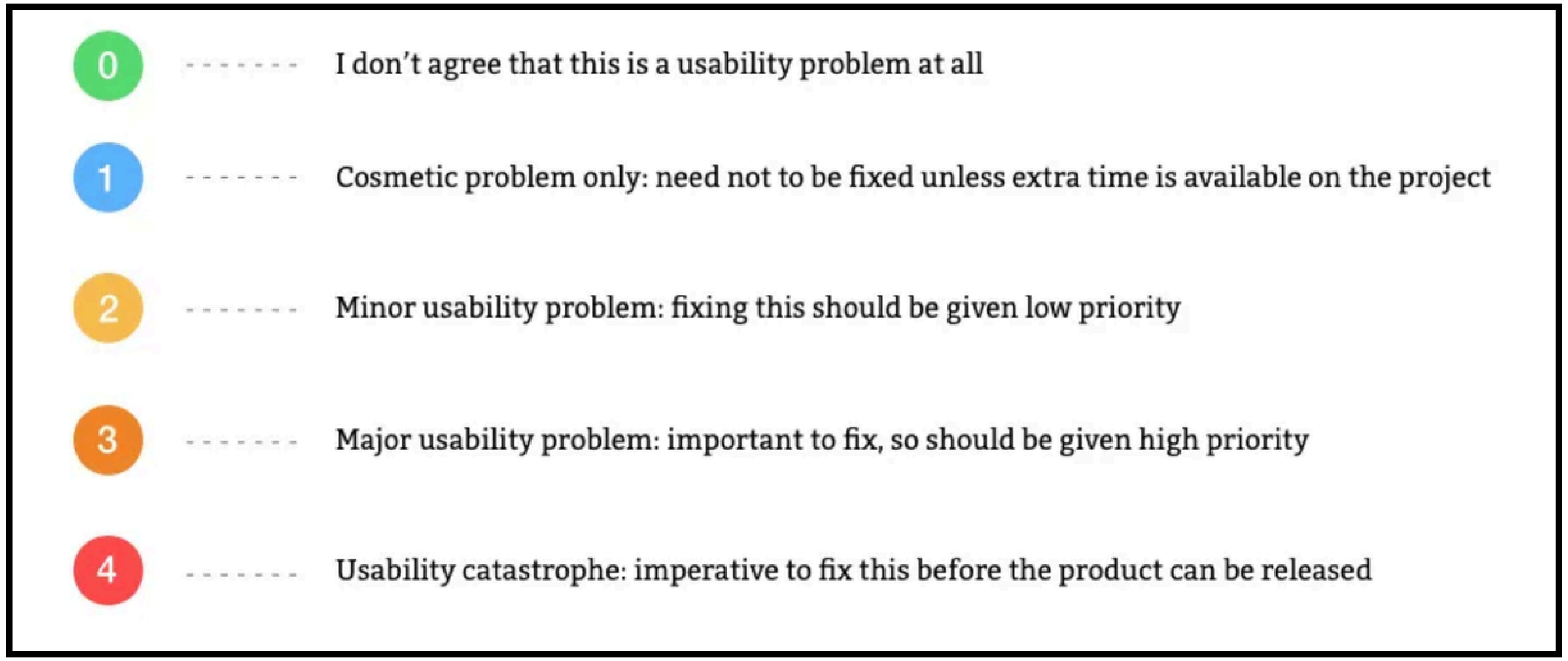
Rate Severity (0 = cosmetic .... 4= critical)

Severity depends on

- How **frequently** does the problem occur: Very often or rarely?

- The **impact** of the problem on users: Would this problem be a huge headache for the users or they can easily overcome the situation?

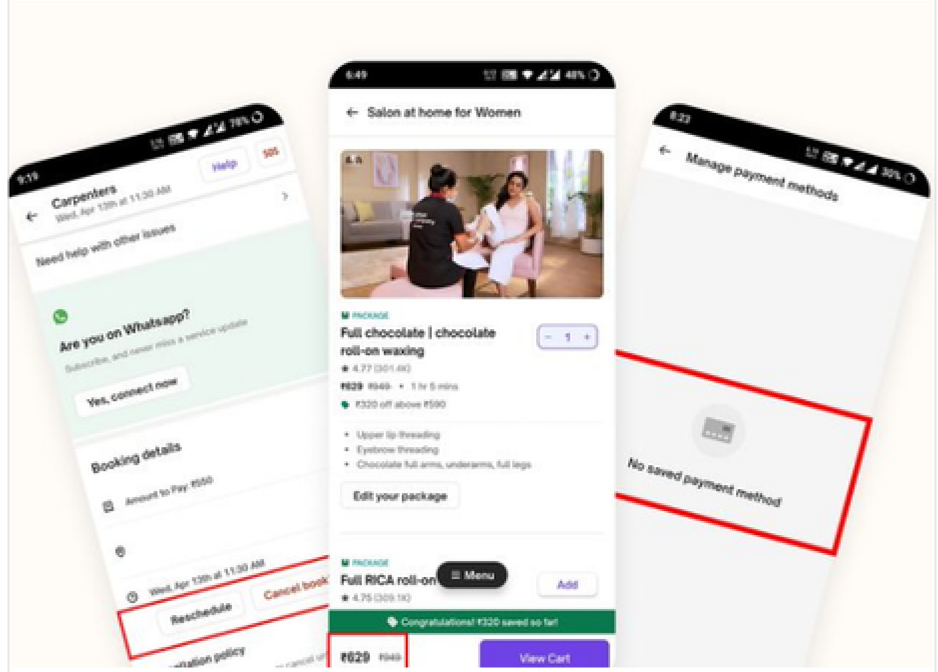
- Persistence of the problem
Is this something that the user can overcome once they know about it, or would this be a big concern for the user again and again?



References: Video Links

H1: Visibility of System Status https://youtu.be/cTtc90jCULU	H6 Recognition rather than recall https://youtu.be/6glQPp6q4Jc
H2: Match Between the System and the Real World https://youtu.be/0TAt9Pln51g	H7 Flexibility and efficiency of use https://youtu.be/LoTdRTBB8BQ
H3: User Control & Freedom https://youtu.be/MXuk-fdbr0A	H8 Aesthetic and minimalist design https://youtu.be/ZgbRmeWDgd0
H4: Consistency and Standards https://youtu.be/Ibndy9KLOSQ	H9 Help users recognize, diagnose, recover from errors https://youtu.be/cCun-ReLTFI
H5: Error Prevention https://youtu.be/imS9s1DUY-I	H10 Help and documentation https://youtu.be/iIQVRzatb50

References : Relevant Articles

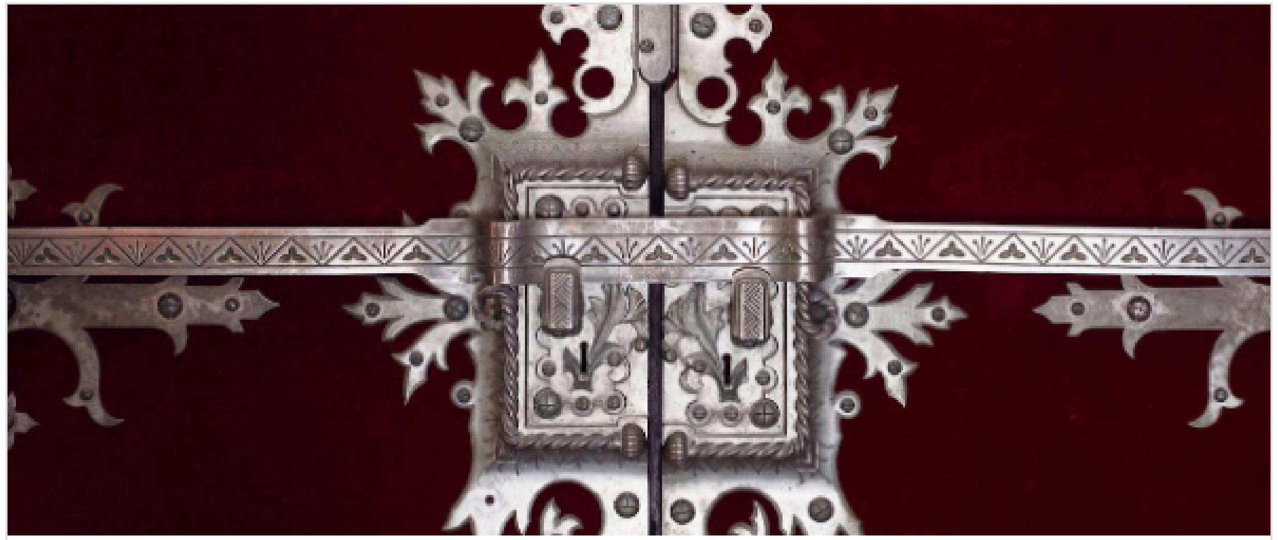


Heuristics Evaluation of Urban Company — A UX Case Study

Urban Company is Asia's largest online home services platform. Its services range from beauty treatments to home cleaning to appliances...

Medium / Nov 8, 2022

https://tinyurl.com/heuristicReading-1

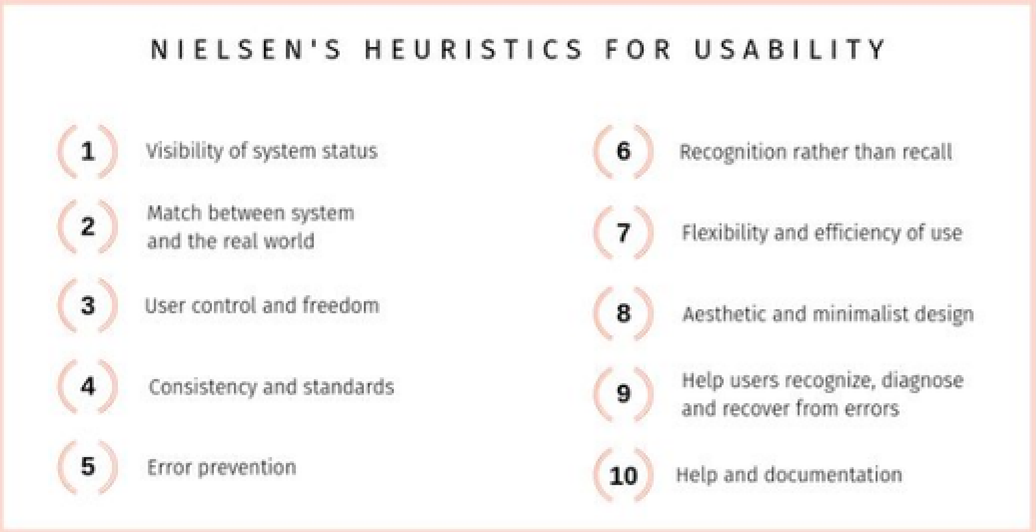


10 Usability Heuristics and How to Apply Them to App Design - Shopify Singapore

This article will review the 10 usability heuristics coined by Jakob Nielsen so you can create a user-friendly app.

Shopify

https://tinyurl.com/heuristicsReading-2



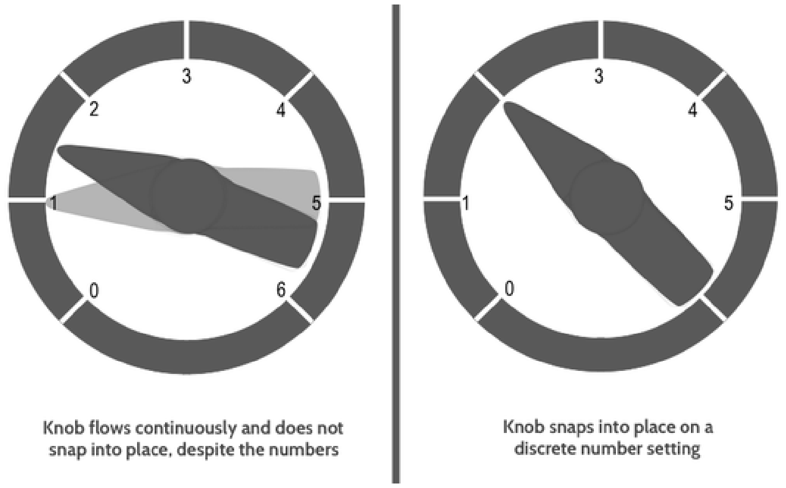
Nielsen's Heuristics as the North Star of User Experience

Have you ever tried out different Apps or sites and noticed that one of them gives you instructions and directs your actions while the...

Medium / Nov 18, 2023

https://tinyurl.com/heuristicsReading-3

Example- Heuristics Spotting



Knob flows continuously and does not snap into place, despite the numbers

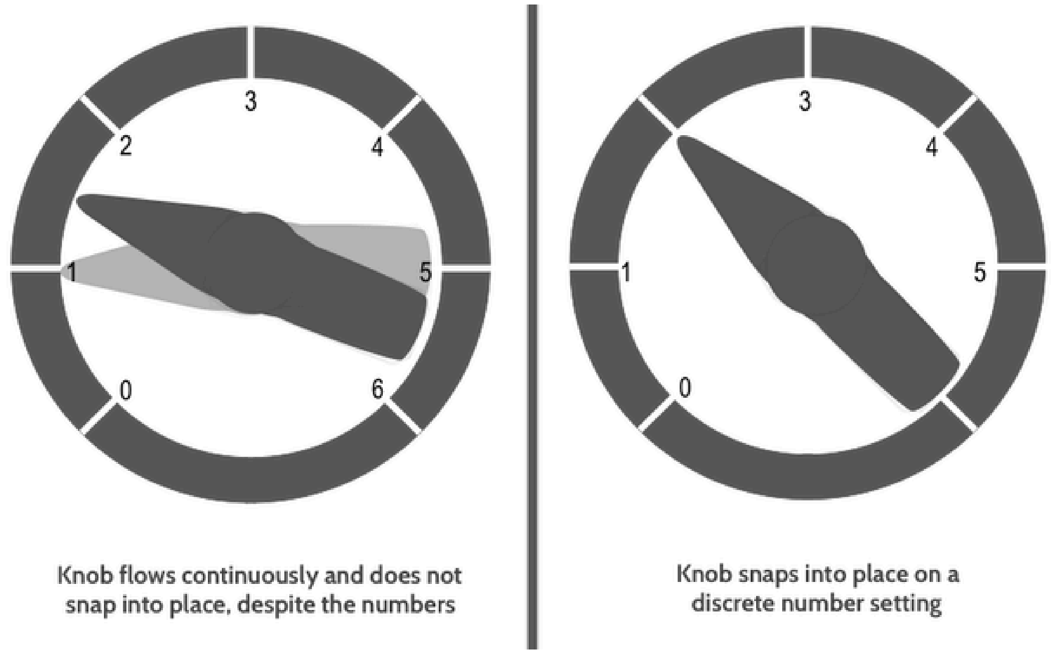
Knob snaps into place on a discrete number setting

Example

A physician treating a patient with oxygen set the flow control knob between 1 and 2 liters per minute, not realizing that the scale numbers represented discrete, rather than continuous, settings. There was no oxygen flow between the settings, yet the knob rotated smoothly, suggesting that intermediate settings were possible.

The patient, an infant, became hypoxic before the error was discovered.

One solution would have been a rotary control that snaps into a discrete setting. Also, some indication of flow also should have been provided



Knob flows continuously and does not snap into place, despite the numbers

Knob snaps into place on a discrete number setting

Observations

The oxygen flow control should have "snapped" from one setting to the next with no chance of pausing between settings.

The physician's expectation of how the system worked was different from how it actually worked.

The equipment should also have been designed with a digital panel displaying the flow rate.

Knob flows continuously and does not snap into place, despite the numbers

The knob rotates smoothly, giving the illusion of continuous control, but oxygen only flows at discrete settings.
Violates:
H-2 Mapping
H-1Visibility

The physician's expectation differed from the system's actual behavior.
Violates:
H-2 Mapping
H-5 Error Prevention

No clear indication of current flow rate or whether oxygen was flowing.
Violates:
H-1 Visibility

See Appendix-1 Slides: for Chatbot Heuristics

See Appendix-2 Slides: for Use of AI for Heuristic EvaluationChatbot Heuristics

Heuristics for evaluating design of AI features

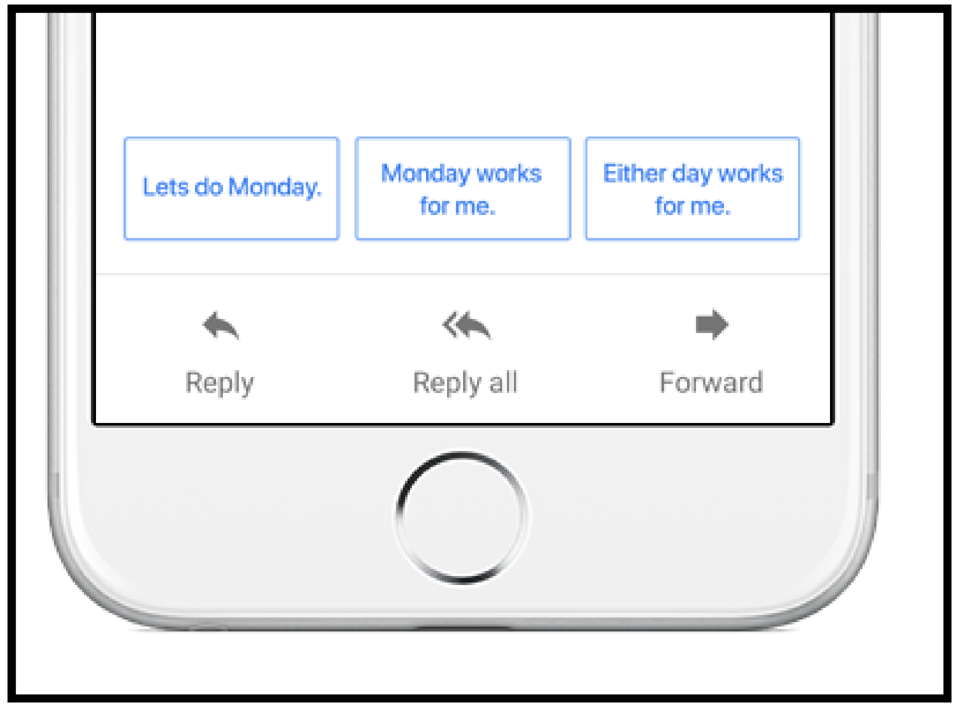
A few key expectations/ principles :

- Visibility of AI Presence
- Transparency - when/ how AI is operating
- Aesthetics associated with AI features
- Protection from Harmful content
- Clarity - what can AI do/ not do
- Delegation and user Control
- Predictability
- Context-Sensitive AI recommendations/suggestions
- Error recovery

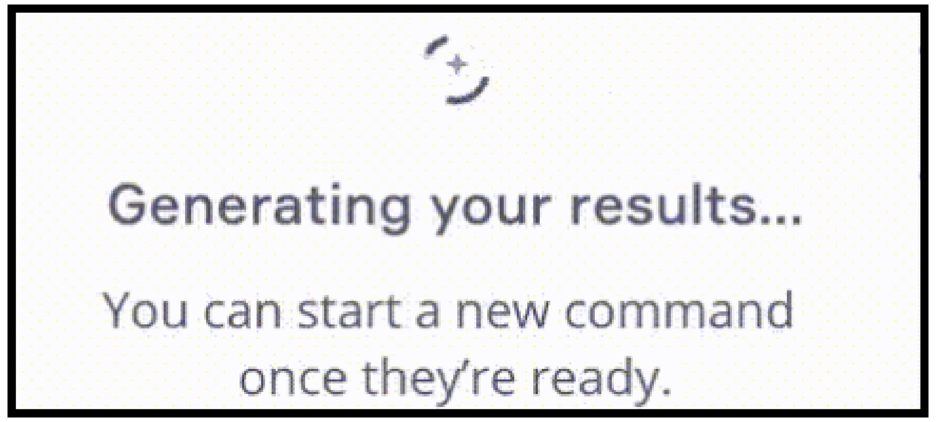
Visibility of AI Presence

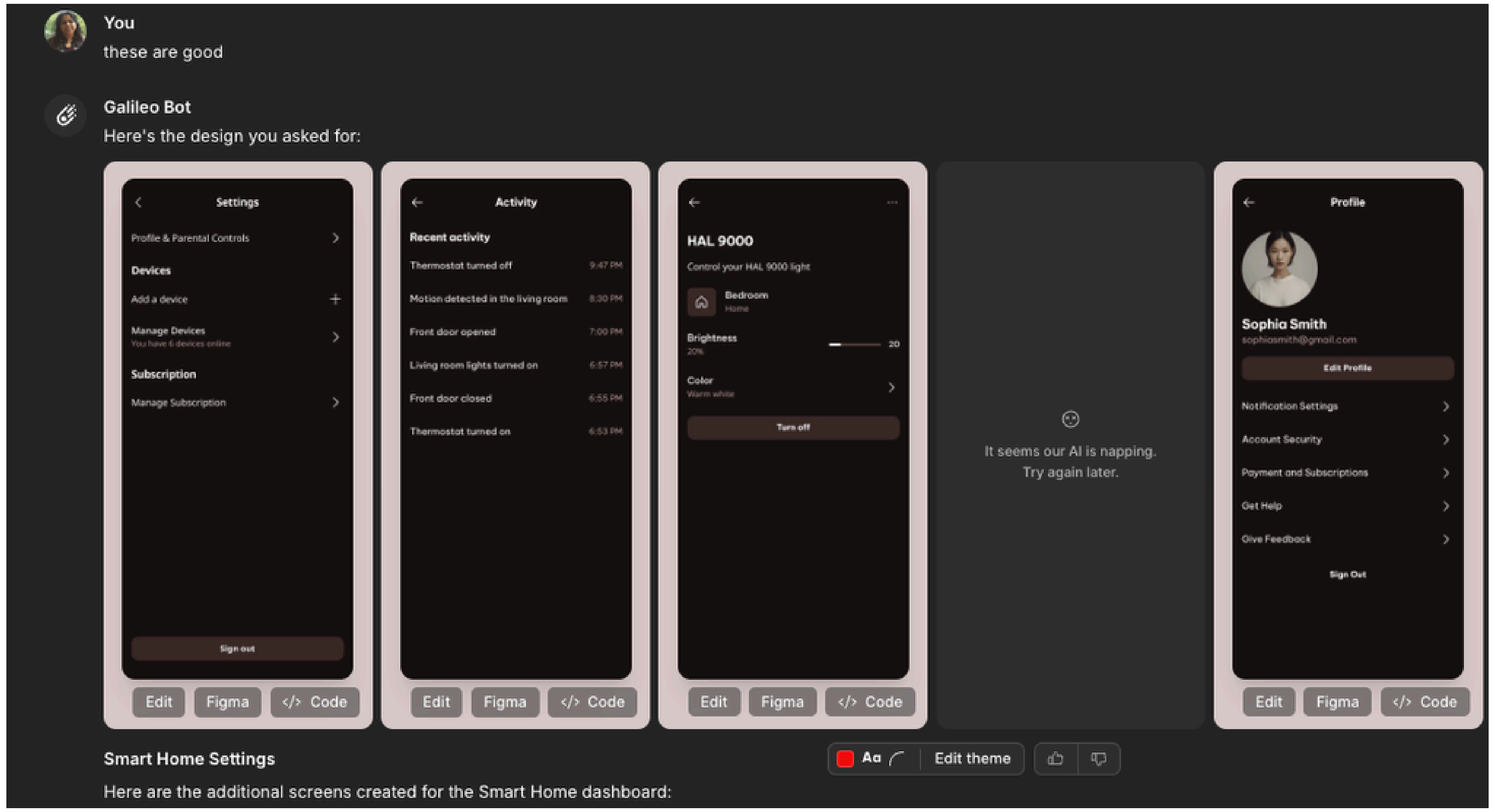
- Clearly indicate when AI-generated content/actions occur.

For example: An icon or label clearly states "Generated by AI."



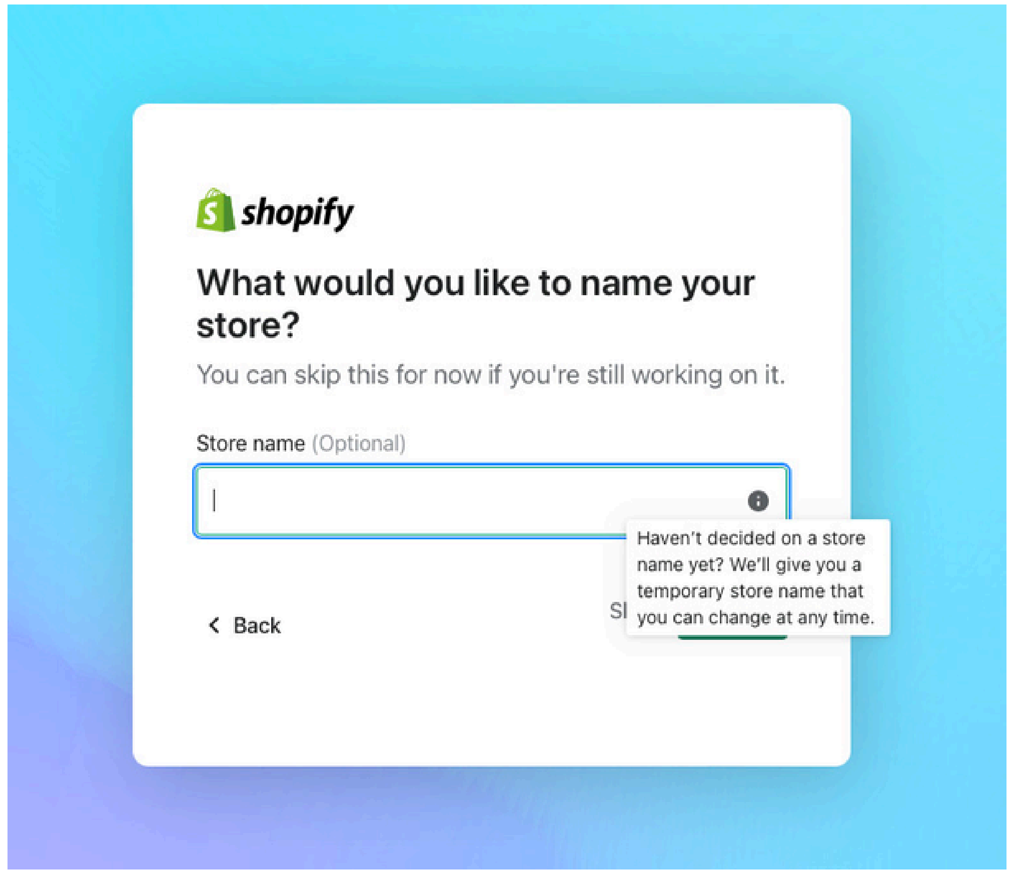
Gmail clearly shows suggested Smart replies

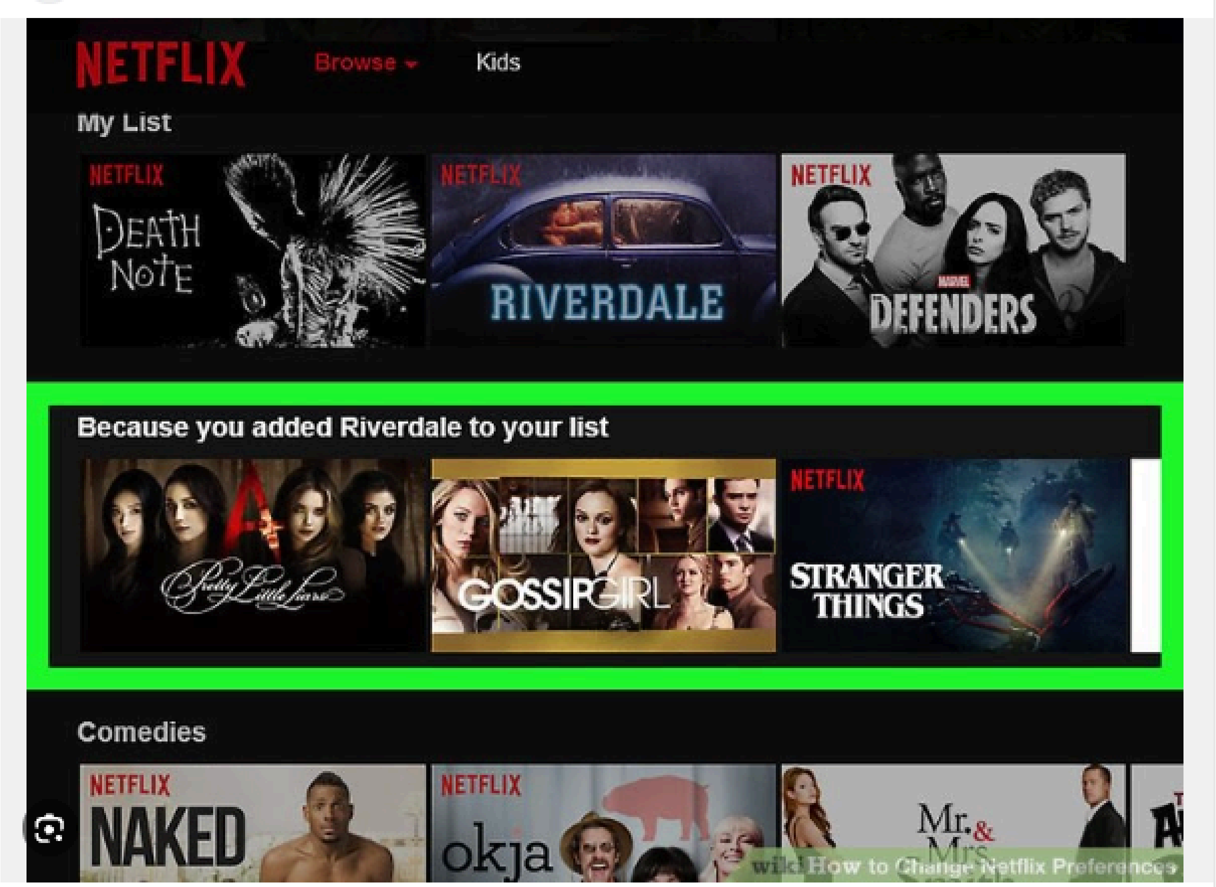




Transparency of AI Operations

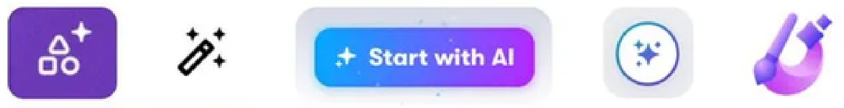
- Clearly communicate when/how AI is operating.





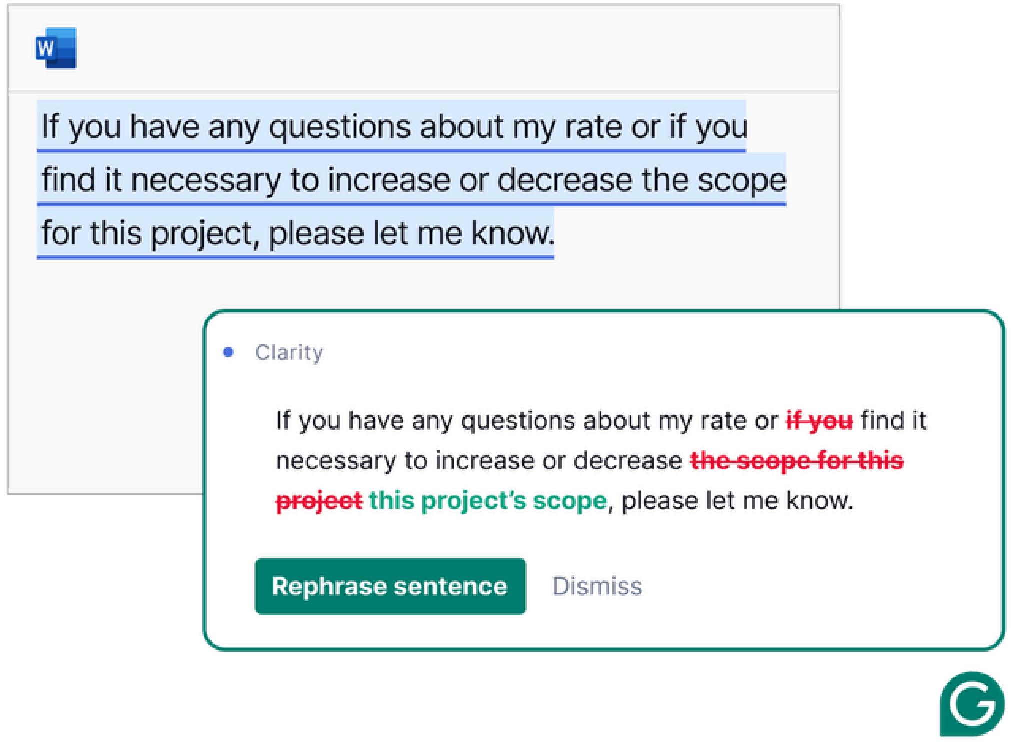
Netflix: Recommendations based on your recent activity.

Aesthetic of AI



- Differentiate AI-driven content visually (color, icon, styling).

For example : AI suggestions highlighted in a specific color



Protection from Harmful or Unwanted Content

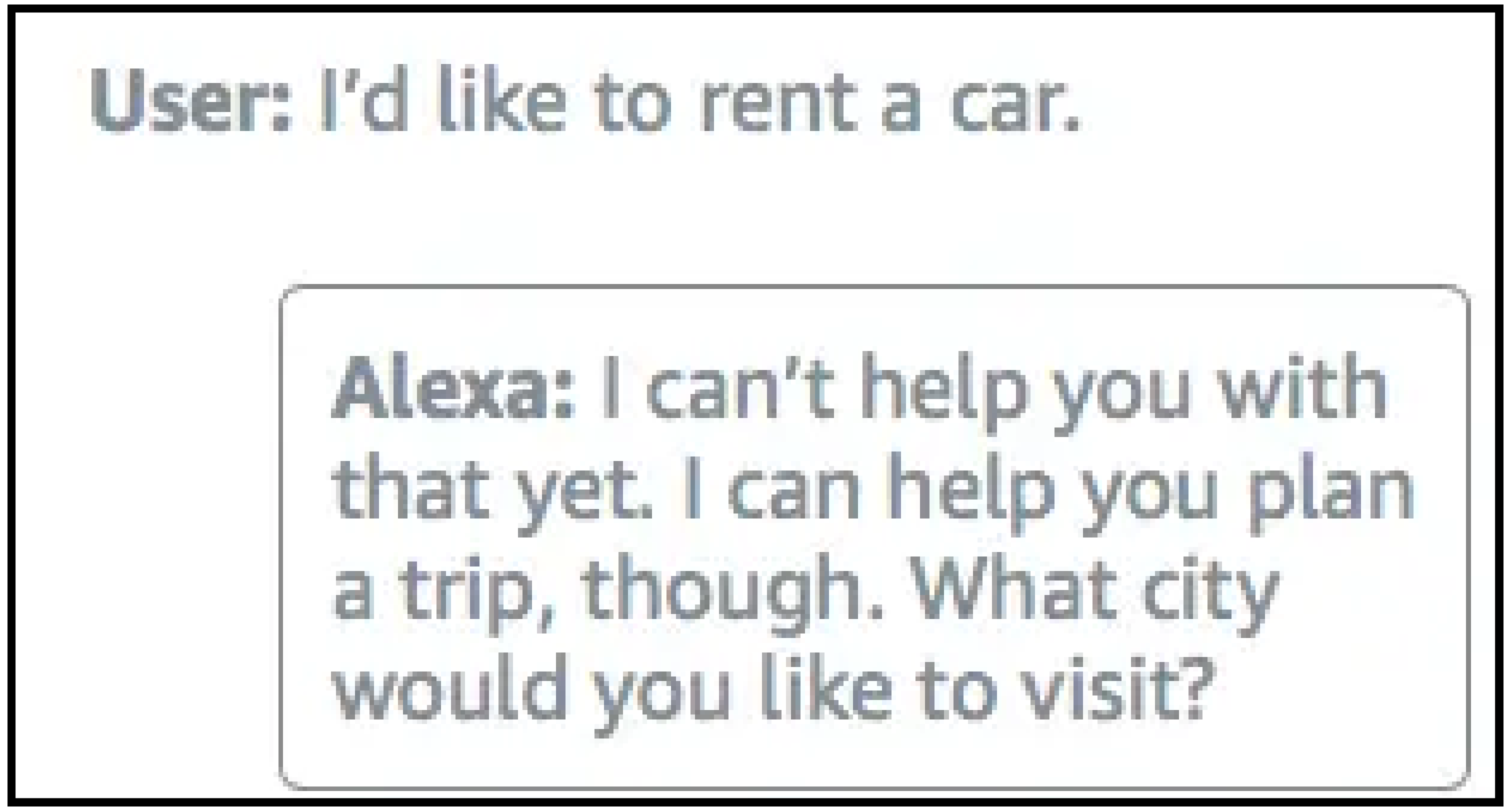
Protection from Harmful or Unwanted Content

- Safeguard against inappropriate, misleading, or harmful content.

e.g. blocking offensive language or inappropriate images

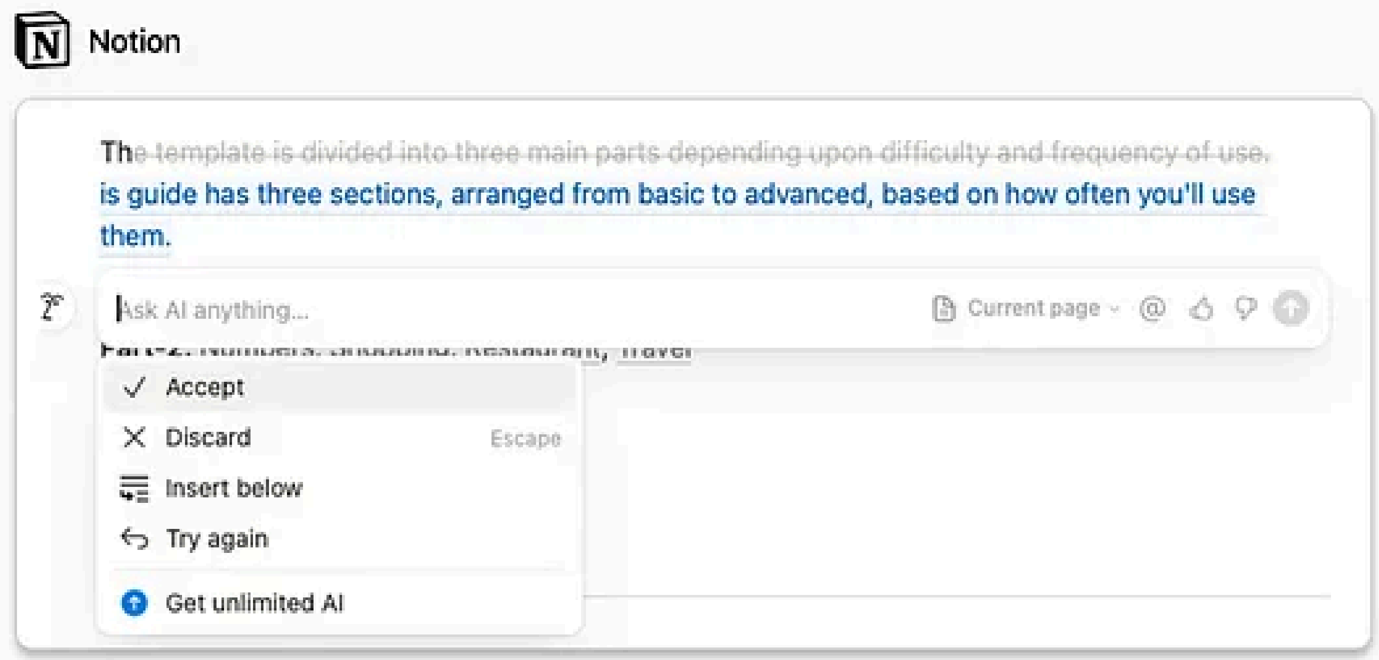
Clarity of AI Capabilities & Limitations

- Clearly state what AI can/cannot do.



Appropriate Delegation and User Control

- Users can easily delegate tasks to AI or reclaim control.



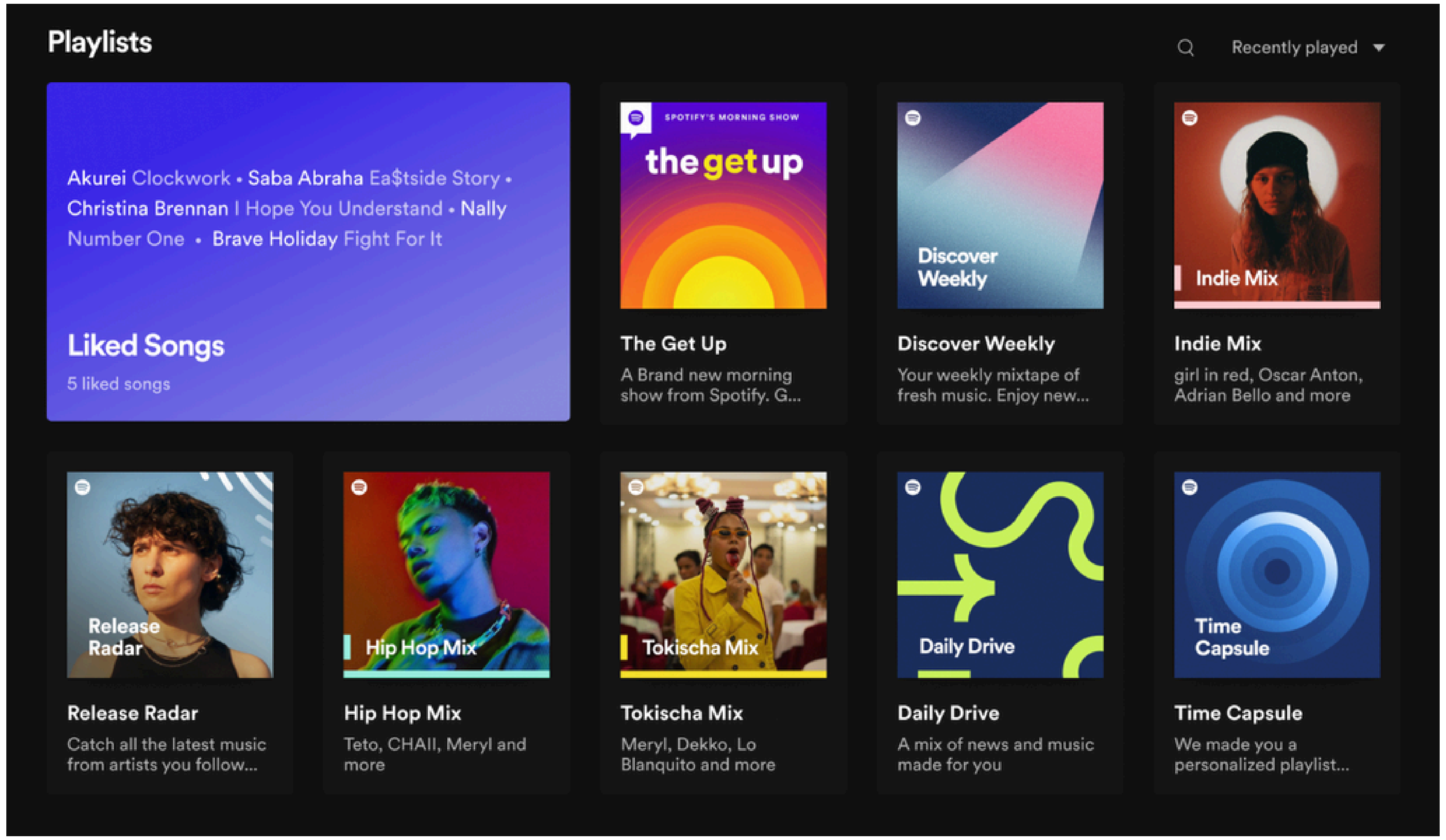
Notion AI gives you contextual suggestions as you draft, summarize and edit content while you control the final version

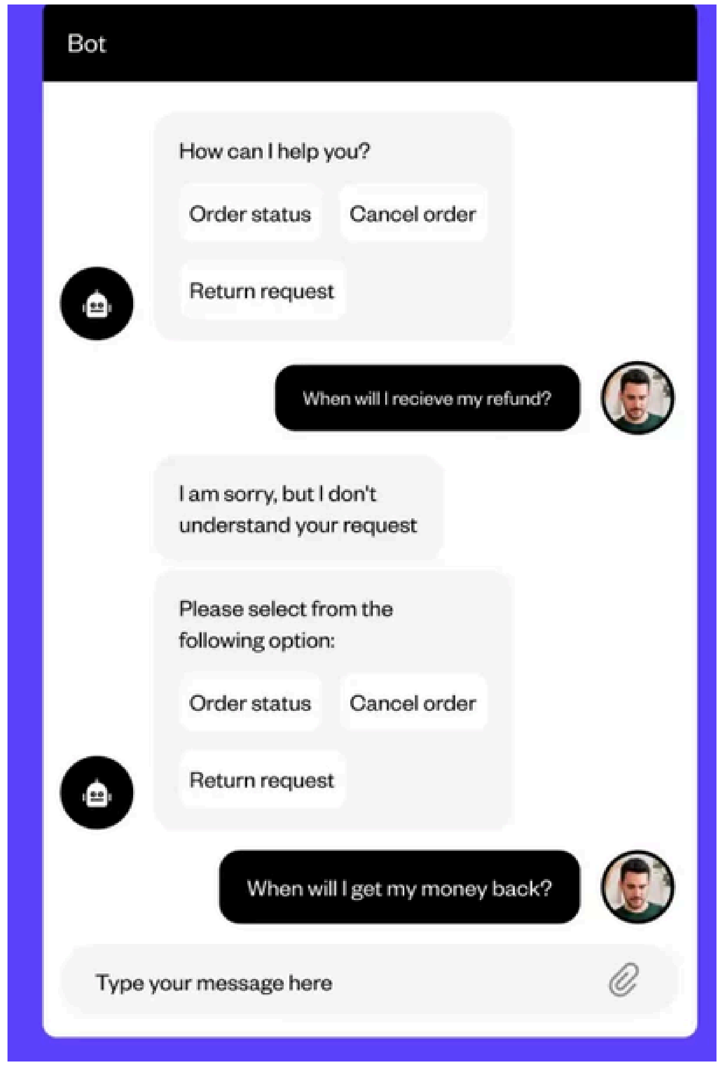
Predictability and Consistency of AI Behavior

- AI behaves consistently, allowing users to predict responses.

e.g. Spotify's consistent music recommendations based on listening history.

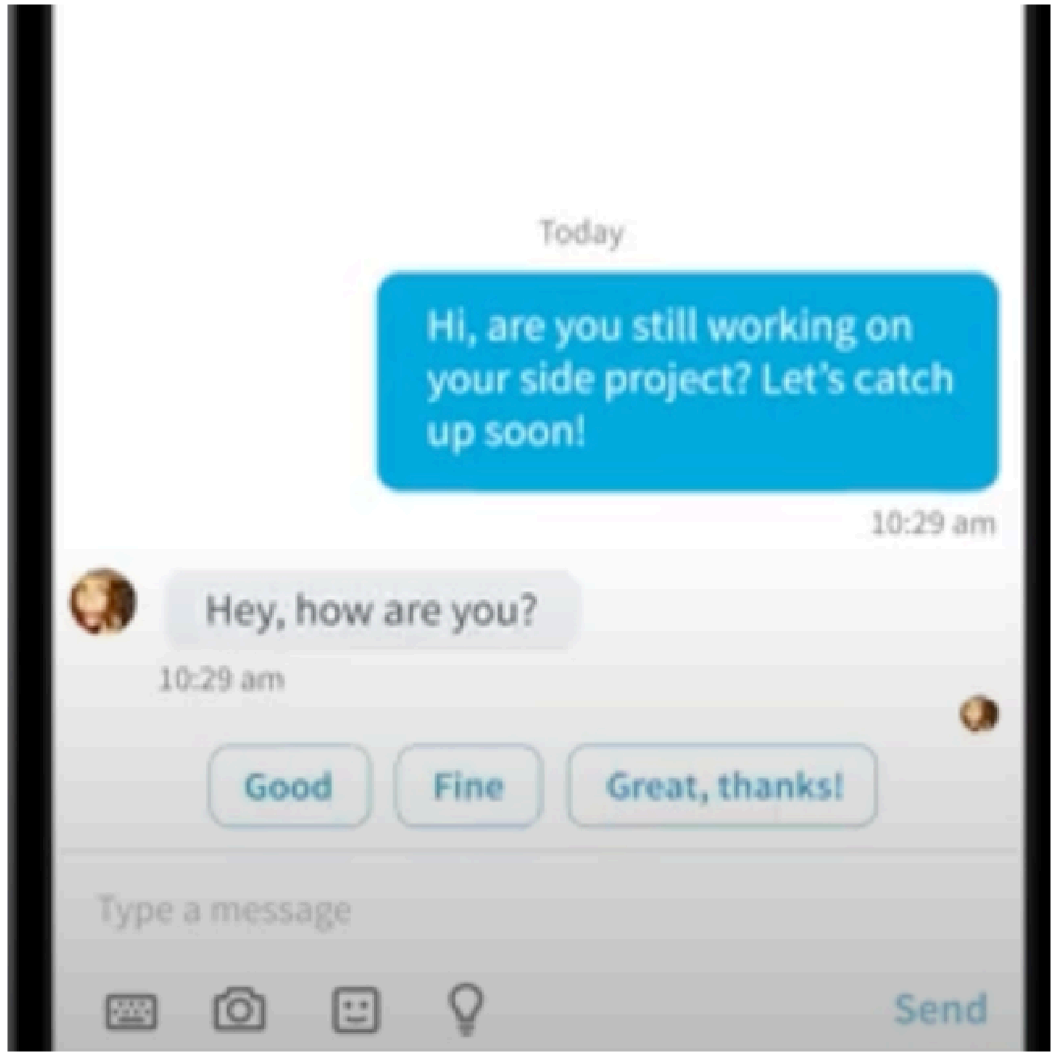
Consistent patterns in chatbot replies to similar queries.

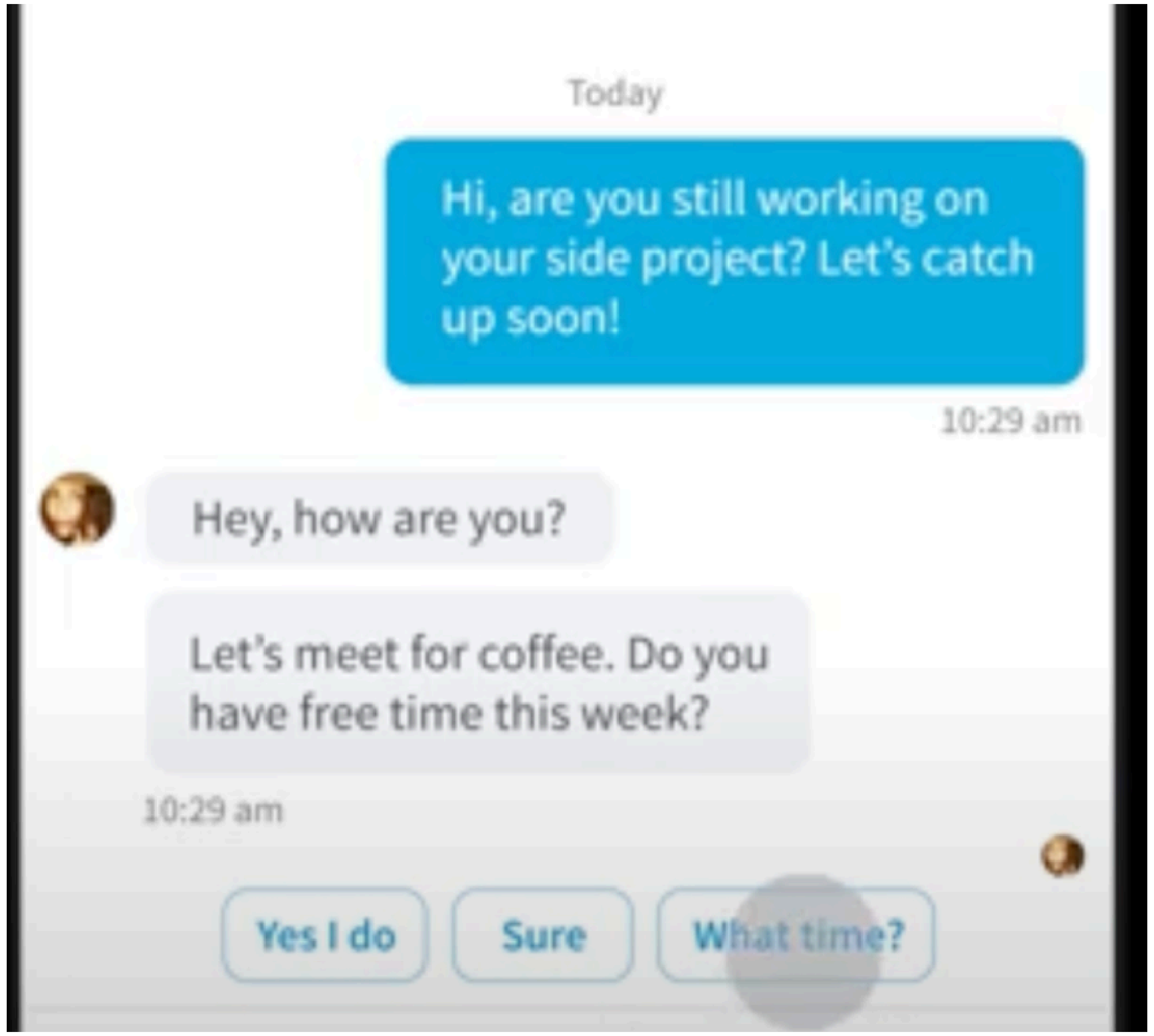




Context-Sensitive AI recommendations/suggestions

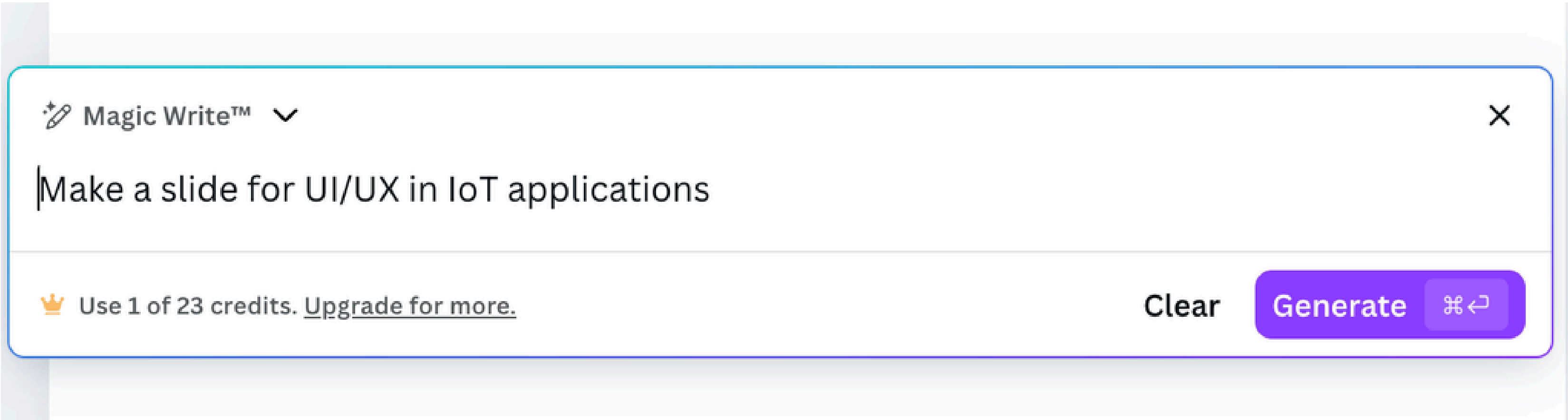
e.g. AI auto-complete providing relevant message responses based on context.





Error Recovery and Fallback Mechanisms

- Provide Regenerate / Retry options
- Allow users to edit inputs or outputs
- Maintain history/versioning when possible
- Offer fallback to manual control



MagicWrite in Canva

Example:
In ChatGPT, users can refine outputs through follow-up prompts, but this is not a true undo mechanism
– "undo-like" affordances
Depending on interface:

- *Edit prompt / regenerate response*
- *Retry / regenerate*
- *Conversation history*

When you say:
"Undo the last changes"
The model:
-Does not remember a structured previous state
-Does not truly revert deterministically
-It reinterprets your request

Summary

Core Ideas

- Usability is not optional. It directly affects user success and adoption.
- Heuristics help you identify problems early , before users are involved.

What you should do as a designer

- Use heuristic evaluation early to catch obvious issues

Heuristics tell you what might be wrong.

Next lecture: Usability Testing

Usability testing reveals real user behavior, often different from expectations.

- Use usability testing later to validate with real user
- Focus on tasks, not just screens
- Good design is not what you intend, it’s what users can successfully do.

Usability testing shows you what actually goes wrong

Appendix-1 Slides :

Chatbot Heuristics

Typical features of a Chatbot

http://chatbottest.com

- **Personality**: Does the chatbot have a clear voice and tone that fits with the users and with the ongoing conversation?
- **Onboarding**: Are users understanding what is the chatbot about? and how to interact with him from the very beginning?
- **Understanding**: Requests, Smalltalk, idioms, emojis… What is the chatbot able to understand?
- **Answering**: What elements does the chatbot send and how well it is doing it? Are they relevant to the moment and context?
- **Navigation**: How easy is to go through the chatbot conversation? Do you feel lost sometimes while speaking with the chatbot?
- **Error management**: How good is the chatbot dealing with all the errors that are going to happen? Is able to recover from them?
- **Intelligence**: Does the chatbot have any intelligence? Is able to remember things? Uses and manages context as a person?

The 12 Heuristics for Conversational UX Analysis

Based on Nielsen[2005] + Moore & Arar [2019] + Shevat [2017]

1. Visibility of system status

(a) Presence of information about the chatbot's state in the entire process
(b) Immediate feedback (did the last user action work?)
(c) Compel user action (what does the chatbot think the user will do next?)

2. Match between system and the real world

(a) Chatbot uses the language familiar to the target users
(b) Visual components (emojis, GIFs, icons) are linked to real-world objects
(c) If metaphors are used, they are understandable for the user

3. User control and freedom

(a) Chatbot supports undo/redo of actions
(b) Chatbot offers a permanent menu
(c) Chatbot provides navigation options
(d) Chatbot understands repair initiations

4. Consistency and standards

(a) Chatbot uses the domain model from the user perspective
(b) Chatbot has a personality, consistency in language and style

The 12 Heuristics for Conversational UX Analysis

Based on Nielsen[2005] + Moore & Arar [2019] + Shevat [2017]

5. Error prevention
(a) Chatbot prevents unconscious slips by meaningful constraints
(b) Chatbot prevents unconscious slips by spelling error detection
(c) Chatbot requests confirmation before actions with significant implications
(d) Chatbot explains consequences of the user actions

6. Recognition rather than recall
(a) Chatbot makes the options clear through descriptive visual elements and explicit instructions
(b) Chatbot shows summary of the collected information before transactions
(c) Chatbot offers a permanent menu and help option

7. Flexibility and efficiency of use

(a) Chatbot understands not only special instructions but also synonyms
(b) Chatbot can deal with different formulations
(c) Chatbot offers multiple ways to achieve the same goal

8. Aesthetic and minimalist design

(a) Chatbot dialogues are concise, only contain relevant information
(b) Chatbot uses visual information in a personality-consistent manner to support the user, not just random decoration

The 12 Heuristics for Conversational UX Analysis

Based on Nielsen[2005] + Moore & Arar [2019] + Shevat [2017]

9. Help users recognise, diagnose, and recover from errors

(a) It clearly indicates that an error has occurred
(b) It uses plain language to explain the error
(c) It explains the actions needed for recovery
(d) It offers shortcuts to fix errors quickly

10. Help and documentation

(a) It provides a clear description of its capabilities
(b) It offers keyword search
(c) It focuses its help on the user task
(d) It explains concrete steps to be carried out for a task

11. Context understanding

Chatbot understands the context within one turn or within a small number of turns (usually 2-3 user-bot turn pairs)

12. Interaction management capabilities

(a) Chatbot understands conversation openings and closings (e.g., 'hello')
(b) Chatbot understands sequence closings (e.g., 'ok' and 'thank you')
(c) Chatbot understands repair initiations and replies with repairs
(d) Chatbot initiates repair to handle potential user errors

Self-explore:
Heuristic evaluation of COVID-19 chatbots

Expert review scores for 24 COVID-19 chatbots on 12 heuristics

0: unsupported,
0.5: partially supported
1: fully supported
- : not applicable

Chatbot	Score	1	2	3	4	5	6	7	8	9	10	11	12
IDEAL BOT	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Suve	**7.8**	0.5	**1.0**	0.15	**1.0**	0.8	0.5	0.8	**1.0**	0.6	0.6	0.3	0.5
Apple	**7.1**	0.7	**1.0**	0.3	**1.0**	**1.0**	0.5	0.0	**1.0**	0.8	0.8	-	-
Infermedica	**7.0**	**1.0**	**1.0**	0.6	**1.0**	**1.0**	0.7	0.0	**1.0**	-	0.8	-	-
Symptoma	**7.0**	**1.0**	0.7	0.6	0.8	0.8	0.8	0.5	**1.0**	0.10	0.8	-	-
HealthBuddy	6.6	0.8	**1.0**	0.1	0.8	0.0	0.5	0.5	0.8	0.9	0.5	0.3	0.5
Docyet	6.5	0.8	**1.0**	0.3	**1.0**	0.5	0.5	0.3	**1.0**	0.8	0.4	-	-
Ada	6.3	0.7	**1.0**	0.3	**1.0**	0.5	0.3	0.0	**1.0**	0.8	0.7	-	-
Germ. Red Cr.	5.9	0.5	**1.0**	0.4	0.5	0.0	0.5	0.8	0.8	0.3	0.6	0.3	0.3
Dub. Dpt. of H.	5.8	0.8	**1.0**	0.0	0.5	0.5	0.3	0.8	0.5	0.4	0.3	0.3	0.5
Corona Bot	5.4	0.5	0.8	0.0	0.8	0.8	0.0	0.3	**1.0**	0.4	0.1	0.3	0.5
Ivan Mask	5.3	0.8	0.8	0.3	0.8	0.3	0.5	0.2	0.5	0.5	0.3	0.3	0.3
Martha	5.3	0.7	**1.0**	0.0	0.8	0.5	0.2	0.2	0.8	0.1	0.5	0.2	0.5
WHO	5.3	0.8	**1.0**	0.4	0.5	0.0	0.8	0.7	0.8	0.1	0.4	0.0	0.0
Bobbi	5.3	0.8	0.7	0.1	0.3	0.3	0.2	0.5	0.8	0.6	0.6	0.2	0.4
MTI	5.1	0.5	0.5	0.1	0.5	0.5	0.2	0.8	**1.0**	0.5	0.4	0.0	0.1
Babylon	4.9	0.7	**1.0**	0.3	0.8	**1.0**	0.0	0.0	0.8	-	0.5	-	-
Covid-19	4.9	0.5	0.7	0.0	**1.0**	**1.0**	0.5	0.0	**1.0**	0.0	0.3	-	-
CDC	4.4	0.5	**1.0**	0.0	0.8	**1.0**	0.2	0.0	0.5	-	0.5	-	-
HSE	4.4	0.3	**1.0**	0.0	0.8	**1.0**	0.0	0.0	**1.0**	-	0.3	-	-
e-bot7	4.1	0.8	0.5	0.1	**1.0**	0.5	0.5	0.0	0.5	0.0	0.3	-	-
Providence	3.9	0.5	0.5	0.0	0.8	**1.0**	0.0	0.0	0.8	0.0	0.4	-	-
Your.MD	3.8	0.7	0.7	0.0	0.8	0.3	0.3	0.2	0.3	0.4	0.4	-	-
Clevel. Clinic	3.7	0.5	0.5	0.0	0.8	**1.0**	0.2	0.0	0.5	-	0.3	-	-
Russ. M. of H.	2.3	0.5	0.8	0.2	0.3	0.0	0.3	0.0	0.0	0.3	0.1	0.0	0.0
Score/heuristic		*64%*	*83%*	*17%*	*74%*	*58%*	*34%*	*28%*	*75%*	*40%*	*44%*	*21%*	*32%*

Höhn, S., Bongard-Blanchy, K. (2021). Heuristic Evaluation of COVID-19 Chatbots. In: Følstad, A., *et al.* Chatbot Research and Design. CONVERSATIONS 2020. Lecture Notes in Computer Science(), vol 12604. Springer, Cham. https://doi.org/10.1007/978-3-030-68288-0_9

Appendix-**2** Slides :

Use of AI for Heuristic Evaluation

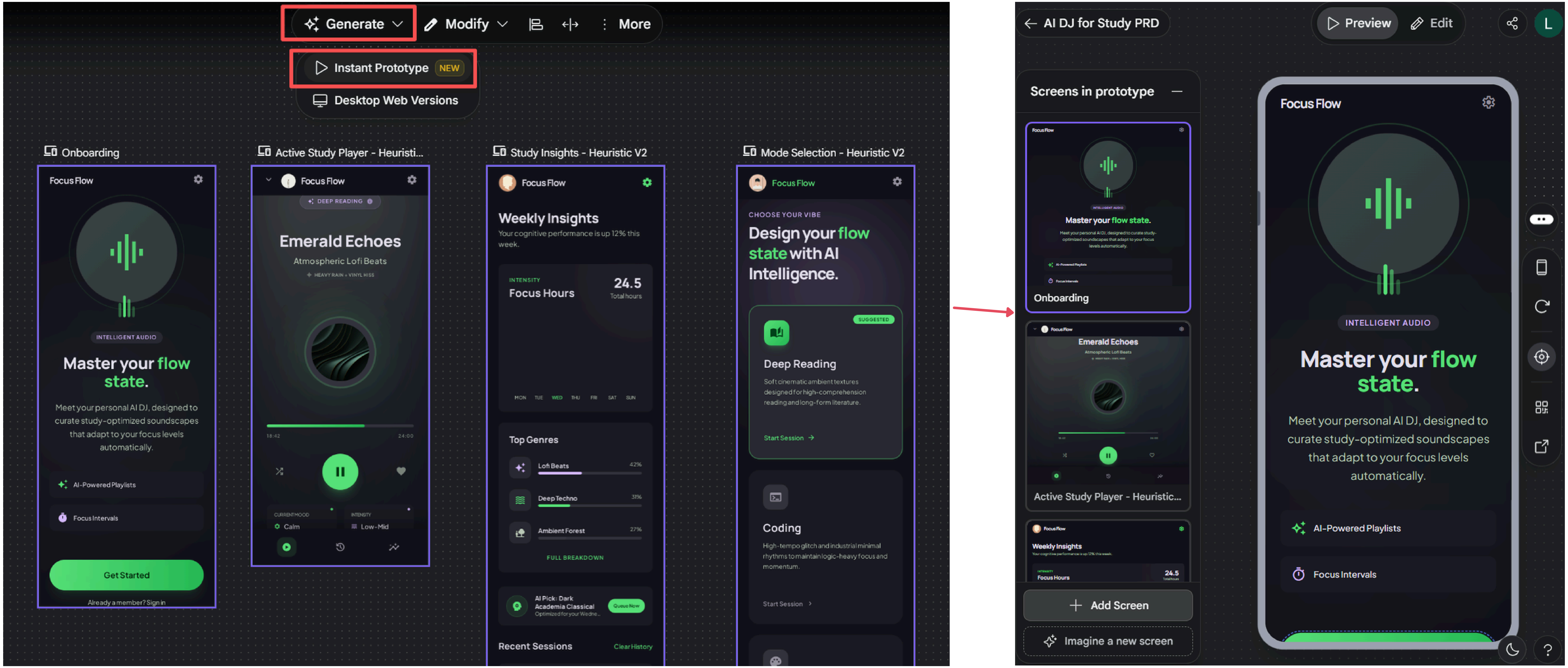
Use AI for heuristic evaluation

Goal:
Use AI to quickly identify usability issues in the interface based on heuristic evaluation principles.

What AI can evaluate:
AI can evaluate the interface based on Nielsen's 10 Heuristics:
Visibility of system status
Match between system and the real world
User control and freedom
Consistency and standards
Error prevention
Recognition rather than recall
Flexibility of use
Minimalist design
Help users recognize, diagnose, and recover from errors
Help and documentation

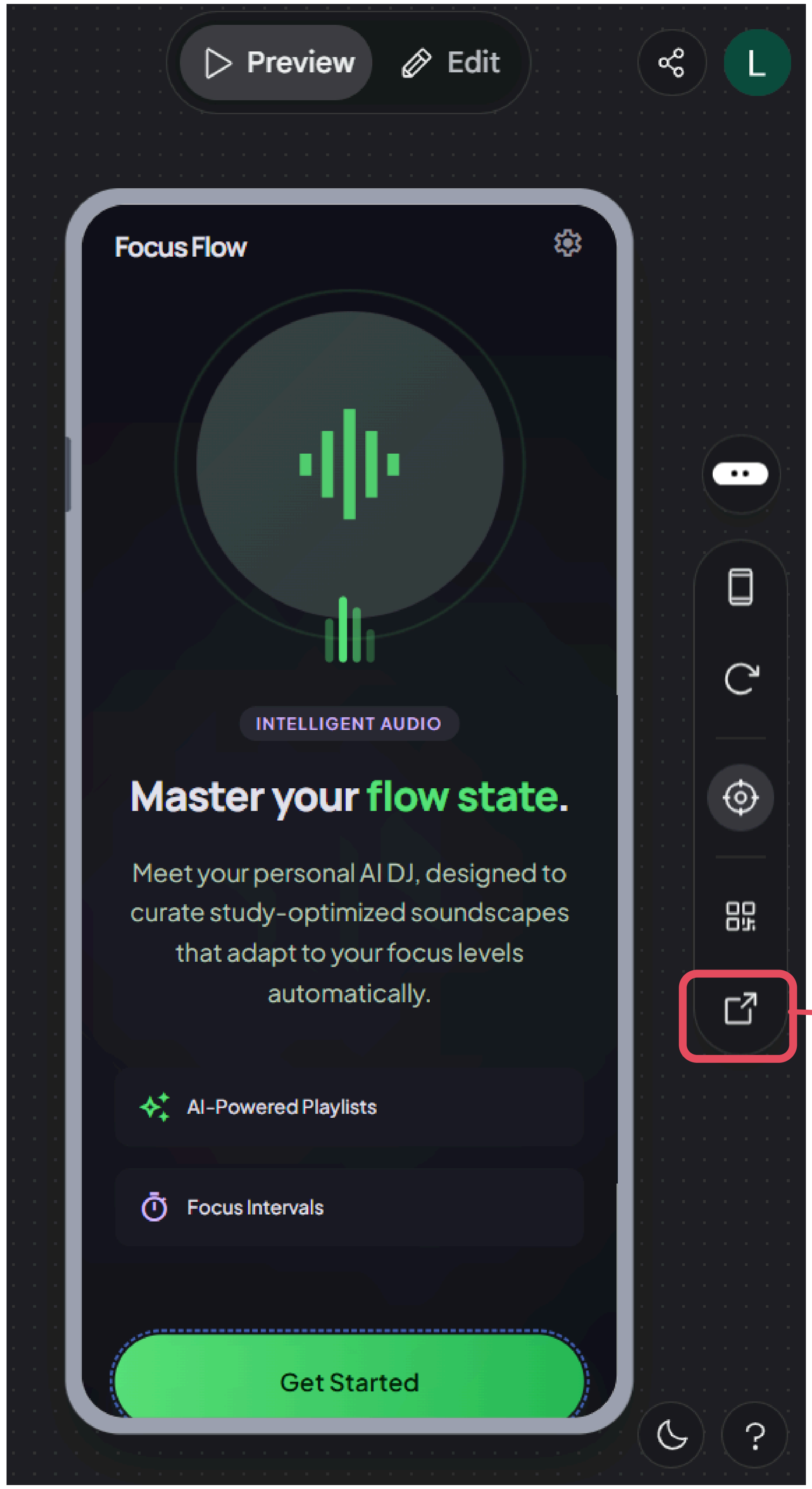
Process followed:
Generate a prototype in Stitch
→ Record the prototype's interaction process using screen-recording software
→ Use Gemini to conduct heuristic analysis
→ Refine the prototype

Generate a prototype in Stitch:



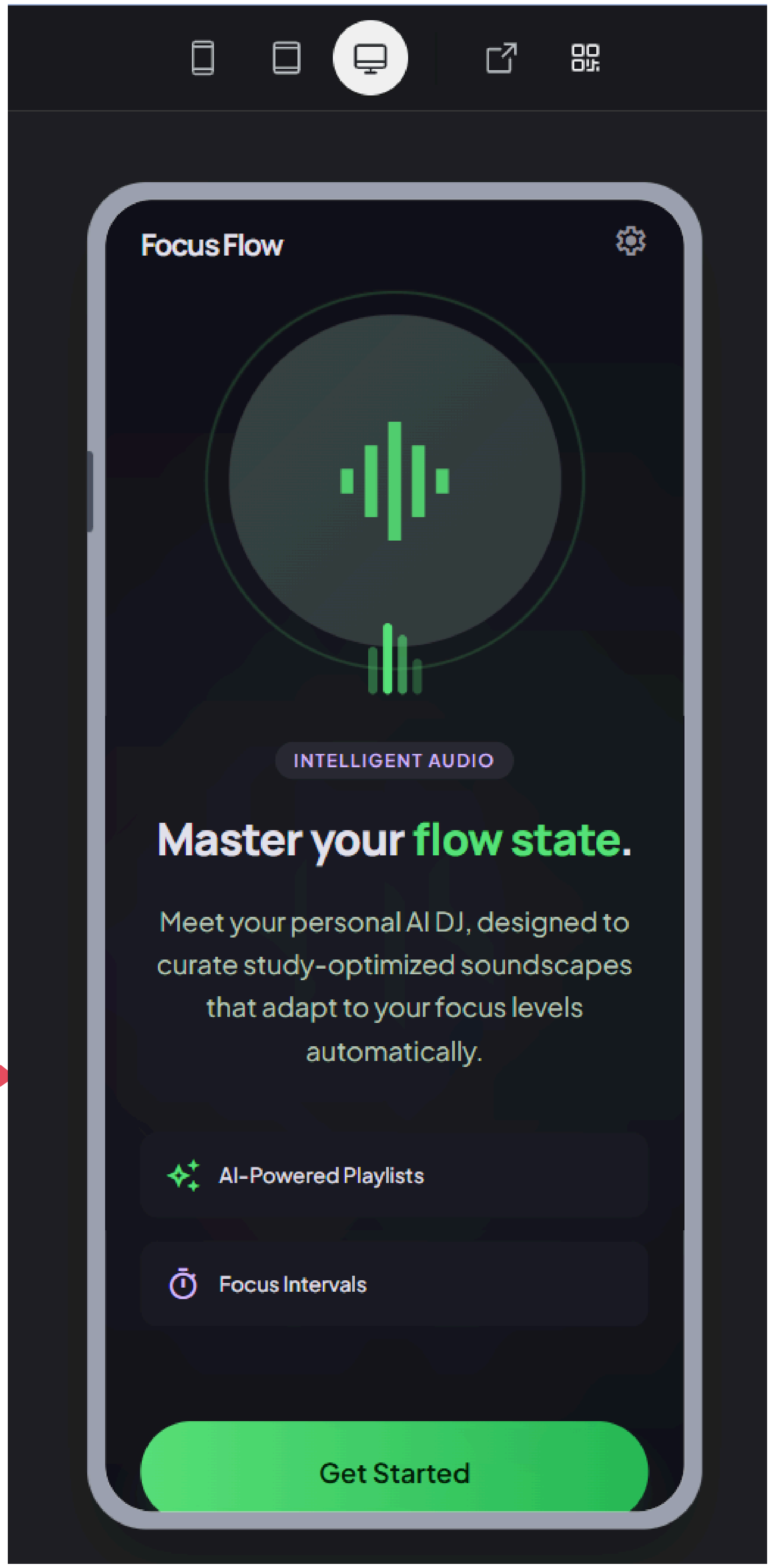
From Stitch

Record the prototype's interaction process:



From Stitch

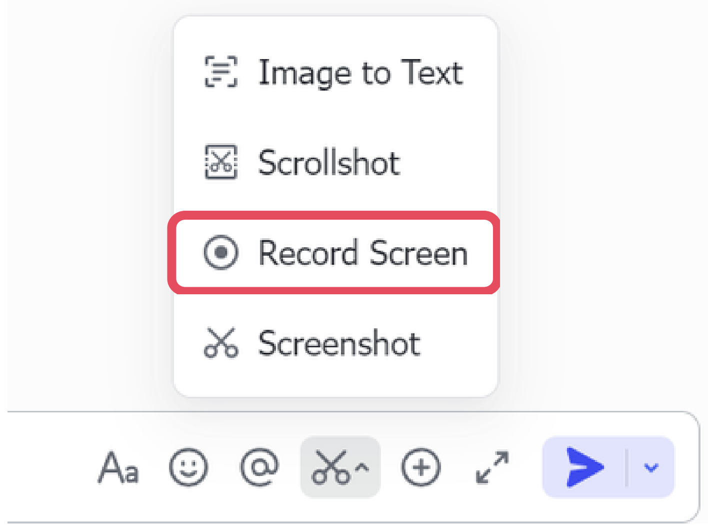
Open in new tab



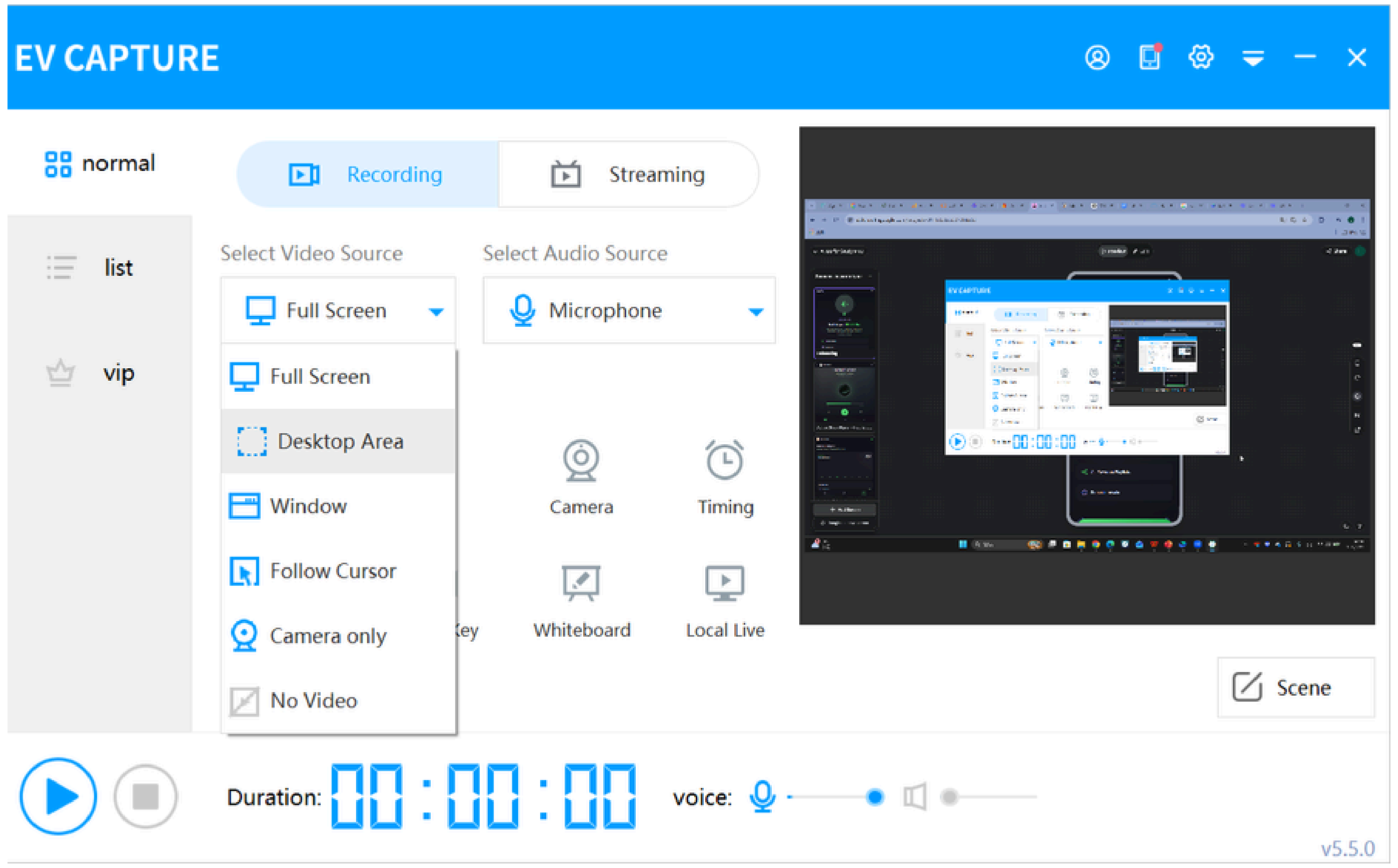
Play Mode

Screen-recording Tools (Free):

Lark (飞书)

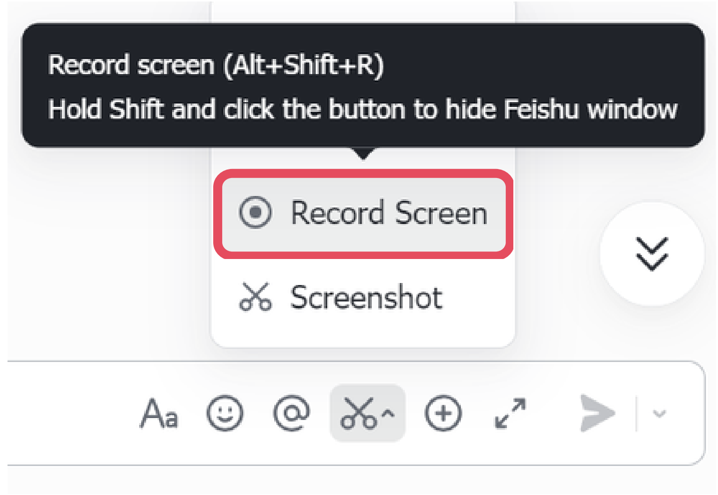


EV capture (EV录屏)

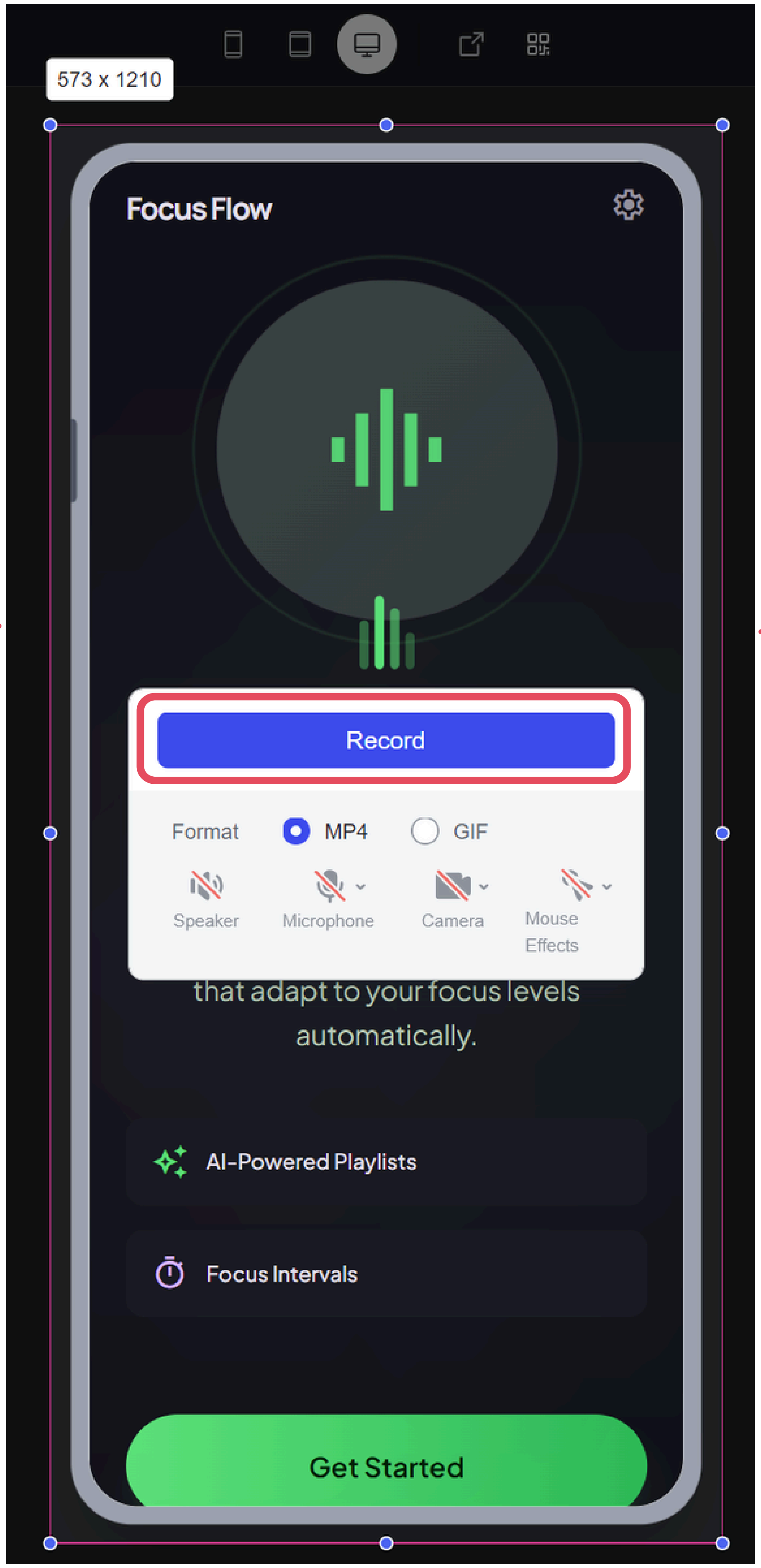


For example:

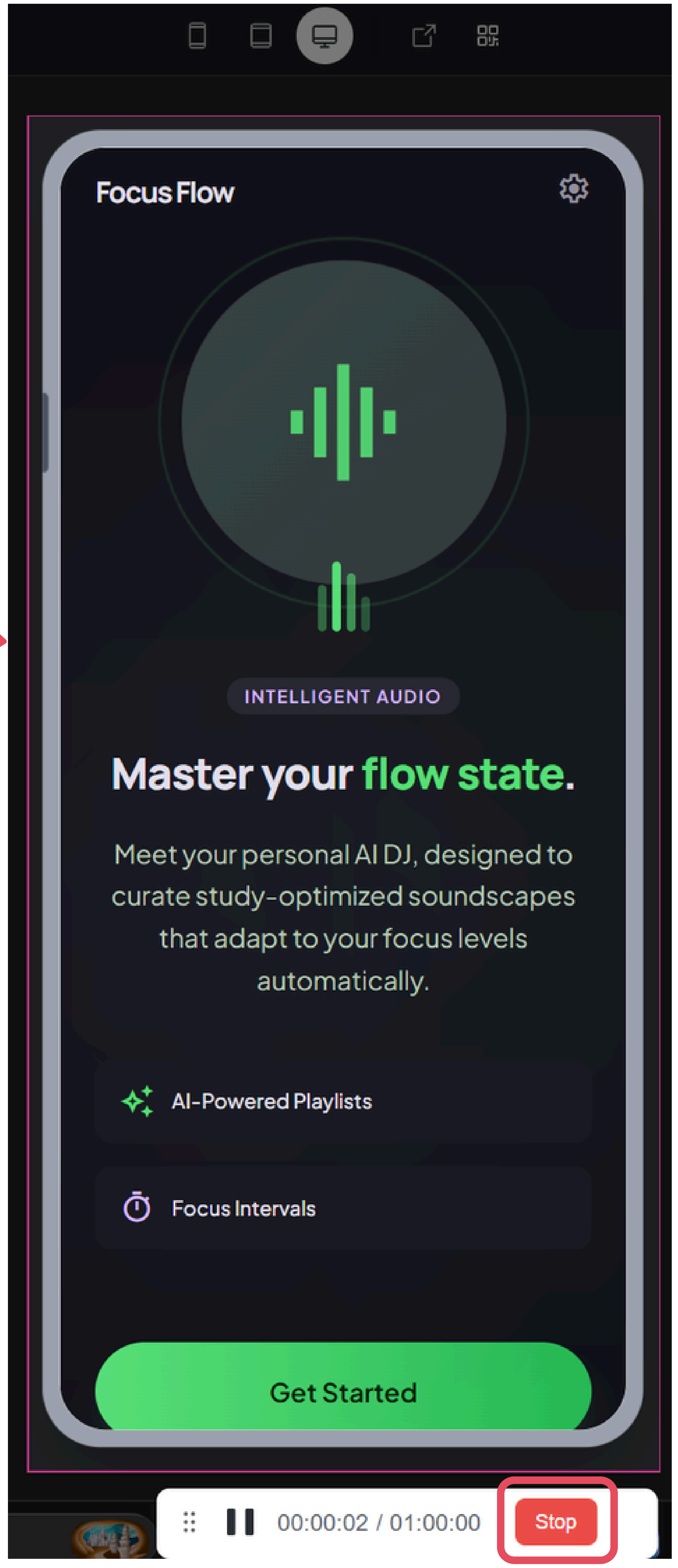
Lark (飞书)



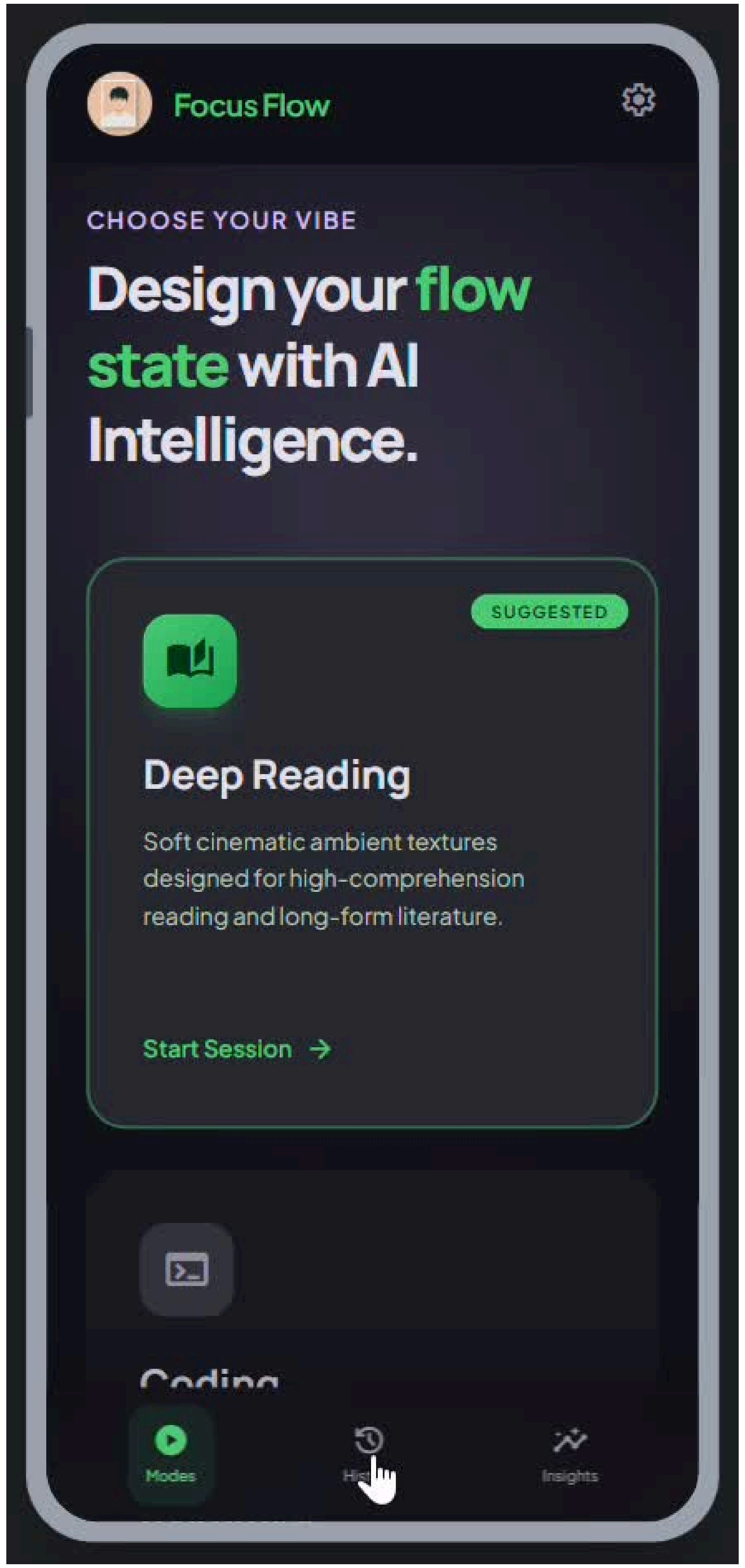
Select the recording area and start recording



Stop recording

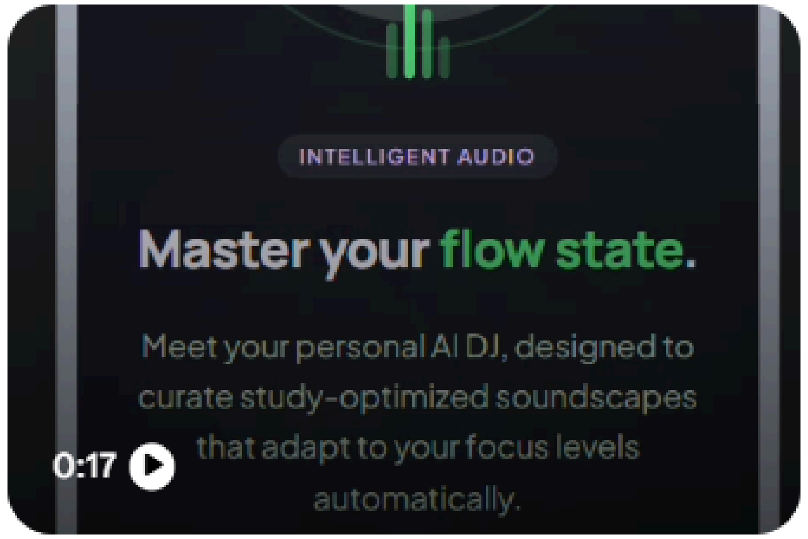


The video has been generated



Use Gemini to conduct heuristic analysis:

Prompt:



Please evaluate this interface using Nielsen's 10 Heuristics:
Visibility of system status, Match between system and the real world, User control and freedom, Consistency and standards, Error prevention, Recognition rather than recall, Flexibility and efficiency of use, Aesthetic and minimalist design, Help users recognize, diagnose, and recover from errors, and Help and documentation.

For each heuristic, identify:
what works well
what usability problems exist
how the design can be improved

Avoid abstract or academic language.

Show thinking

This interface, "Focus Flow," has a strong aesthetic foundation that aligns well with modern "dark mode" productivity tools. Below is an evaluation based on Nielsen's 10 Heuristics.

---

1. Visibility of System Status

- **What works well:** The playback screen clearly shows the track name, progress bar, and active mode ("Deep Reading"). The "Weekly Insights" provides immediate feedback on total focus hours.
- **Usability problems:** It is unclear how "Intensity" (Low-Mid) is determined. Is the system changing it, or is it a status the user set?
- **How to improve:** Add a subtle animation to the "Intensity" or "Current Mood" tags if the AI is actively adjusting them in real-time, so the user knows the "Personal AI DJ" is working.

2. Match Between System and the Real World

- **What works well:** The terminology used—"Lofi," "Deep Reading," "Memorization," and "Flow State"—perfectly matches the mental model of students and professionals.
- **Usability problems:** The "Emerald Echoes" title is poetic but doesn't tell the user what the sound actually *is* (e.g., rain, white noise, or synth).
- **How to improve:** Include a small sub-label or tag on the player screen that identifies the primary sound elements (e.g., "Heavy Rain + Vinyl Hiss").

3. User Control and Freedom

- **What works well:** Users can easily jump between different "vibes" and have a clear "Start Session" entry point.
- **Usability problems:** On the playback screen, there is no visible "Exit" or "Back" button to return to the library without stopping the music.
- **How to improve:** Add a "chevron down" icon at the top of the player to indicate the screen can be minimized, allowing users to browse other sounds while listening.

4. Consistency and Standards

- **What works well:** The use of cards for different modes and a standard bottom navigation bar (Home, History, Insights, Settings) follows established mobile UI patterns.
- **Usability problems:** The "Recall Mode" button for Memorization uses a different naming convention than "Start Session" for Deep Reading, which might confuse users about what happens when they click.
- **How to improve:** Standardize the primary action buttons to "Start Session" across all cards, using sub-labels to explain specific modes like "Recall."

5. Error Prevention

- **What works well:** Categorizing sounds into "Vibes" prevents users from picking a sound that might be counter-productive to their specific task.
- **Usability problems:** The "Clear History" button in the Insights tab is positioned very close to the scrollable list and looks like a single-tap trigger.
- **How to improve:** Add a confirmation pop-up ("Are you sure you want to clear your history?") to prevent accidental data loss.

6. Recognition Rather than Recall

- **What works well:** The "Recent Sessions" list is excellent. It allows users to jump back into a specific environment (e.g., "Midnight Focus") without remembering exactly which settings they used.
- **Usability problems:** The "Custom Session" description is a bit wordy. A user might forget what the "AI DJ" is capable of if they have to read a paragraph every time.
- **How to improve:** Use visual tags or keywords (e.g., "Rain," "Fast-paced," "Noisy Office") as suggestions near the "Generate with AI" button.

7. Flexibility and Efficiency of Use

- **What works well:** The "Suggested" tag on Deep Reading helps new users get started immediately, while "Custom Session" provides a "power user" path.
- **Usability problems:** There is no way to "Favorite" a specific AI-generated track from the playback screen for quick access later.
- **How to improve:** Add a "Heart" or "Save" icon on the playback screen so users can build a library of their favorite generated soundscapes.

8. Aesthetic and Minimalist Design

- **What works well:** The high-contrast green on dark gray is very readable. The layout is clean, and the spacing between cards prevents the screen from feeling cluttered.
- **Usability problems:** The "AI Recommendation" card in the Insights tab is quite large and pushes the "Recent Sessions" off-screen.
- **How to improve:** Condense the Recommendation card or use a horizontal carousel for insights to keep the most important data (recent history) visible.

9. Help Users Recognize, Diagnose, and Recover from Errors

- **What works well:** Not explicitly shown in the video, but the clear "Generate with AI" button suggests a structured flow that minimizes input errors.
- **Usability problems:** If the AI fails to generate a custom session (e.g., due to no connection), the video doesn't show how the user is notified.
- **How to improve:** Ensure there is a clear, friendly error message if a generation fails, with a "Try Again" or "Use Offline Preset" option.

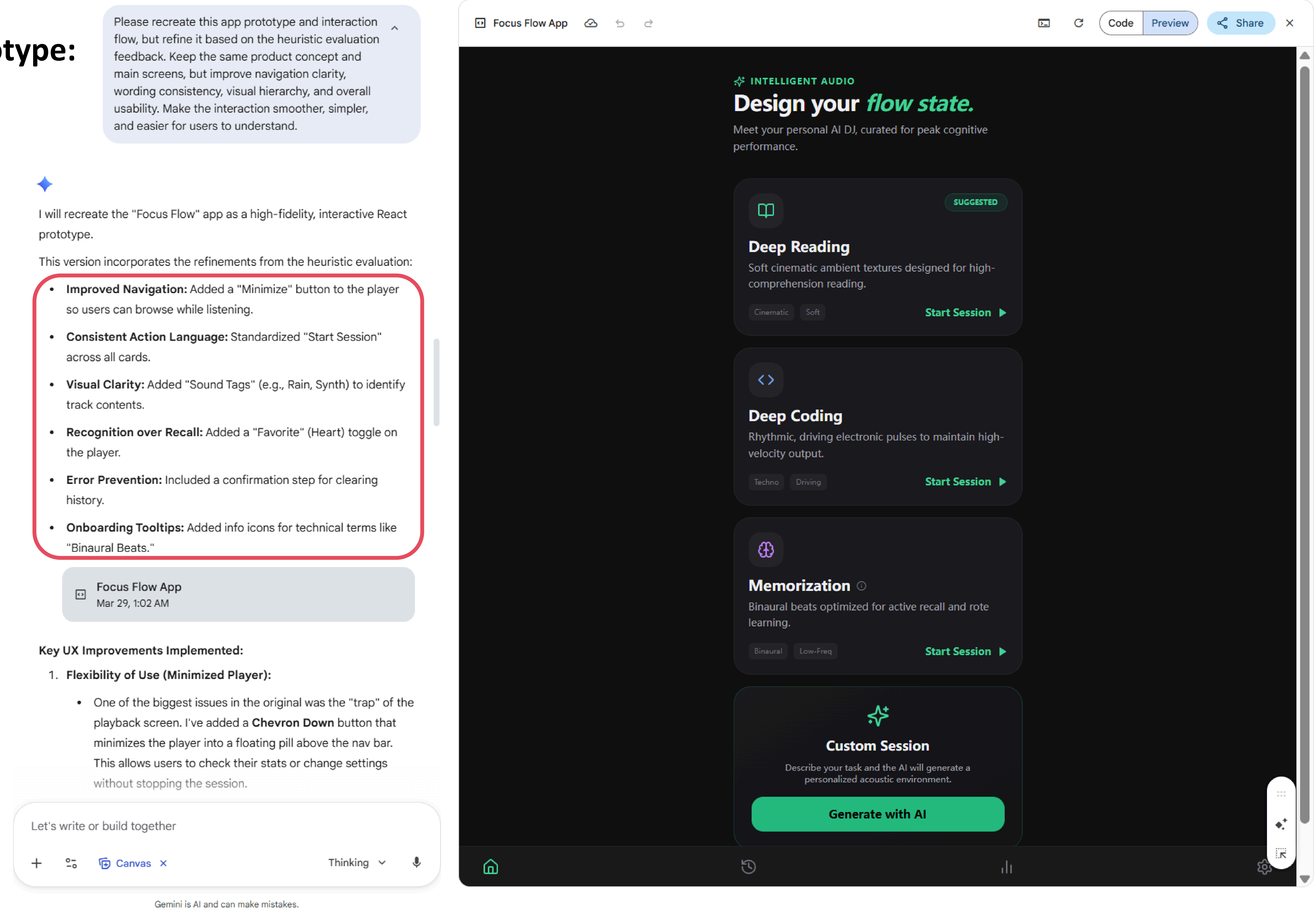
10. Help and Documentation

- **What works well:** The onboarding screen ("Master your flow state") briefly explains the value proposition.
- **Usability problems:** Complex concepts like "Binaural beats" or "Non-linear patterns" are mentioned but not explained for a layperson.
- **How to improve:** Add a small "info" icon next to technical terms like "Binaural beats" that triggers a short tooltip explaining how it helps with focus.

Would you like me to help you draft a user testing script to validate these findings with actual students?

[RA] Gemini's evaluation was useful, but its main weakness was that it was too **text-heavy and included too many suggestions at once**. This made the feedback time-consuming to read and difficult to match with the prototype efficiently. Users had to keep switching between the design and the long list of comments, which reduced clarity and usability. So while the response was comprehensive, it was not very convenient for quick design review and still needed human filtering and prioritization.

Refine the prototype:



Mar 29, 1:02 AM

Key UX Improvements Implemented:

1. **Flexibility of Use (Minimized Player):**
   - One of the biggest issues in the original was the "trap" of the playback screen. I've added a **Chevron Down** button that minimizes the player into a floating pill above the nav bar. This allows users to check their stats or change settings without stopping the session.
2. **Recognition vs. Recall (Sound Tags):**
   - Instead of just "Emerald Echoes," I added descriptors like **"Synth + Rain"** or **"Atmospheric Lofi."** This helps users recognize the sound texture before hitting play.
3. **Error Prevention (Clear History):**
   - I added a high-contrast **Confirmation Modal** for the "Clear History" action to prevent accidental loss of performance data.
4. **Consistency (Standardized CTA):**
   - The original used "Recall Mode" for memorization and "Start Session" for others. I updated all cards to use **"Start Session"** for a consistent mental model, while using descriptive icons to differentiate the *type* of work.
5. **Aesthetic Hierarchy (Insights):**
   - The Insights tab was redesigned to prioritize the **Weekly Stat (24.5 hrs)** and use the **AI Recommendation** as a secondary "upgrade" card, rather than letting it dominate the visual space.
6. **Help and Documentation (Info Tooltips):**
   - Added a subtle **Info icon** next to technical terms like "Memorization" (which uses binaural beats). This provides on-demand help without cluttering the UI for expert users.

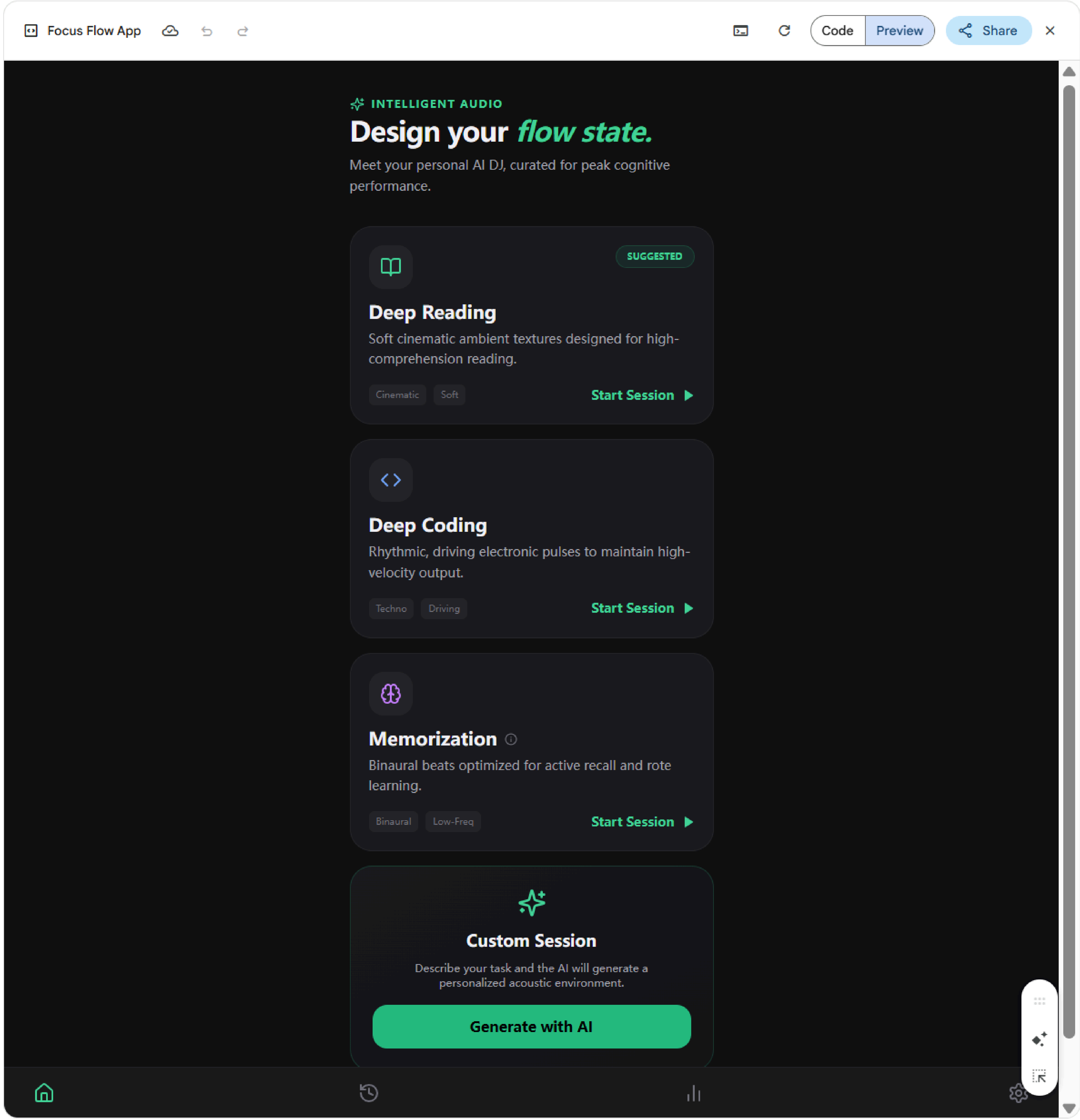
The interaction flow is now smoother: clicking "Start Session" triggers a bottom-up transition, and the player can be toggled between full-screen focus and a minimized utility state.

Let's write or build together

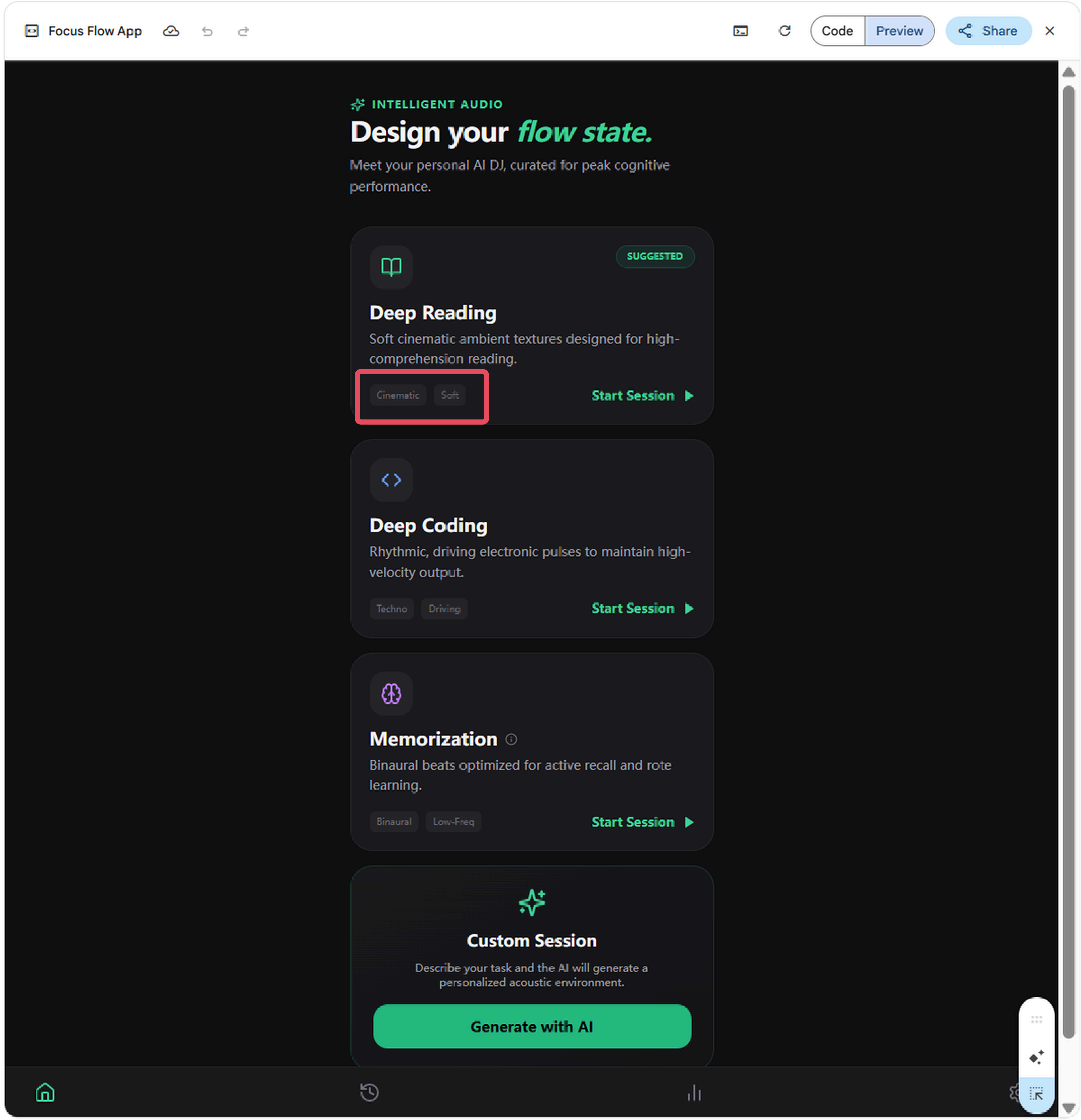
Canvas

Thinking

Gemini is AI and can make mistakes.

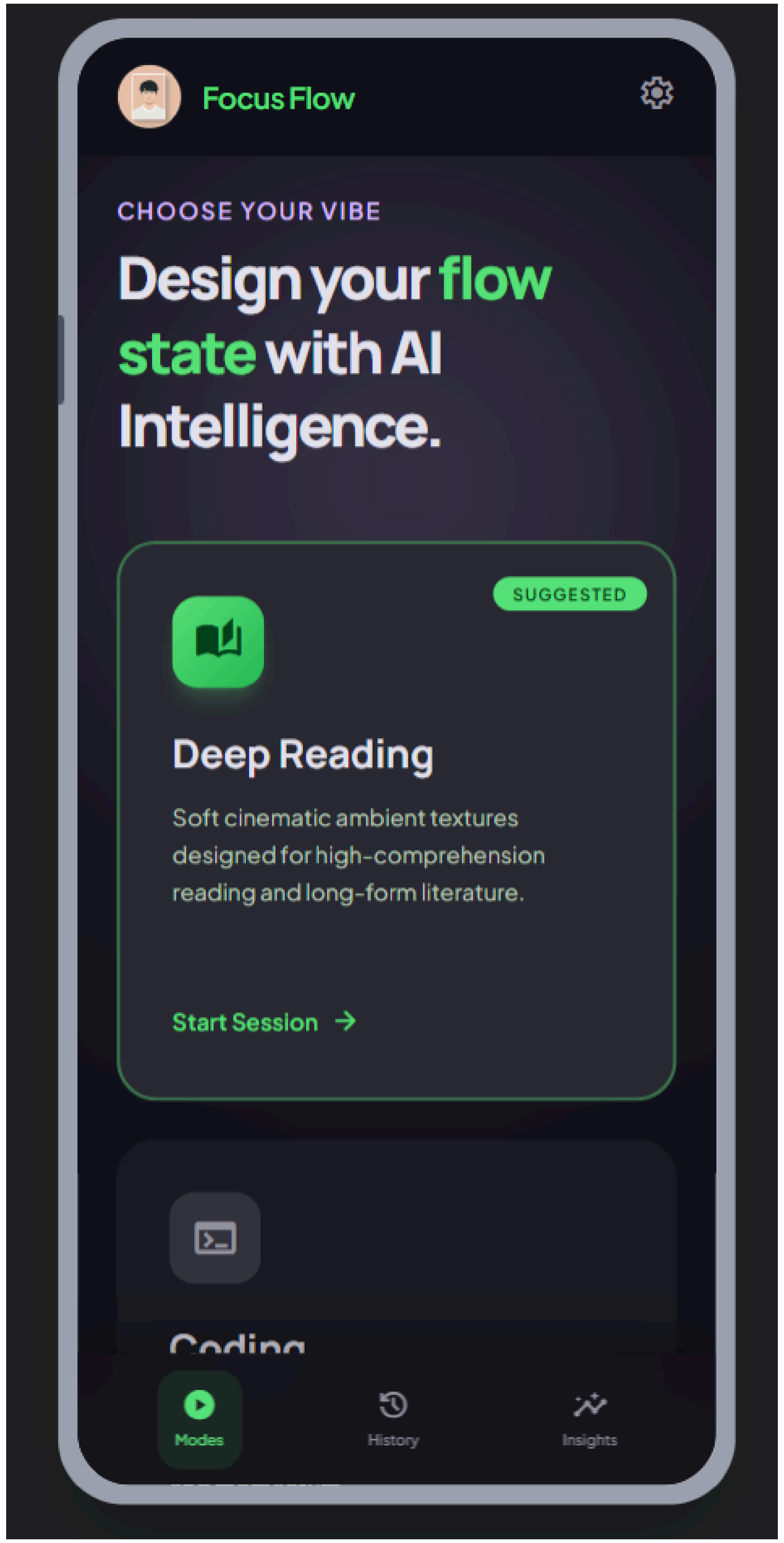


Gemini's version:

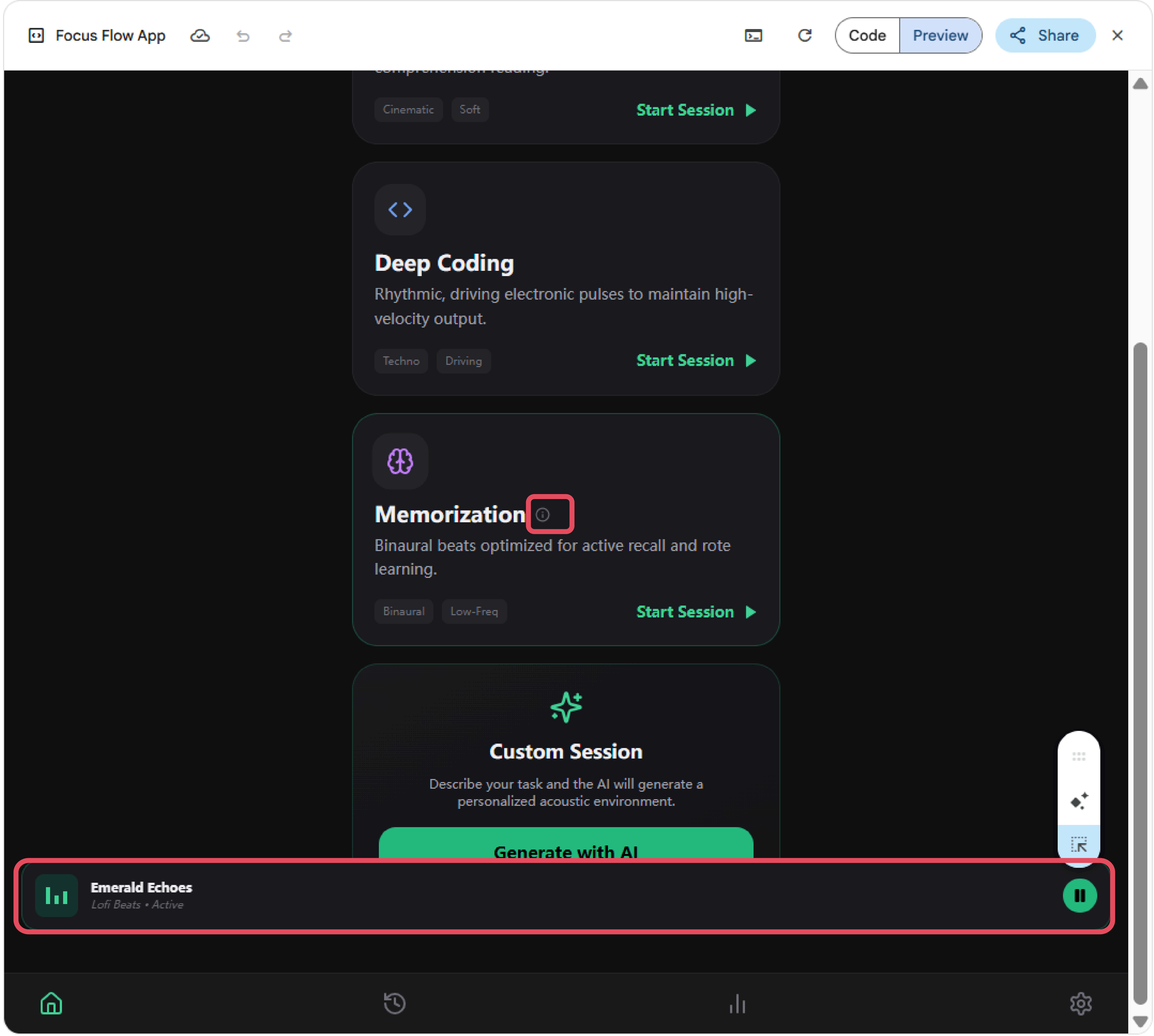


VS

The original version:

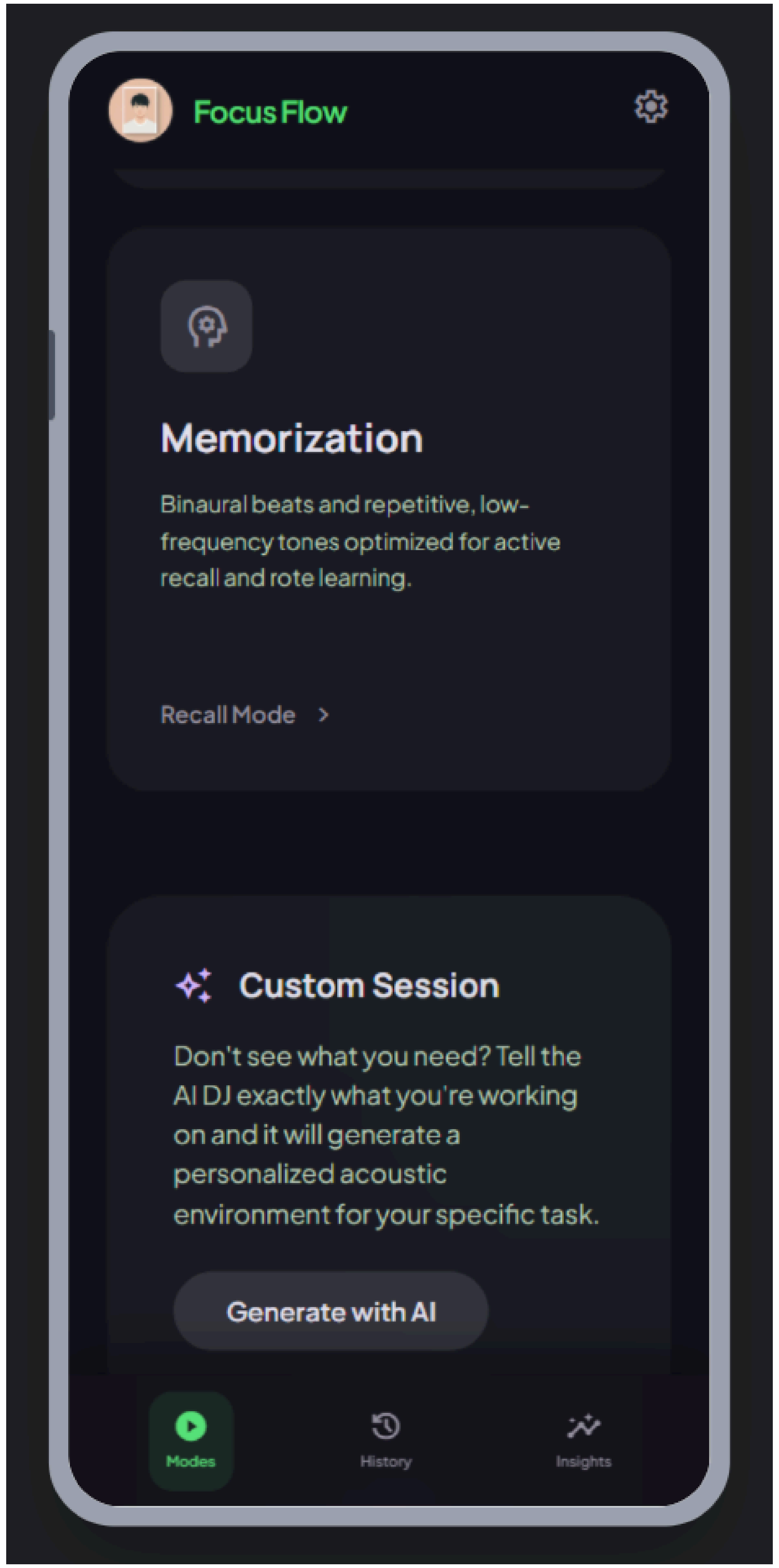


Gemini's version:

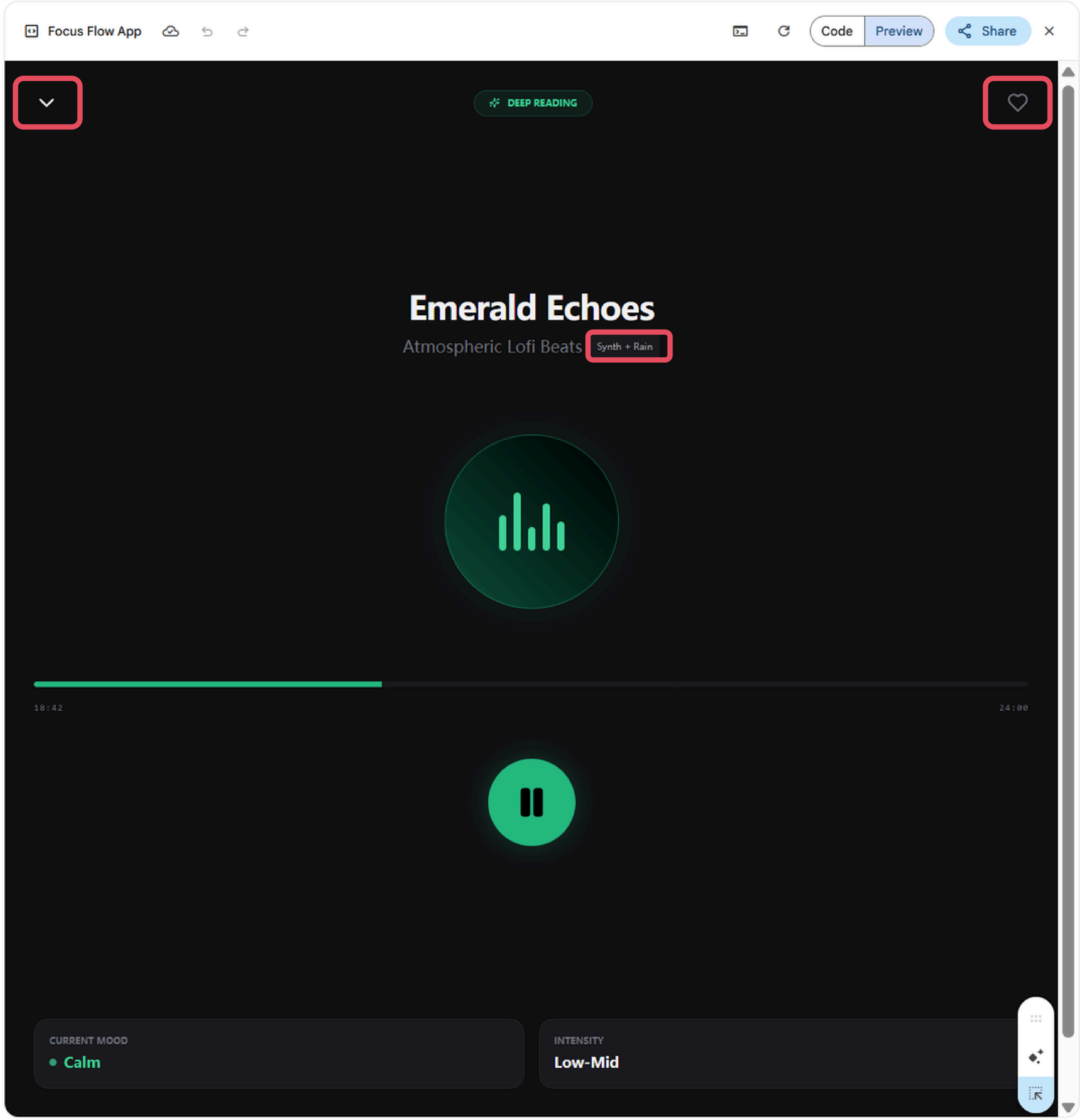


VS

The original version:

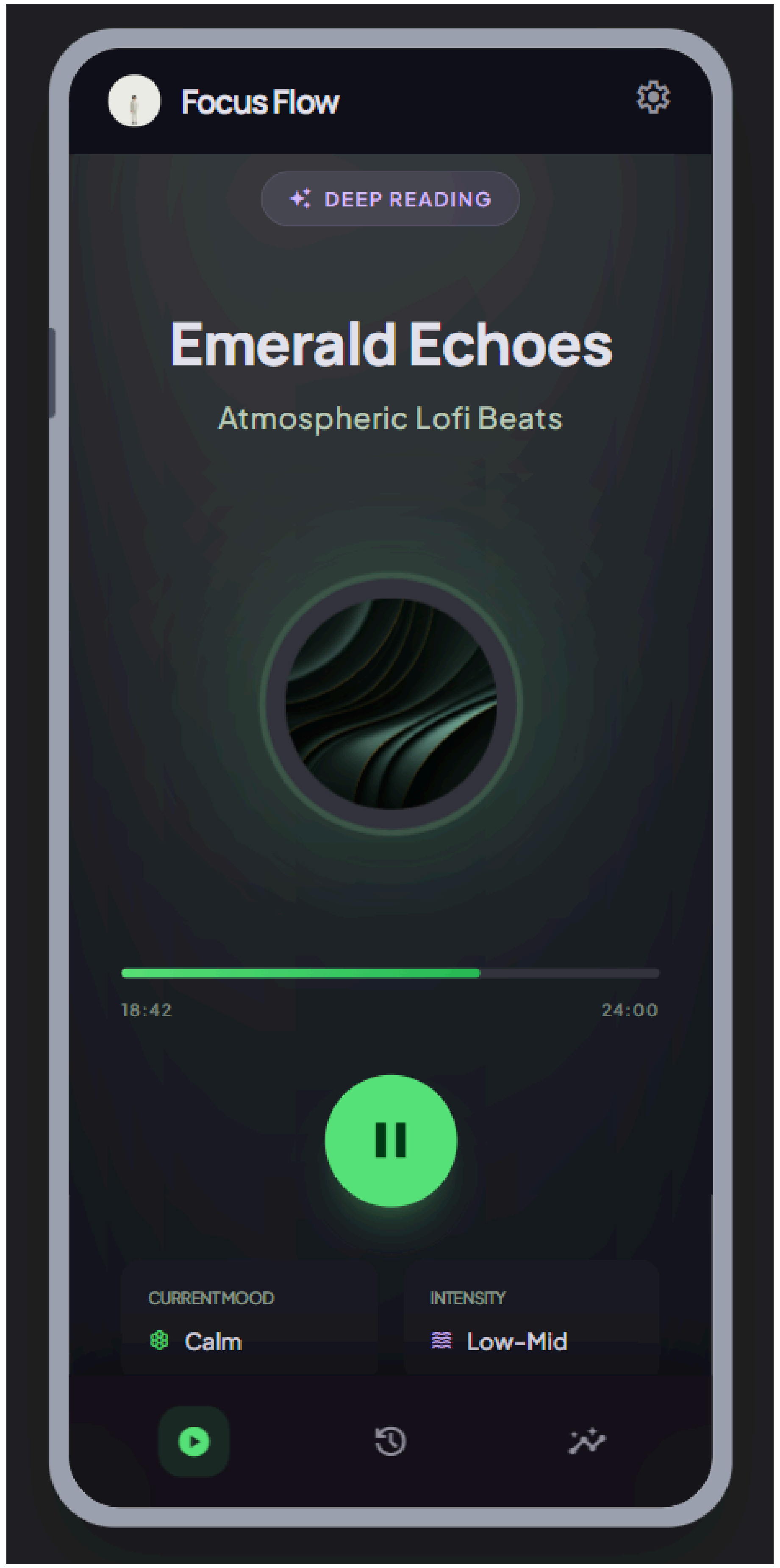


Gemini's version:

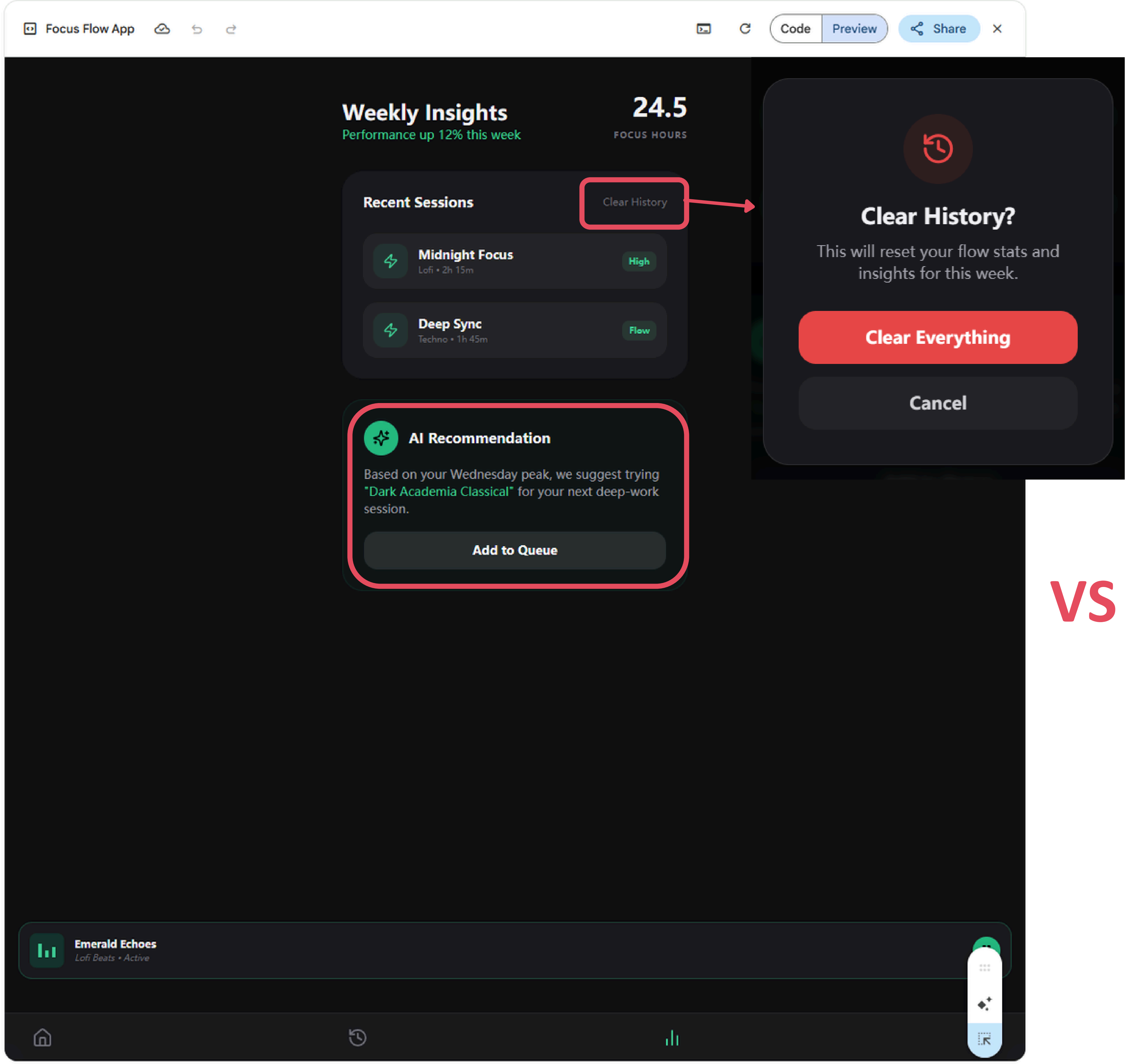


VS

The original version:

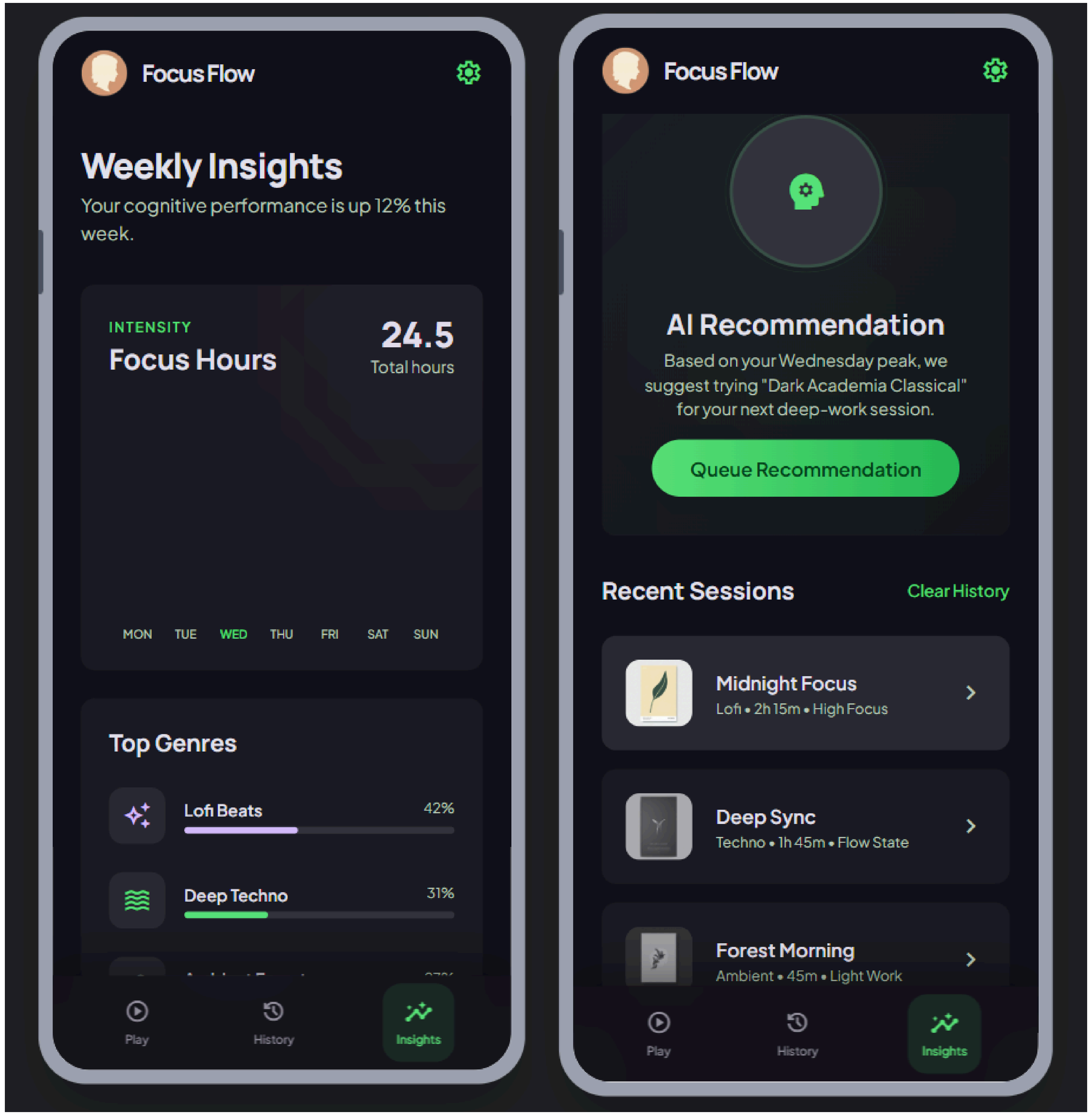


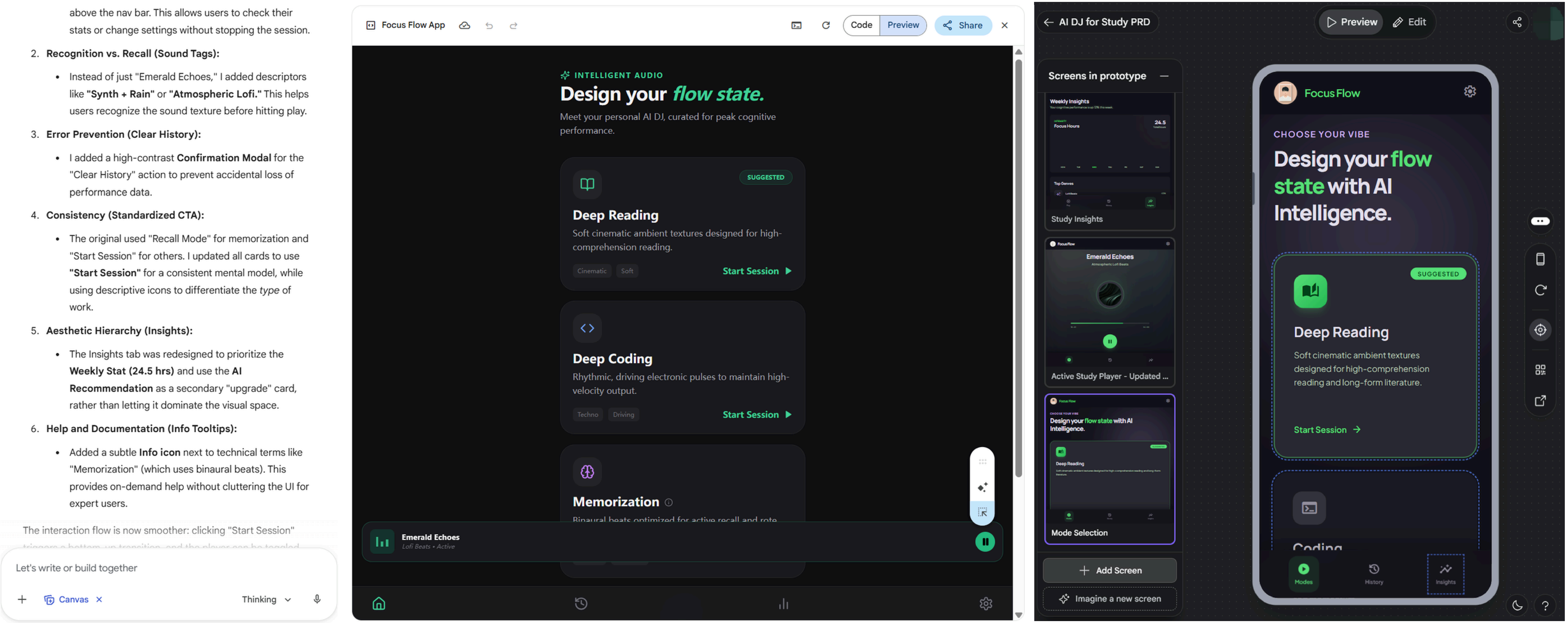
Gemini's version:



VS

The original version:

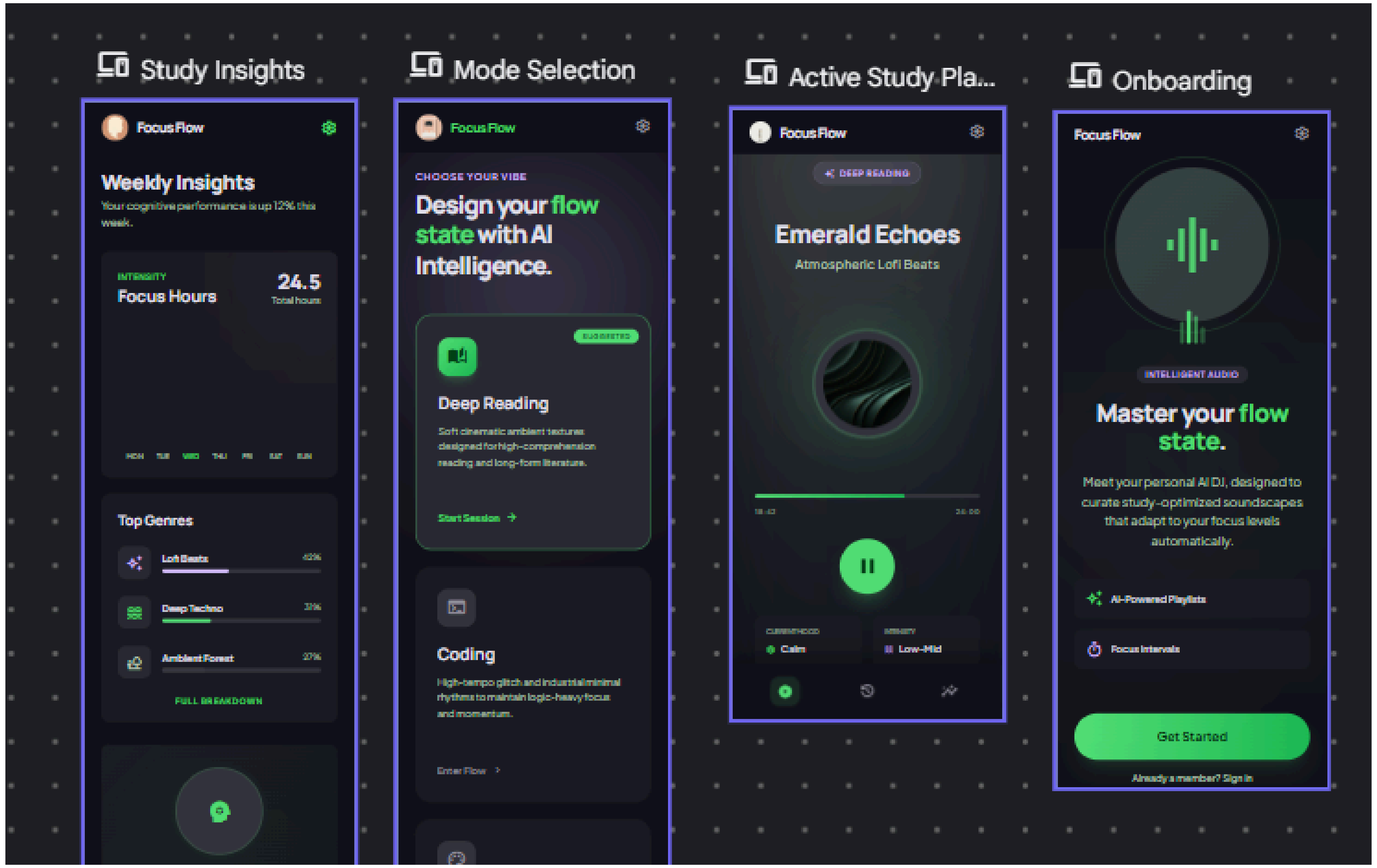




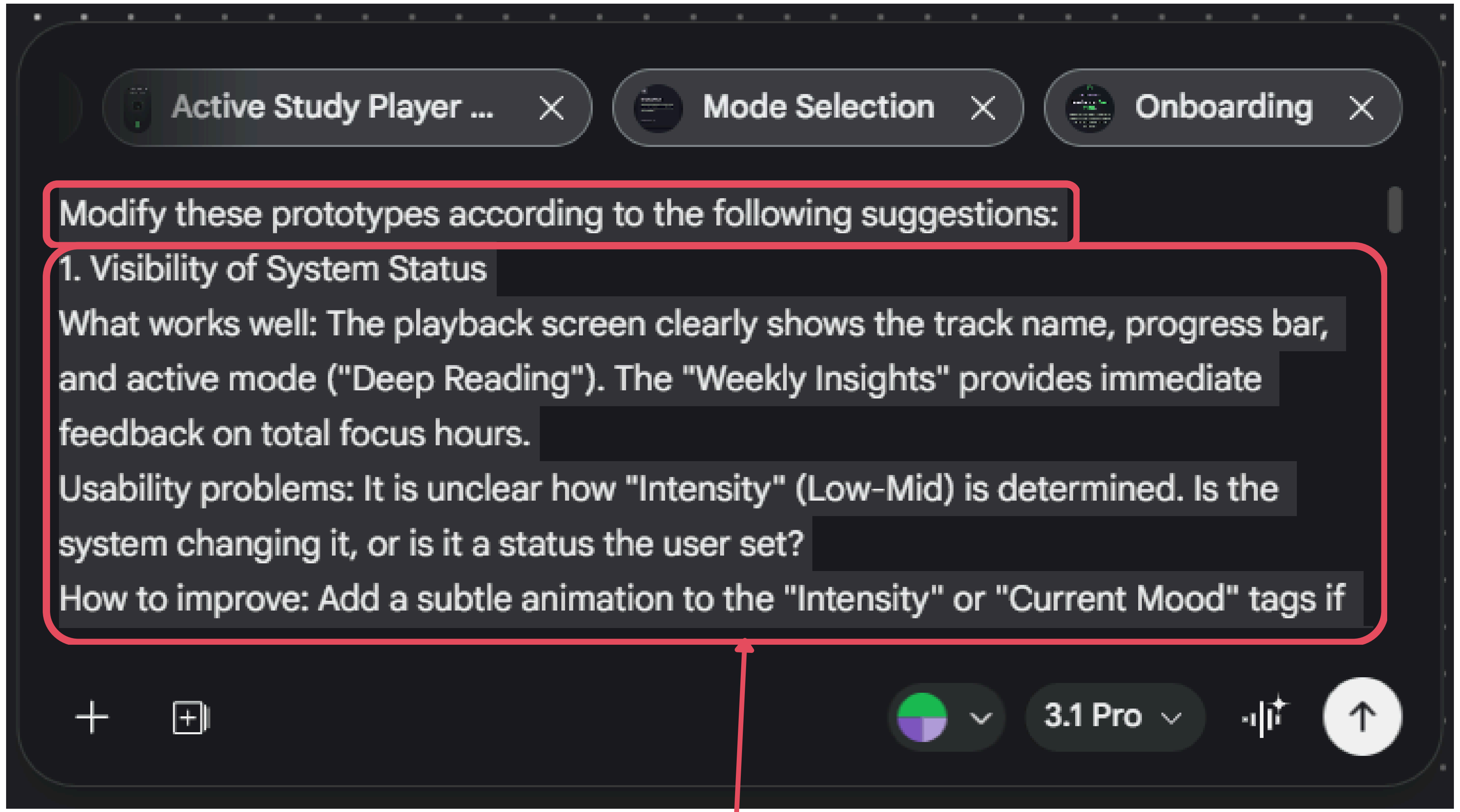
[RA] Gemini's suggestions were helpful in some areas and followed Nielsen's Heuristic principles well. For example, features like a confirmation modal for "Clear History" and subtle info icons for technical terms were **reasonable improvements**. However, the overall recreated result was not very successful. The **sizing felt inconsistent**, because the bottom tab bar looked like a web interface while the middle screens looked more like an app. The **layout also felt a bit rigid**, and it did not have the same rich colors, detail, or refinement as the Stitch version. Even so, it still offered some useful ideas that were worth keeping and applying selectively.

A better way to refine the prototype

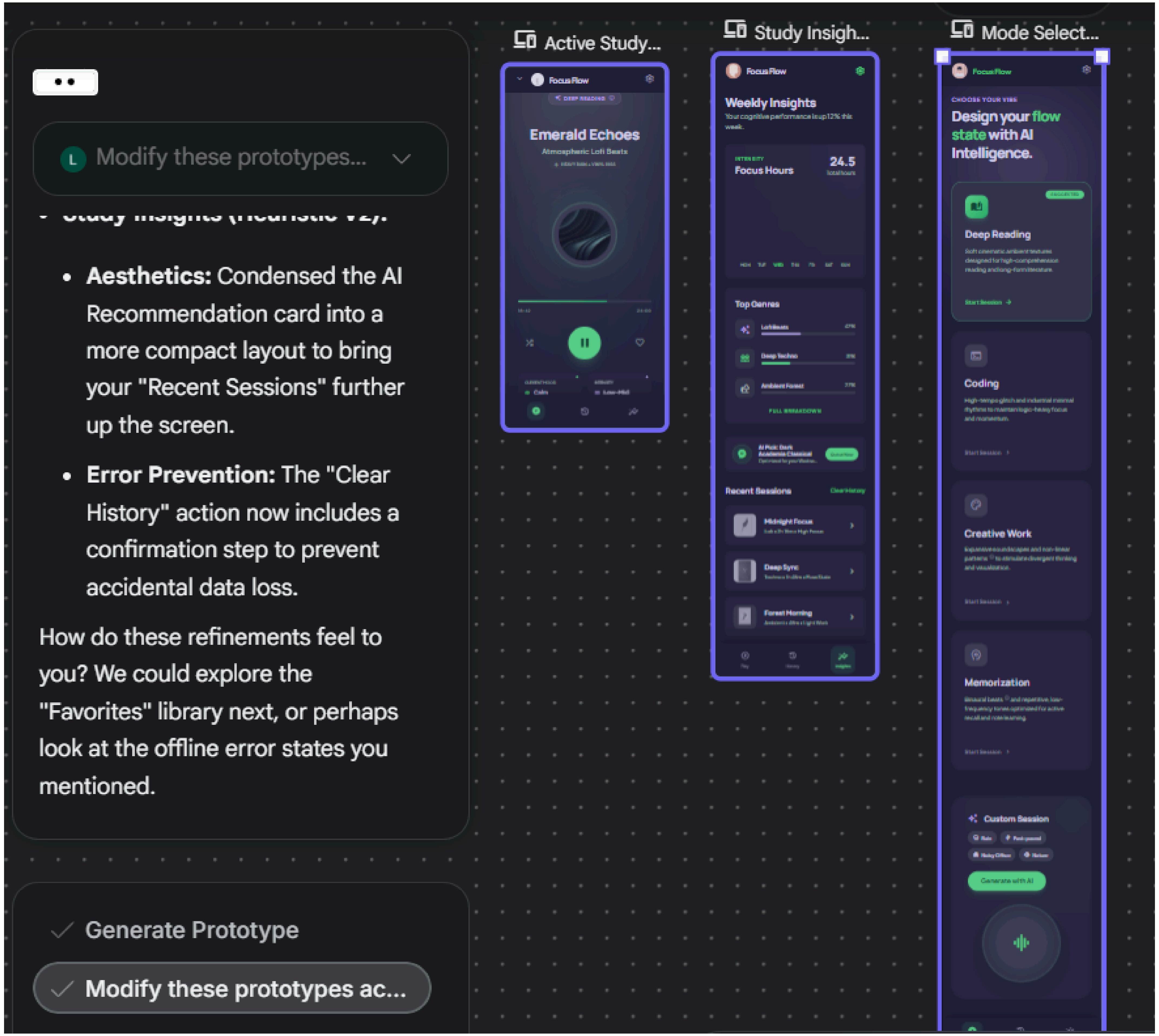
Return to Stitch, select pages:

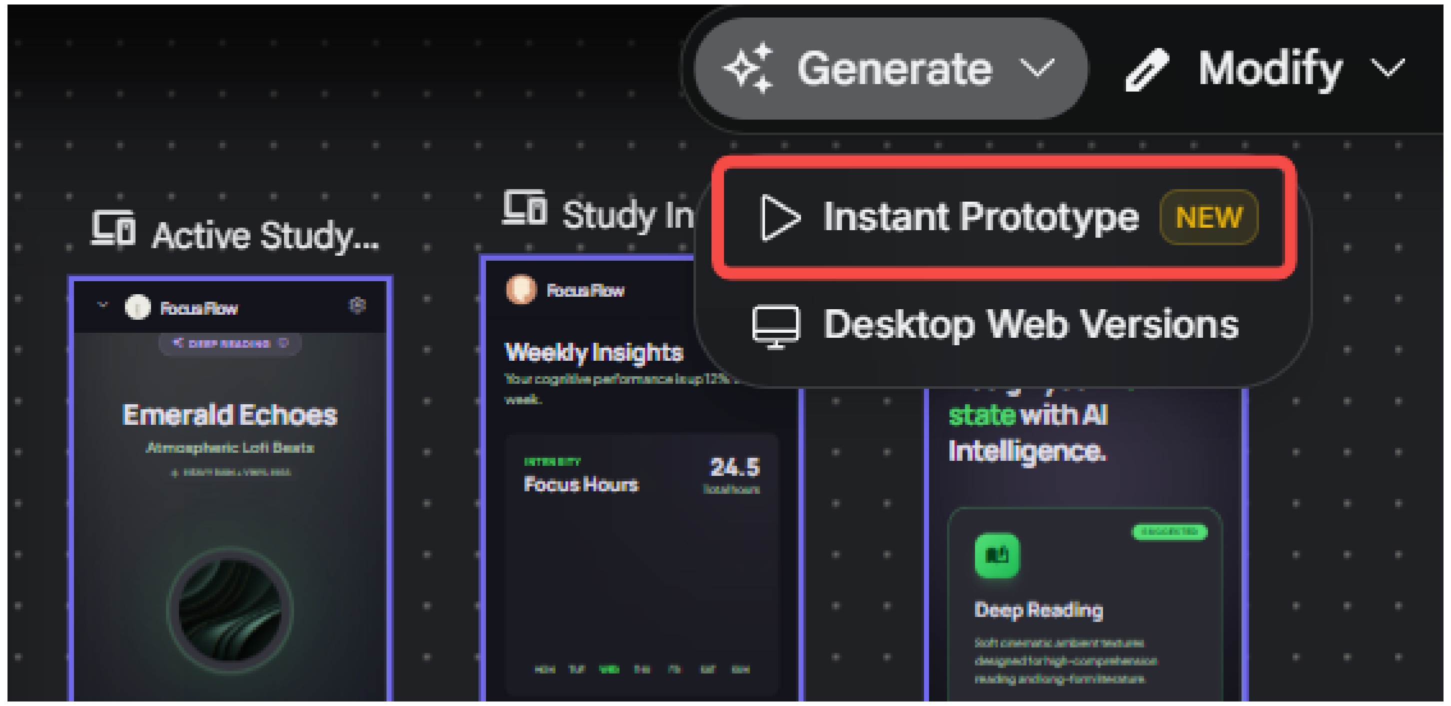


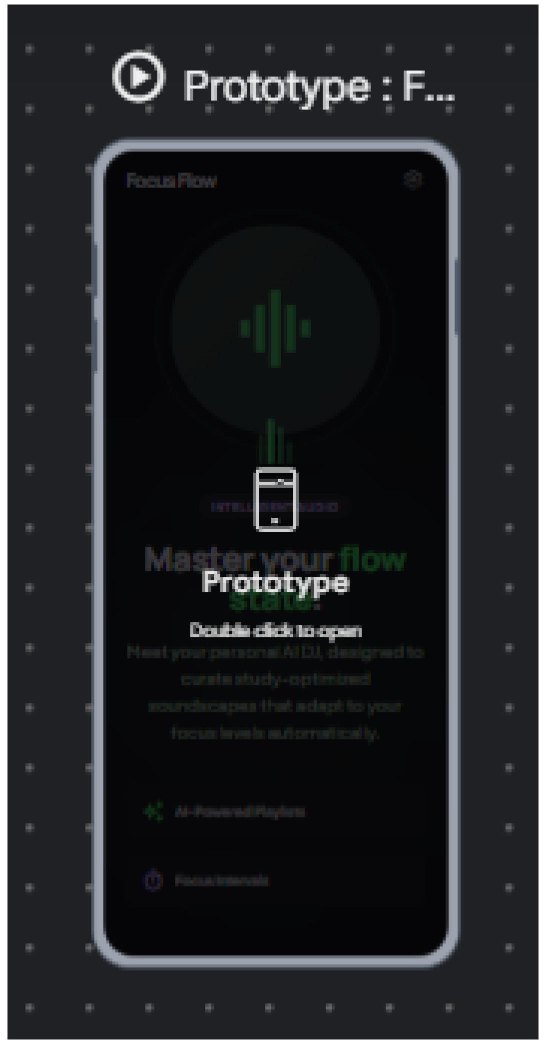
Prompt:

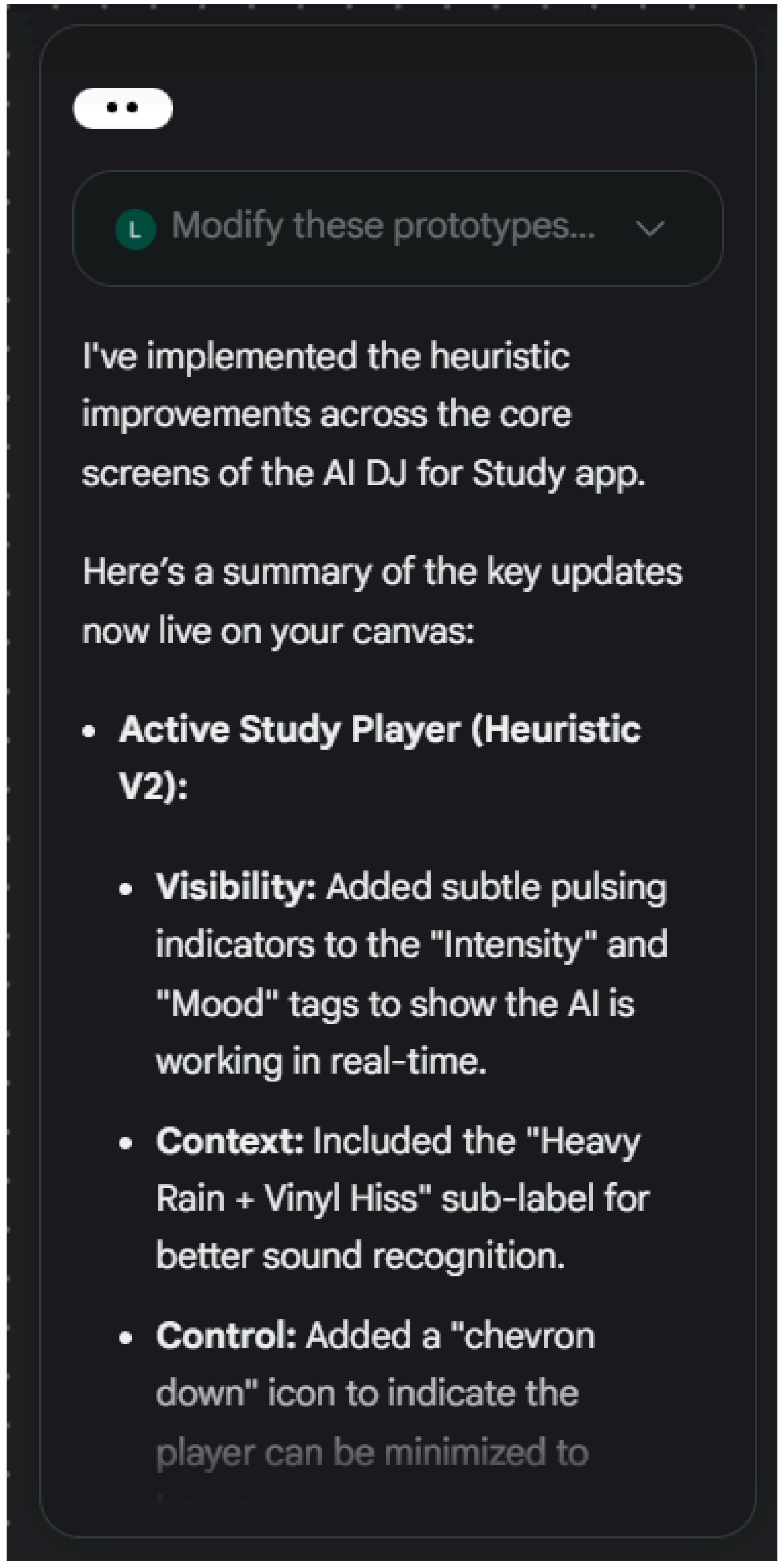


Copy from Gemini


Modify these prototypes...
Aesthetics: Condensed the AI Recommendation card into a more compact layout to bring your "Recent Sessions" further up the screen.
Error Prevention: The "Clear History" action now includes a confirmation step to prevent accidental data loss.
How do these refinements feel to you? We could explore the "Favorites" library next, or perhaps look at the offline error states you mentioned.
Generate Prototype
Modify these prototypes ac...
Active Study...
Study Insigh...
Mode Select...
Emerald Echoes
Weekly Insights
Focus Hours
24.5
Top Genres
Recent Sessions
Design your flow state with AI Intelligence.
Deep Reading
Coding
Creative Work
Memorization
Custom Session
Generate with AI


Generate
Modify
Active Study...
Study In
Instant Prototype
NEW
Desktop Web Versions
Emerald Echoes
Weekly Insights
Focus Hours
24.5
state with AI Intelligence.
Deep Reading


Prototype : F...
Prototype
Double click to open

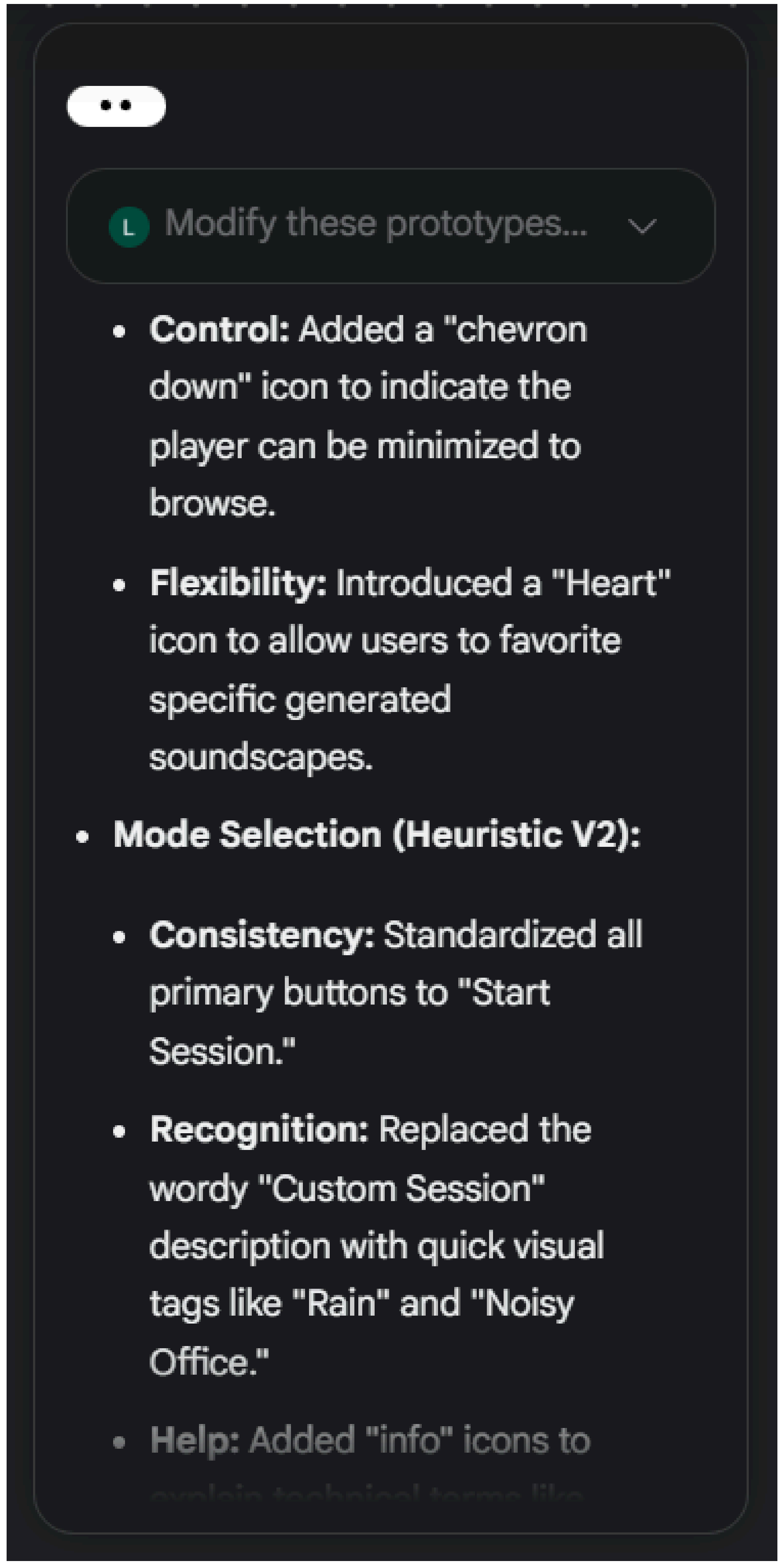


Modify these prototypes...

I've implemented the heuristic improvements across the core screens of the AI DJ for Study app.

Here's a summary of the key updates now live on your canvas:

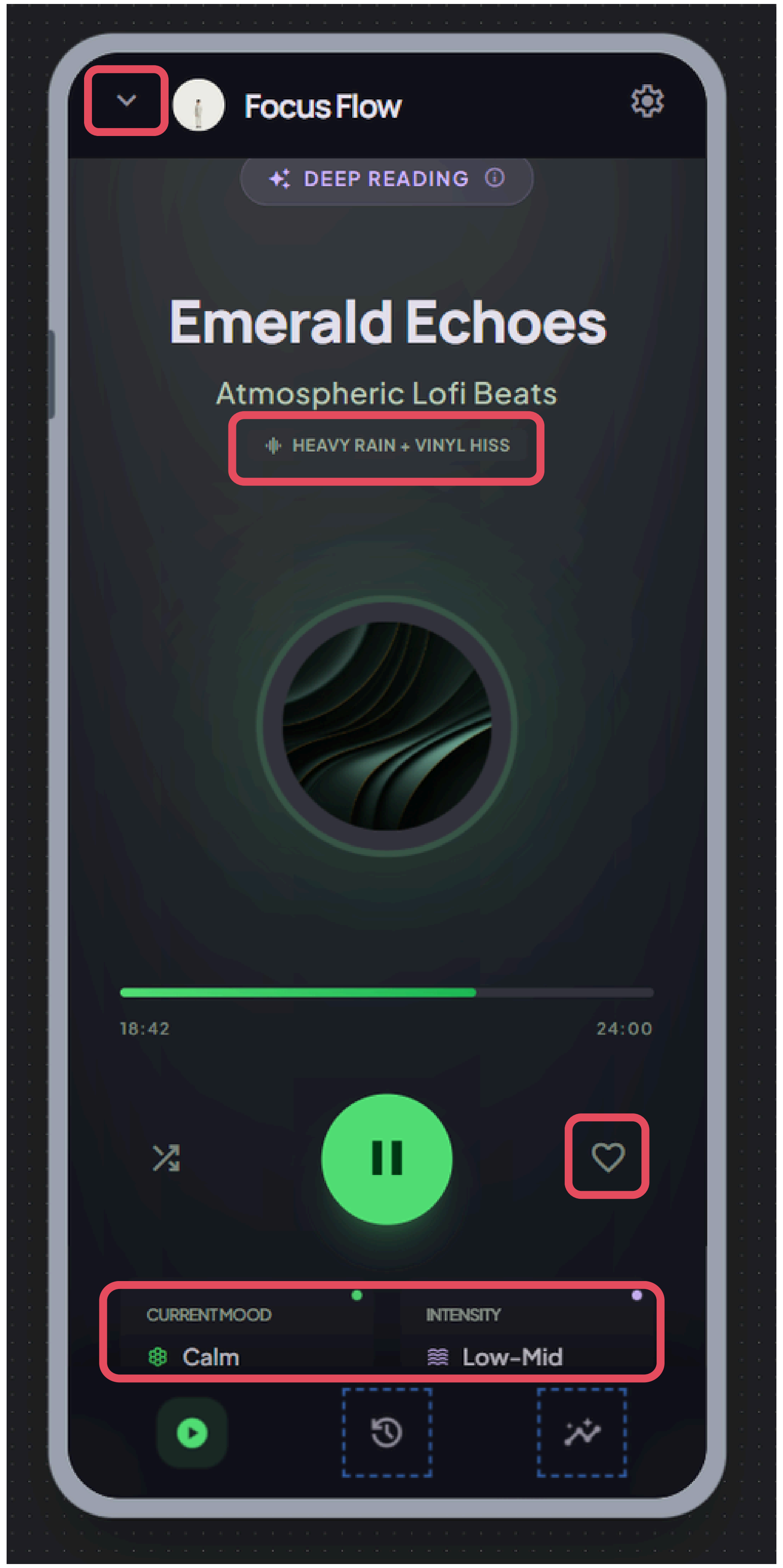
- **Active Study Player (Heuristic V2):**
  - **Visibility:** Added subtle pulsing indicators to the "Intensity" and "Mood" tags to show the AI is working in real-time.
  - **Context:** Included the "Heavy Rain + Vinyl Hiss" sub-label for better sound recognition.
  - **Control:** Added a "chevron down" icon to indicate the player can be minimized to

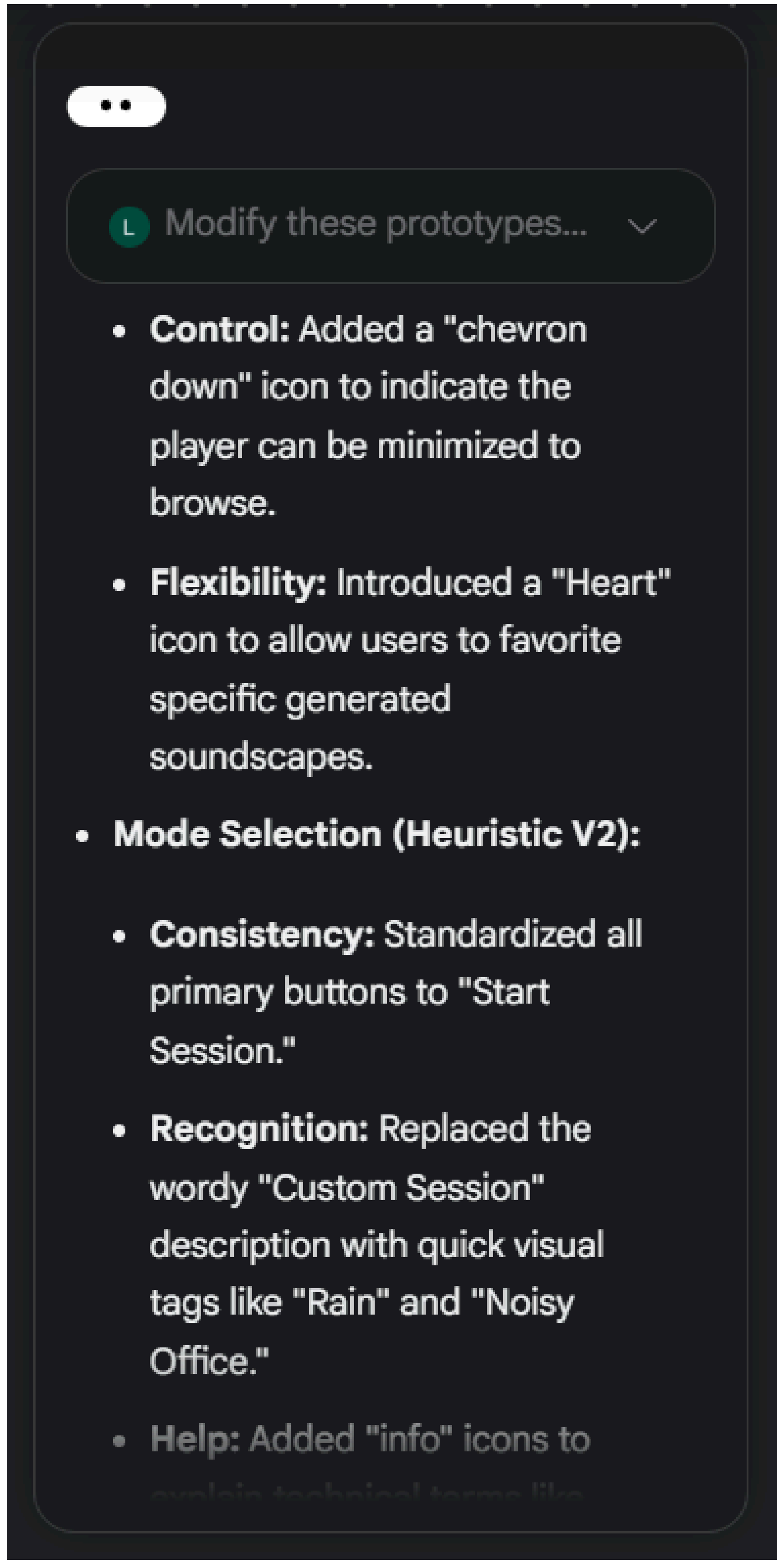


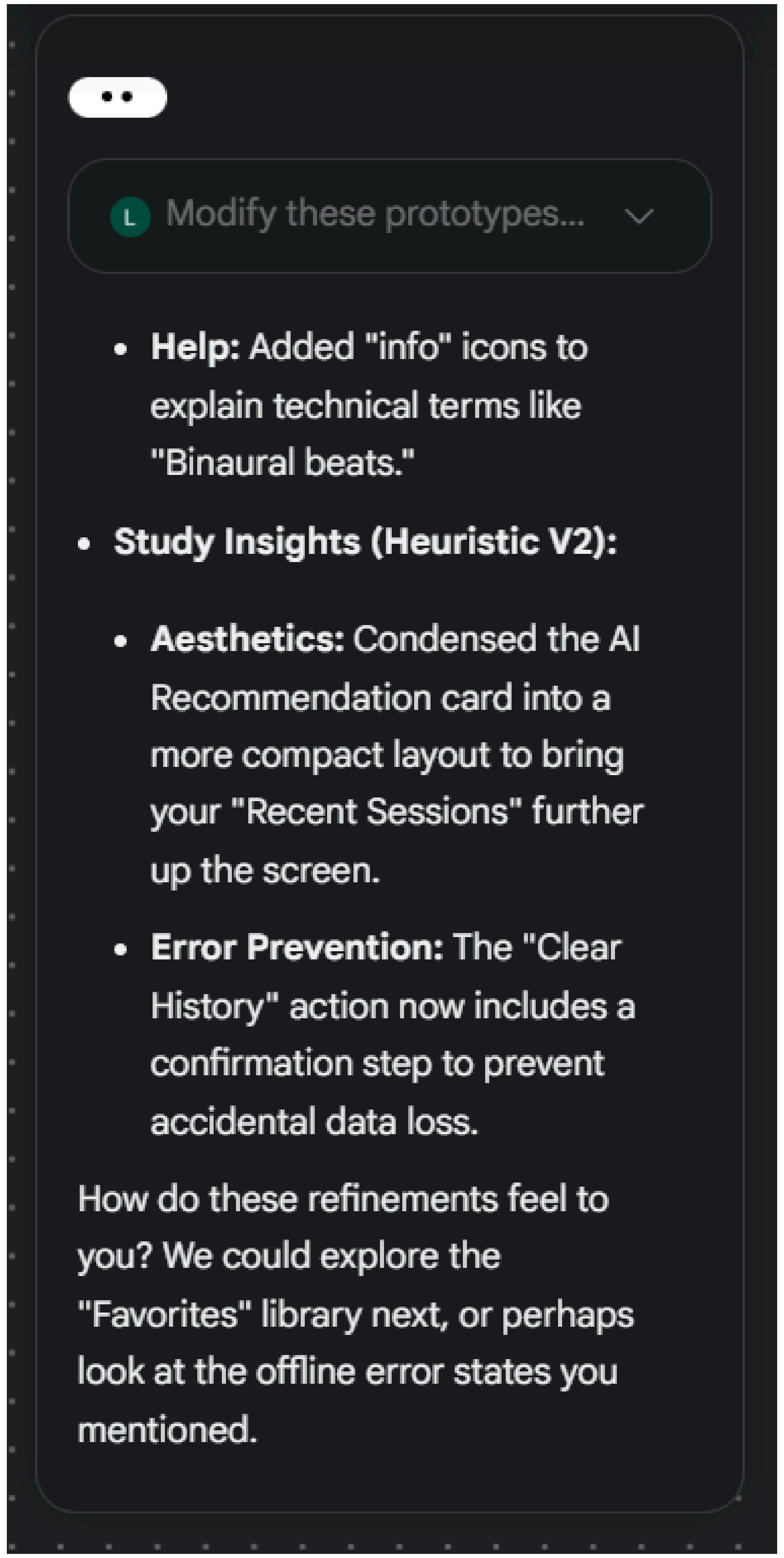
Modify these prototypes...

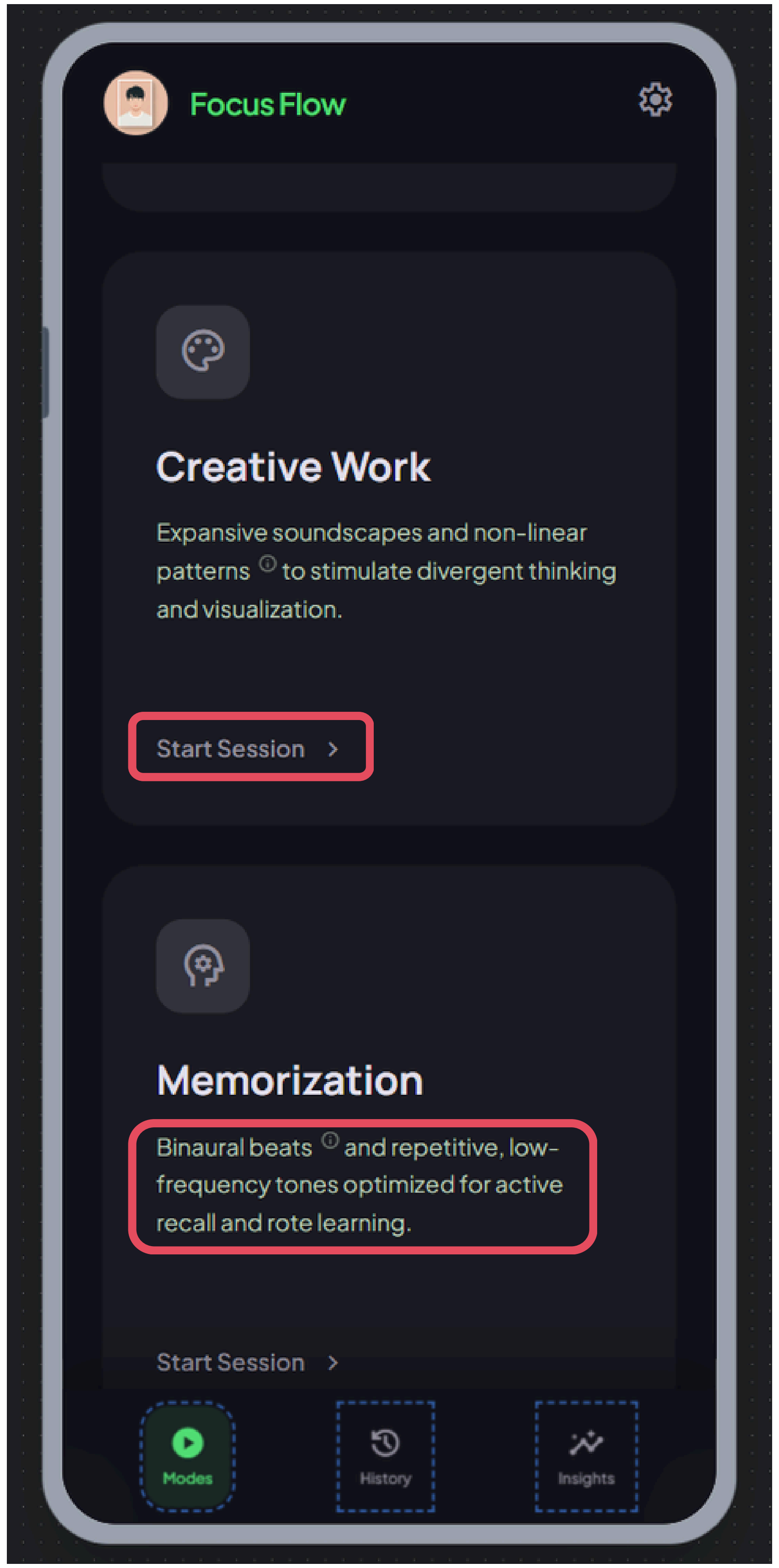
- **Control:** Added a "chevron down" icon to indicate the player can be minimized to browse.
- **Flexibility:** Introduced a "Heart" icon to allow users to favorite specific generated soundscapes.

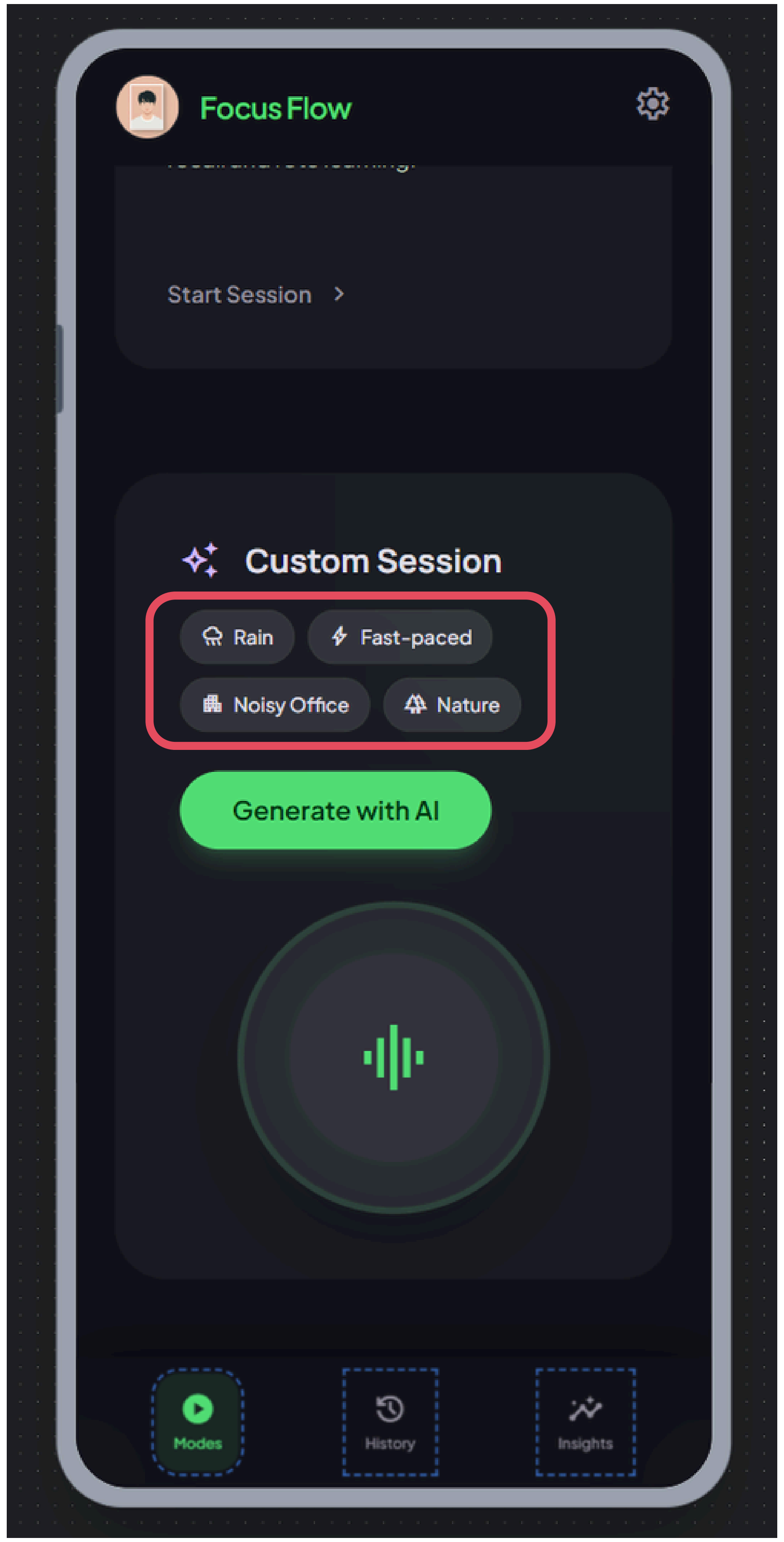
- **Mode Selection (Heuristic V2):**
  - **Consistency:** Standardized all primary buttons to "Start Session."
  - **Recognition:** Replaced the wordy "Custom Session" description with quick visual tags like "Rain" and "Noisy Office."
  - **Help:** Added "info" icons to

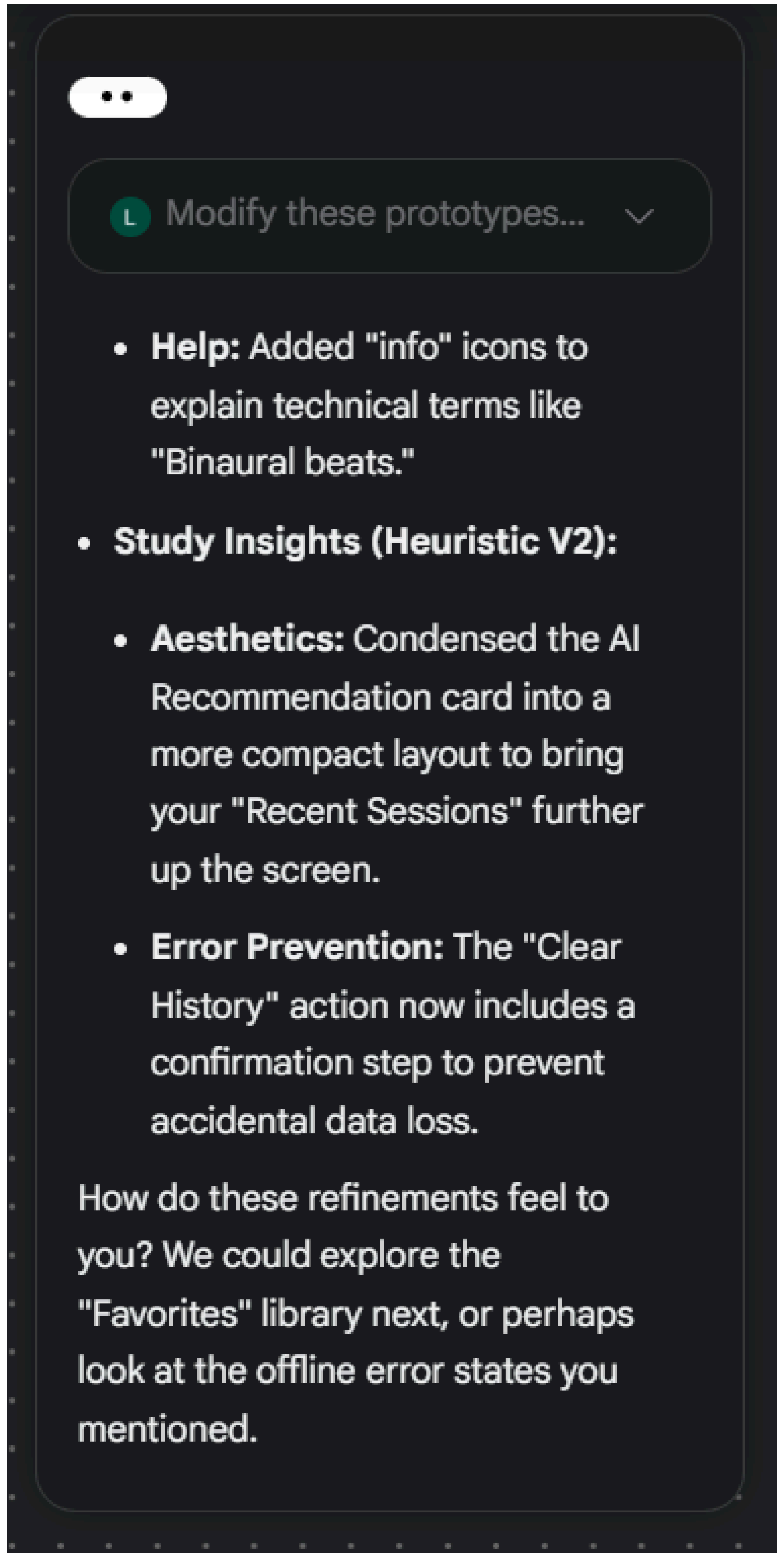


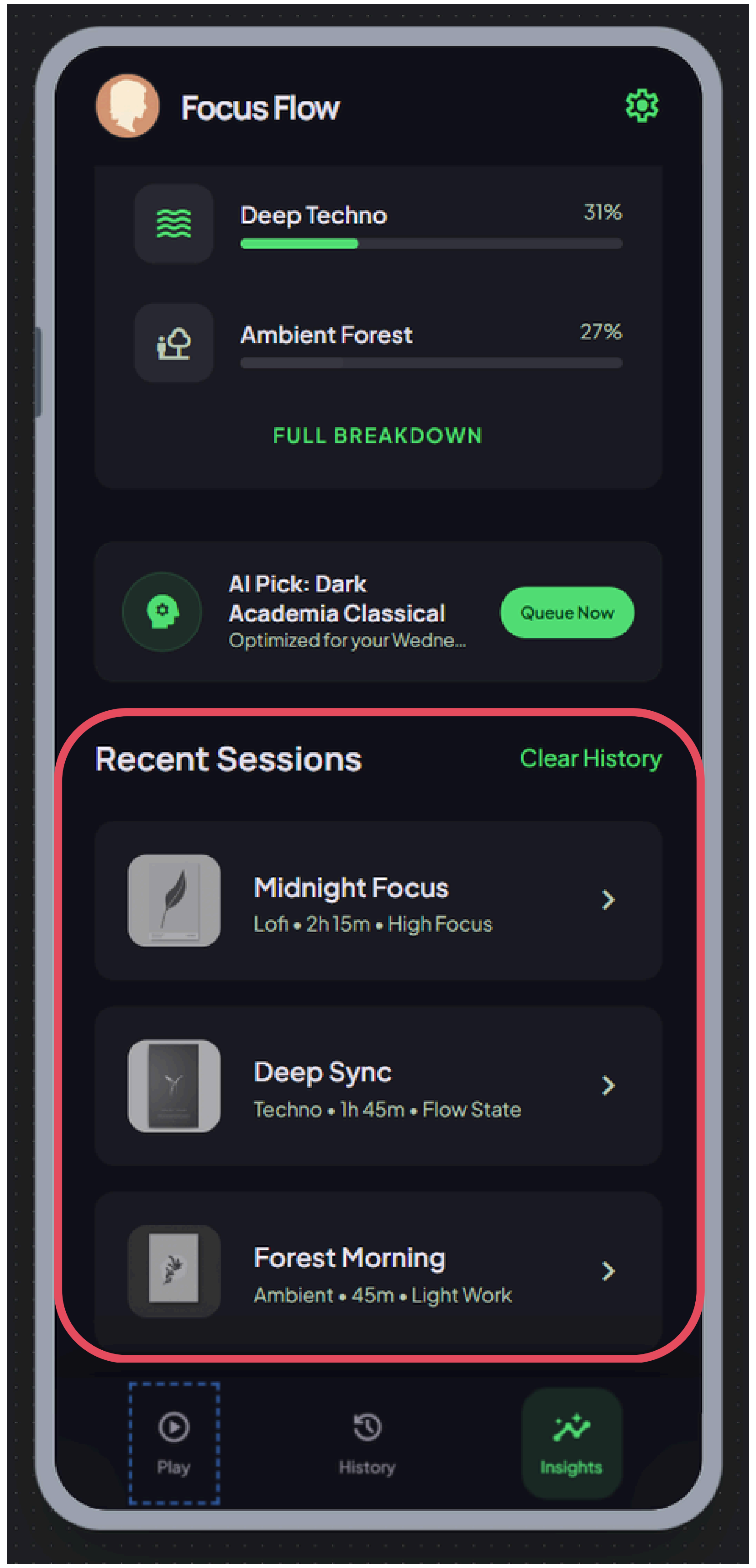

Modify these prototypes...
Control: Added a "chevron down" icon to indicate the player can be minimized to browse.
Flexibility: Introduced a "Heart" icon to allow users to favorite specific generated soundscapes.
Mode Selection (Heuristic V2):
Consistency: Standardized all primary buttons to "Start Session."
Recognition: Replaced the wordy "Custom Session" description with quick visual tags like "Rain" and "Noisy Office."
Help: Added "info" icons to

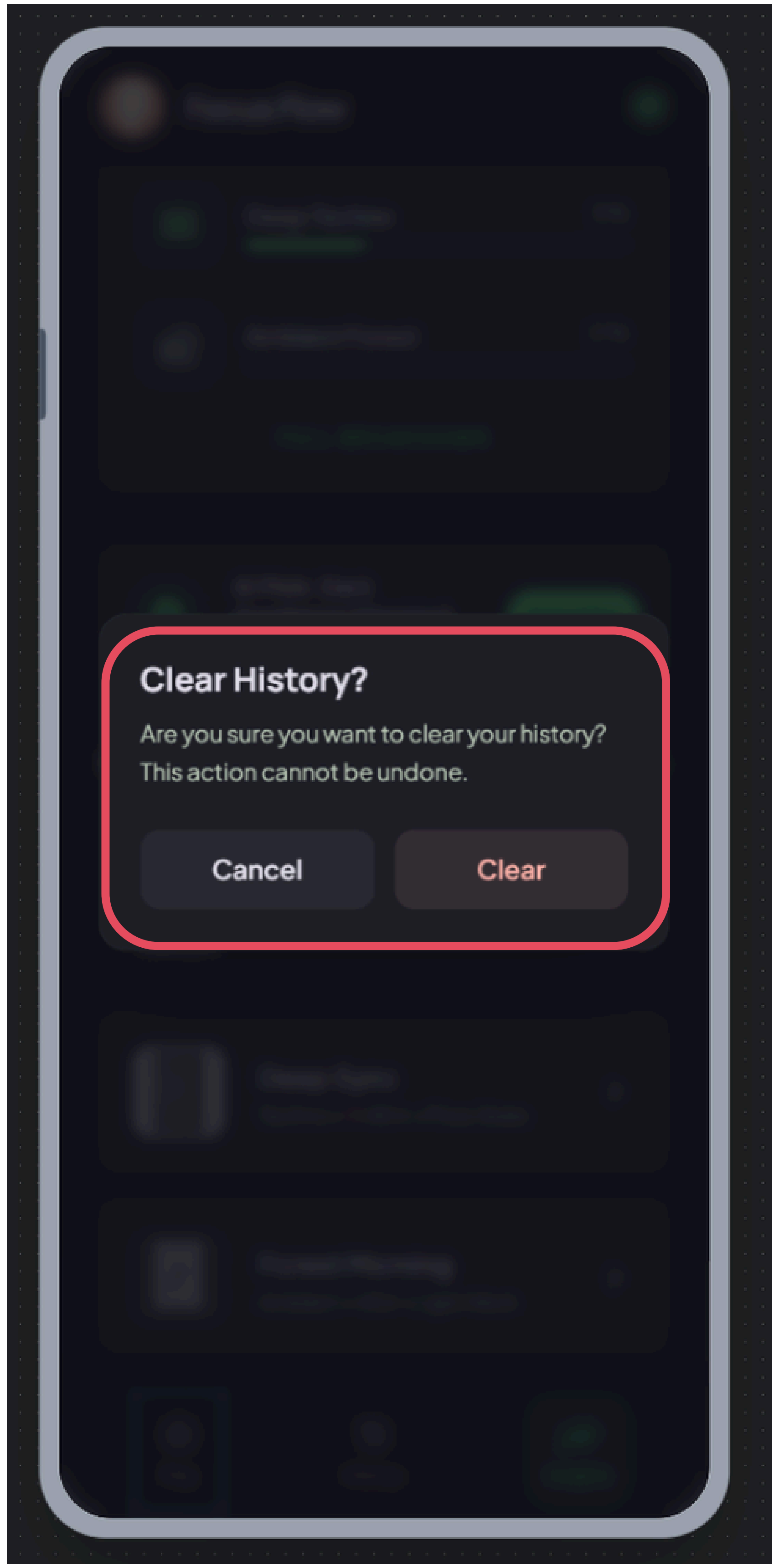

Modify these prototypes...
Help: Added "info" icons to explain technical terms like "Binaural beats."
Study Insights (Heuristic V2):
Aesthetics: Condensed the AI Recommendation card into a more compact layout to bring your "Recent Sessions" further up the screen.
Error Prevention: The "Clear History" action now includes a confirmation step to prevent accidental data loss.
How do these refinements feel to you? We could explore the "Favorites" library next, or perhaps look at the offline error states you mentioned.


Focus Flow
Creative Work
Expansive soundscapes and non-linear patterns to stimulate divergent thinking and visualization.
Start Session
Memorization
Binaural beats and repetitive, low-frequency tones optimized for active recall and rote learning.
Start Session
Modes
History
Insights

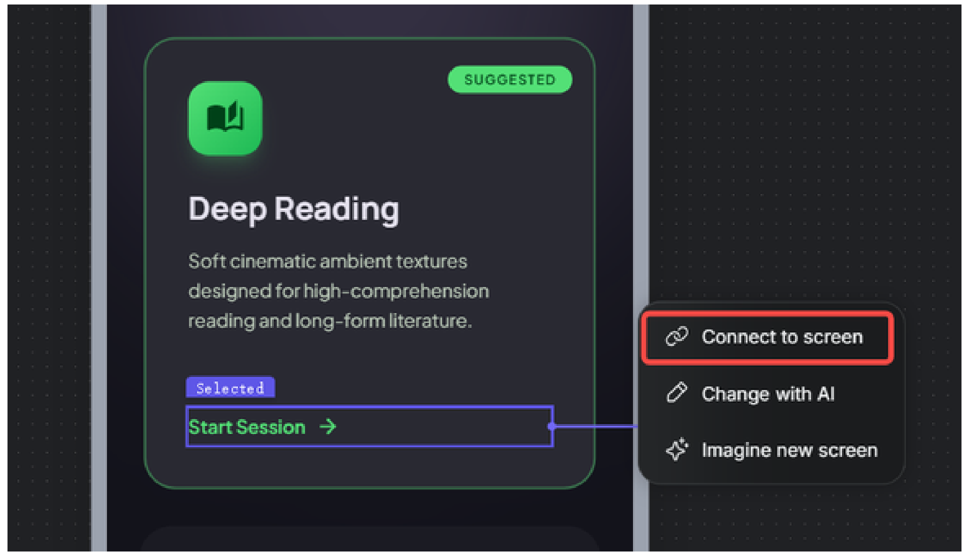

Focus Flow
Start Session
Custom Session
Rain
Fast-paced
Noisy Office
Nature
Generate with AI
Modes
History
Insights



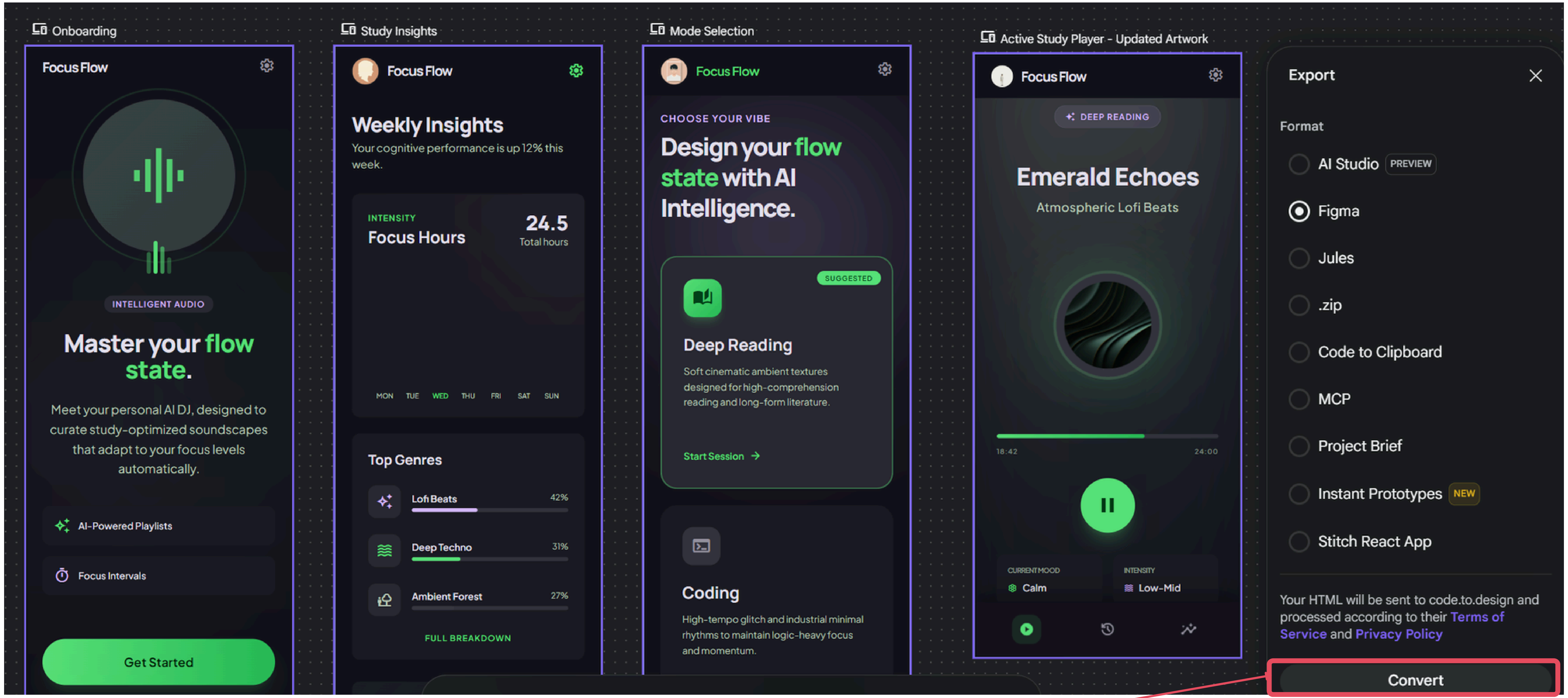




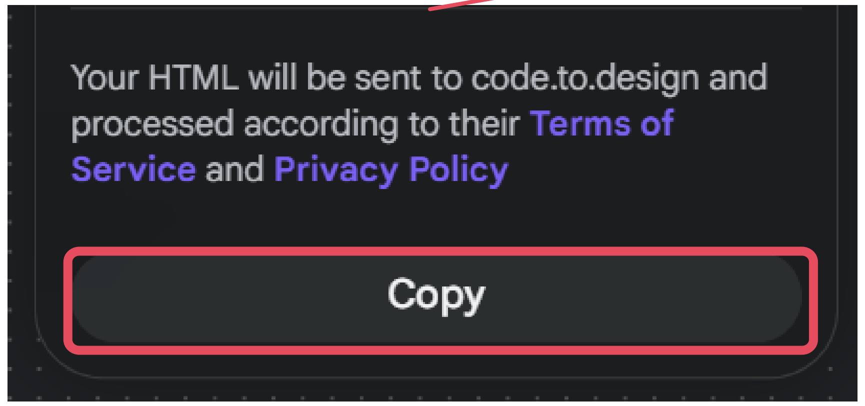
[RA] If Gemini's suggestions are applied directly in Stitch, the result can be much better, because the prototype was originally built there. However, Stitch's built-in interactions are still somewhat limited, and the links between pages are not always complete. We can still fix some of these issues manually, so it is sufficient for showing basic interactions. Since Stitch does not currently support importing interaction prototypes into Figma, it works better as a reference tool, while more complete and polished interactions can be developed later in Figma.

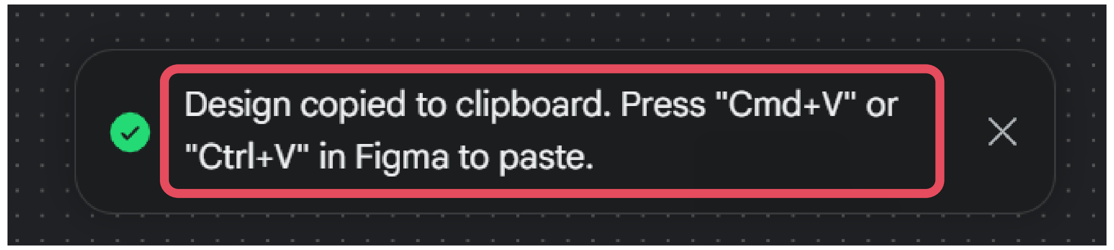


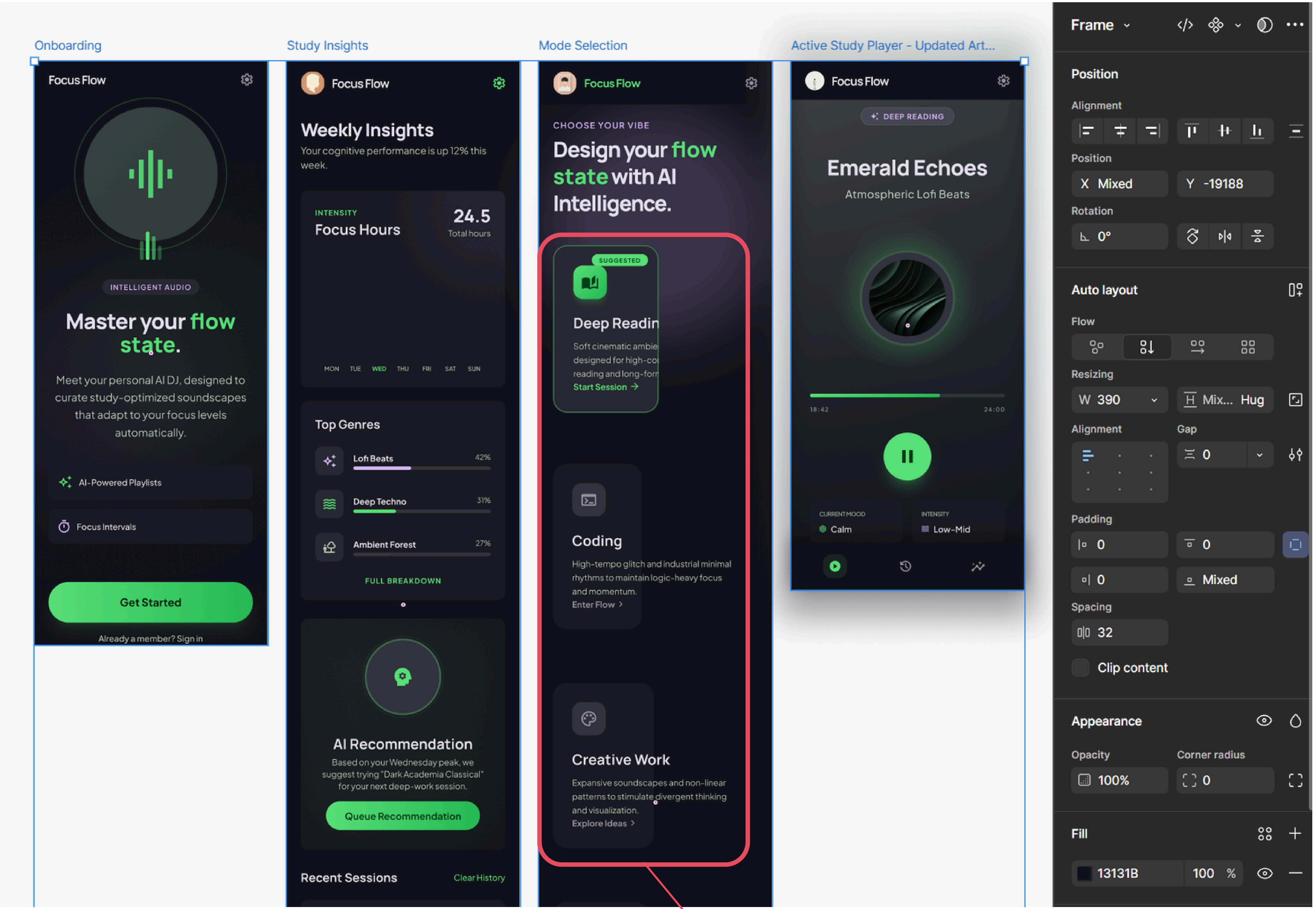
Export to Figma to refine the prototype



Stitch



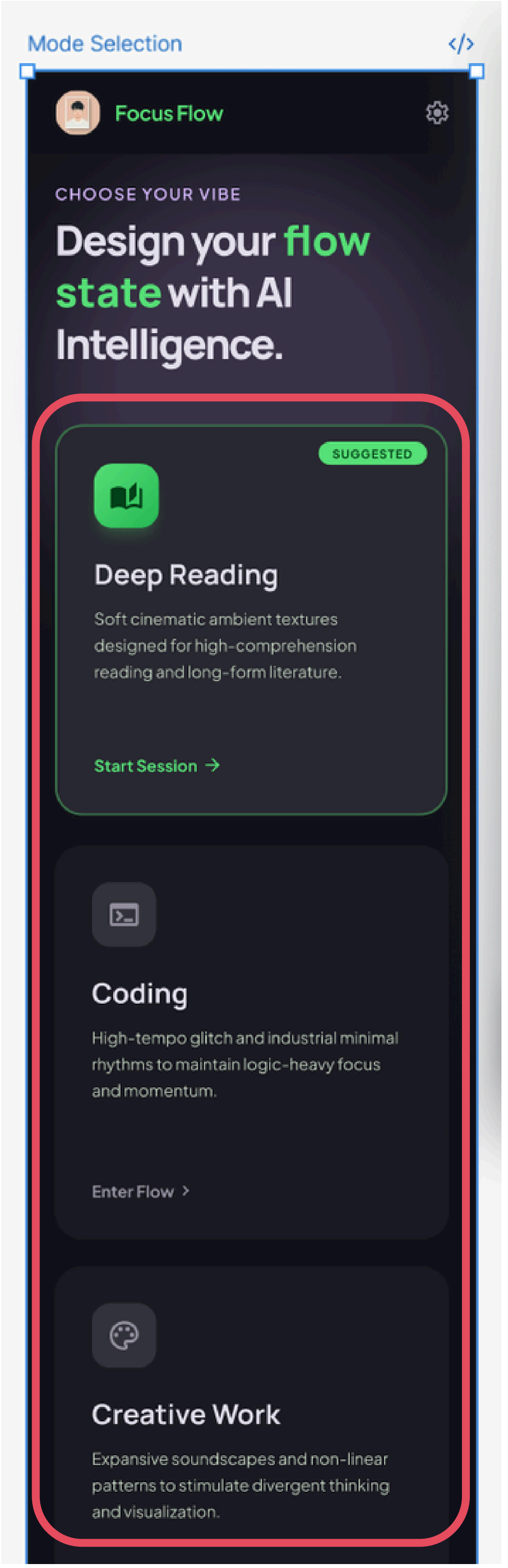




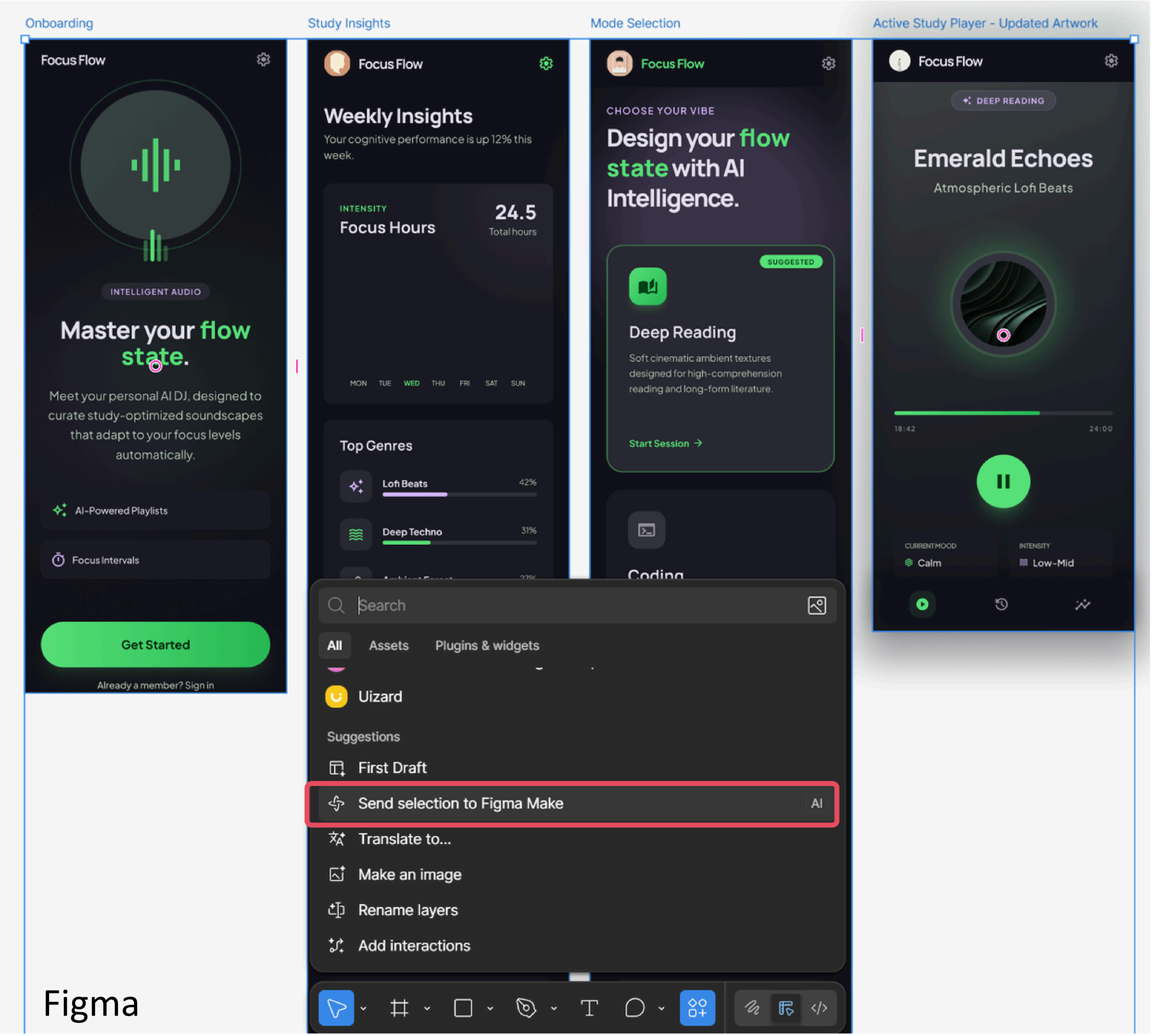
Figma

After pasting, there is a problem with the border, which needs to be adjusted manually

The modified interface

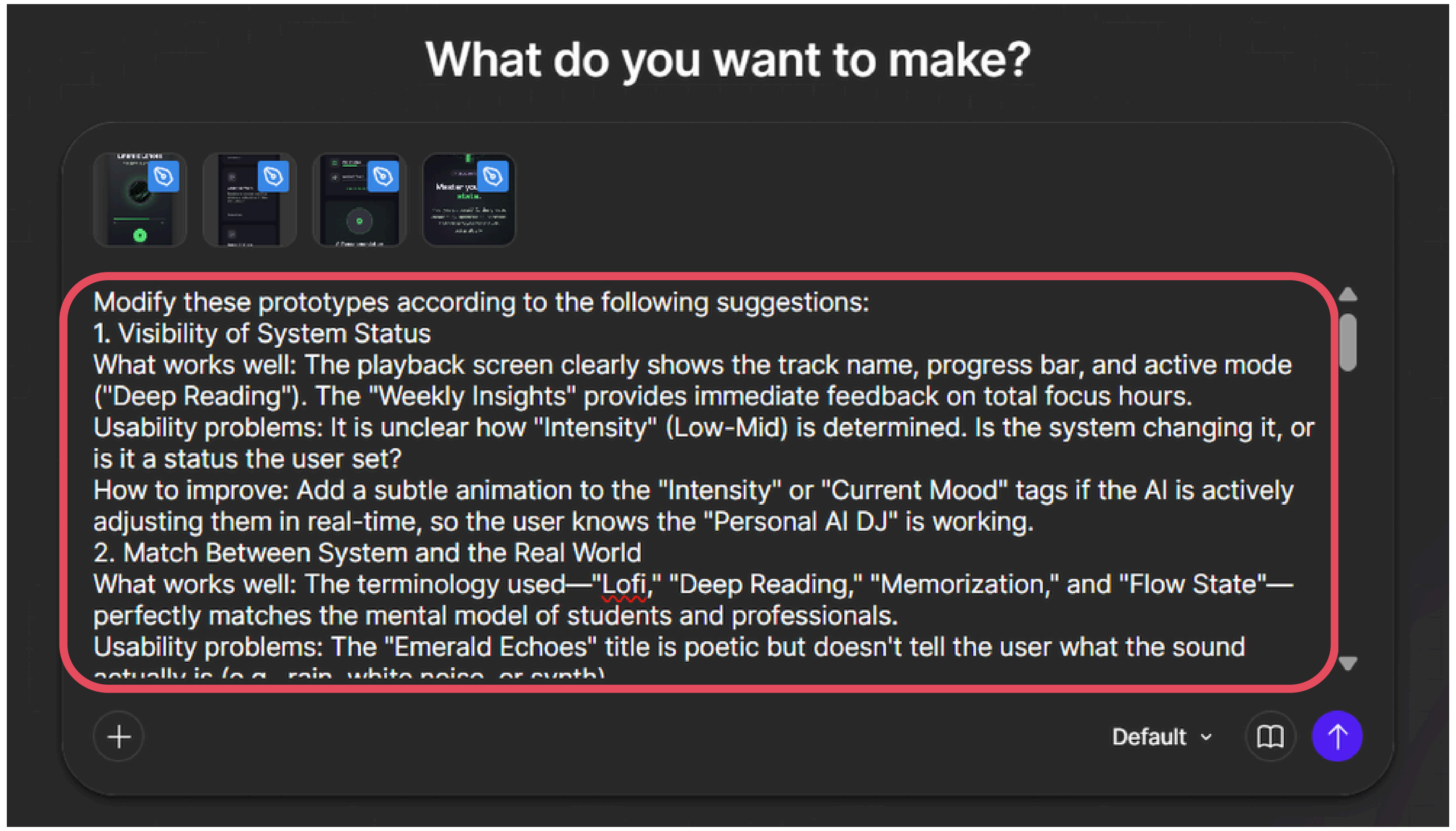


Send the interfaces to Figma Make:



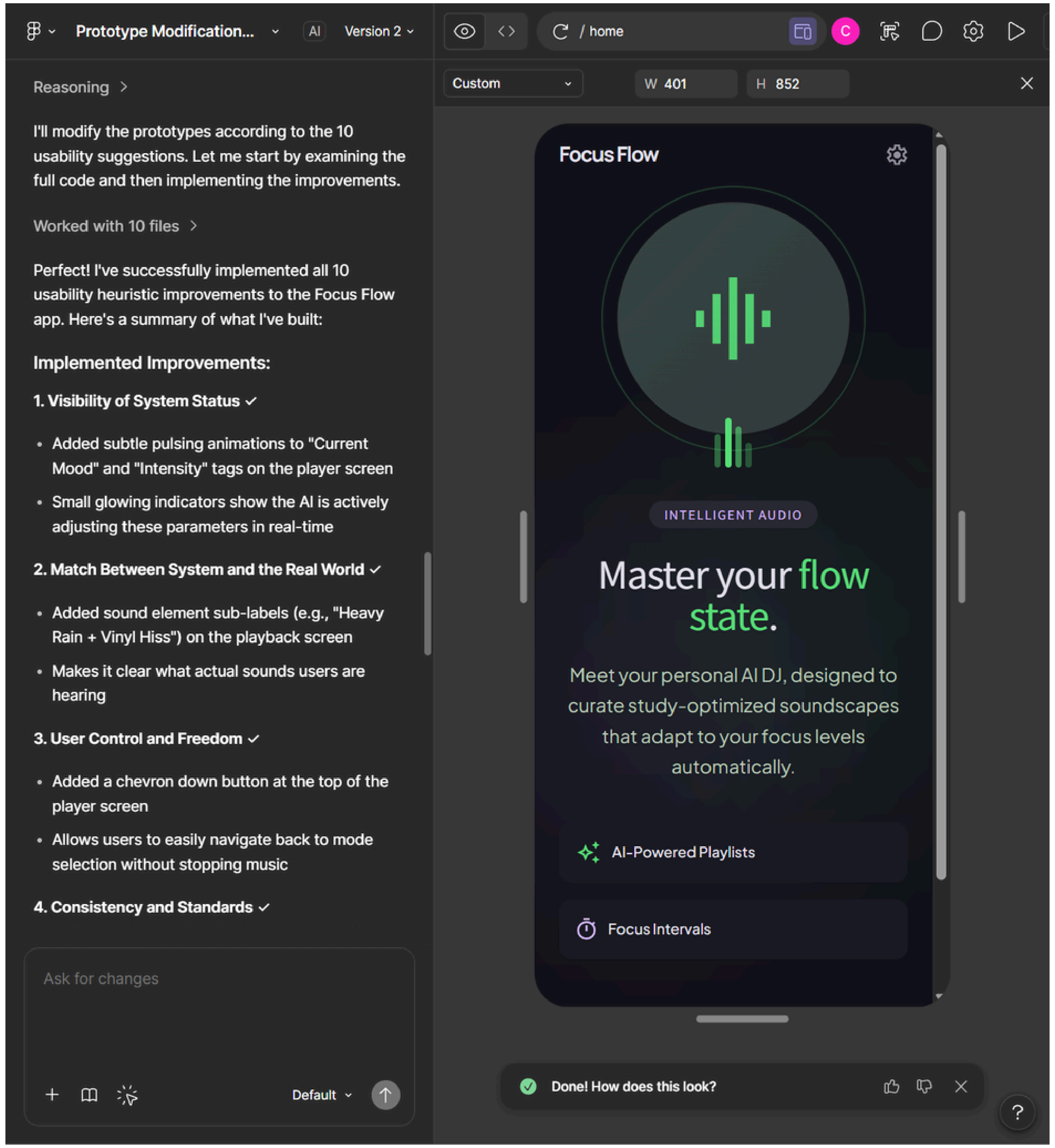
Figma

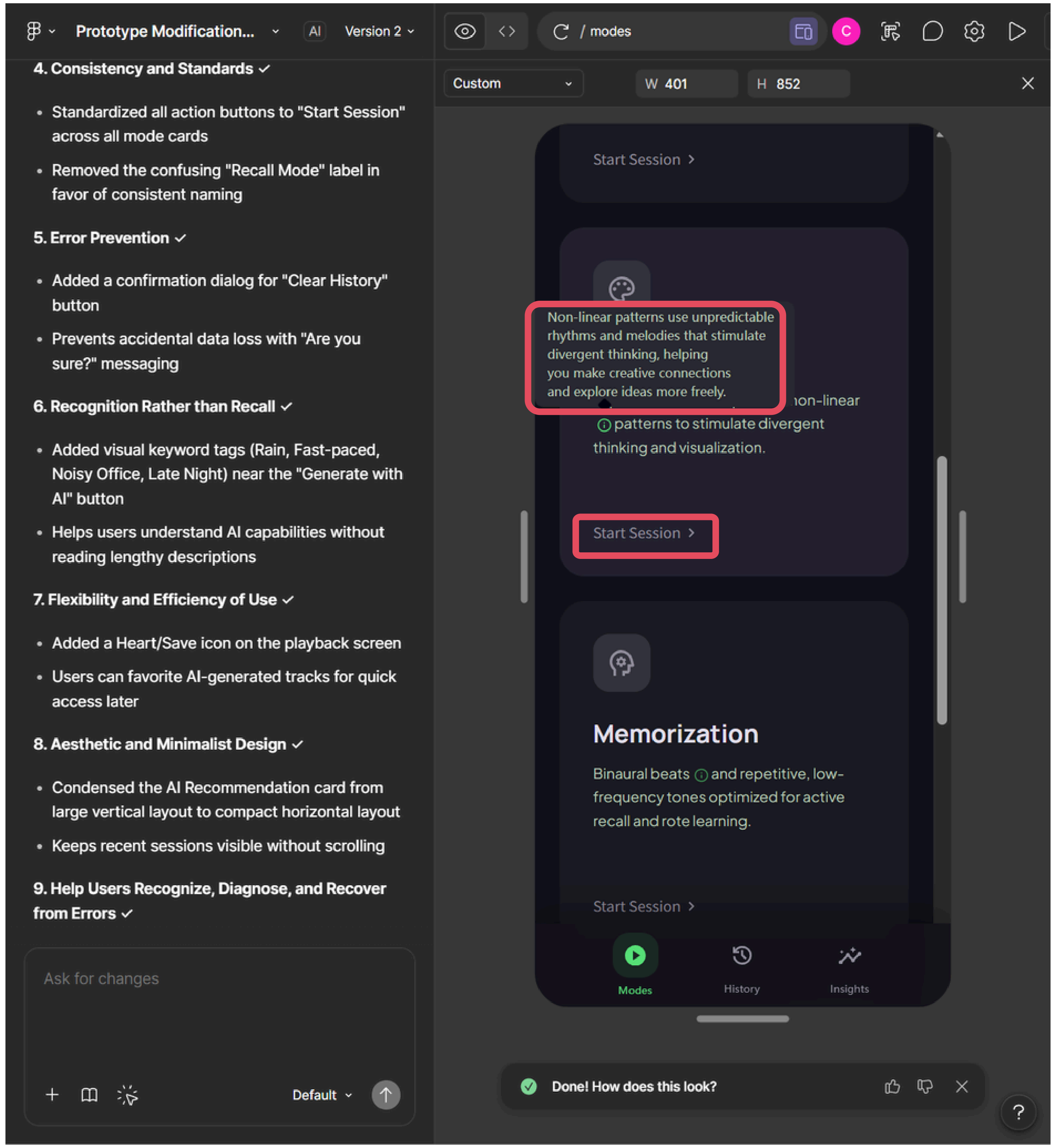
The same prompt as in stitch:

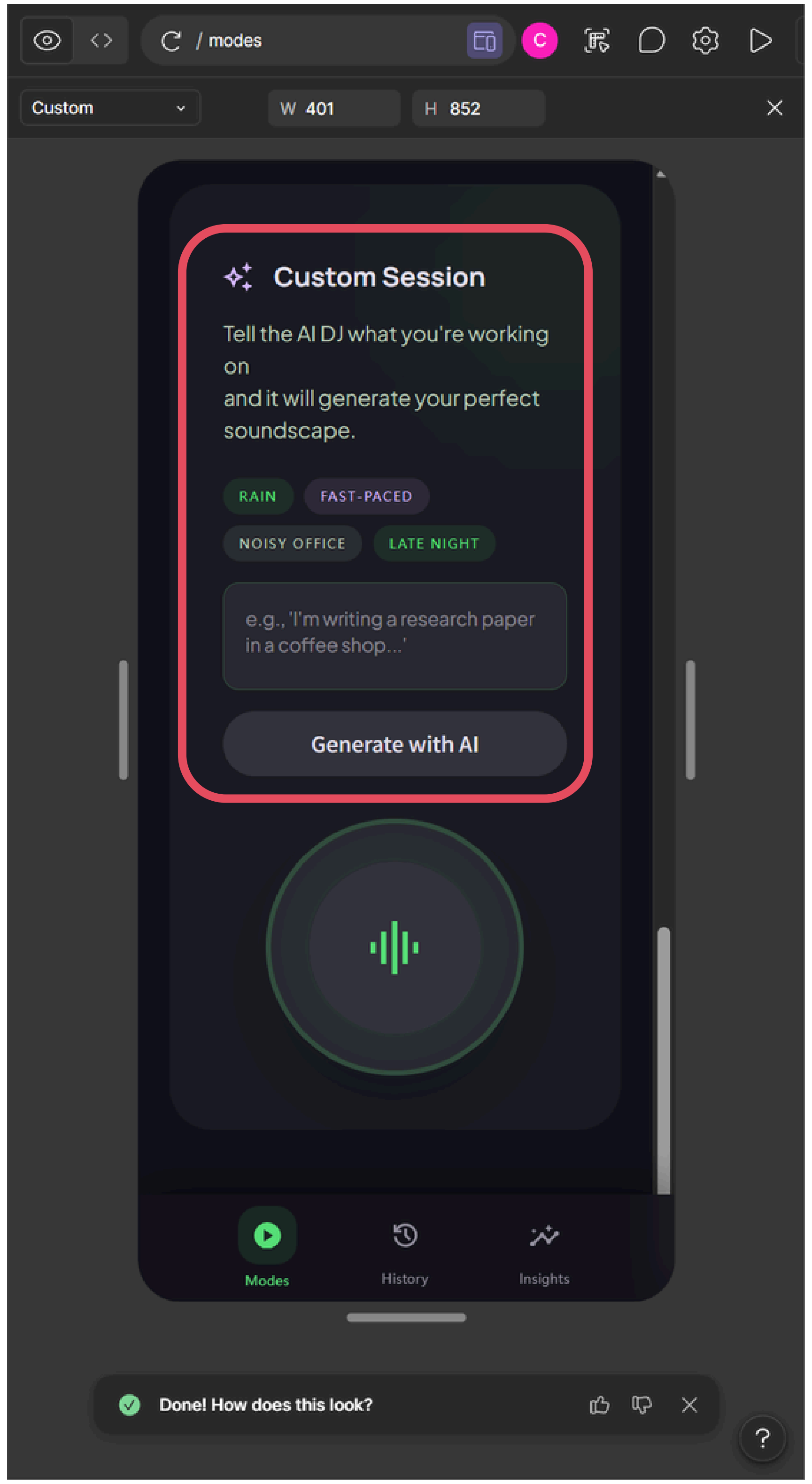


Figma Make

Result:

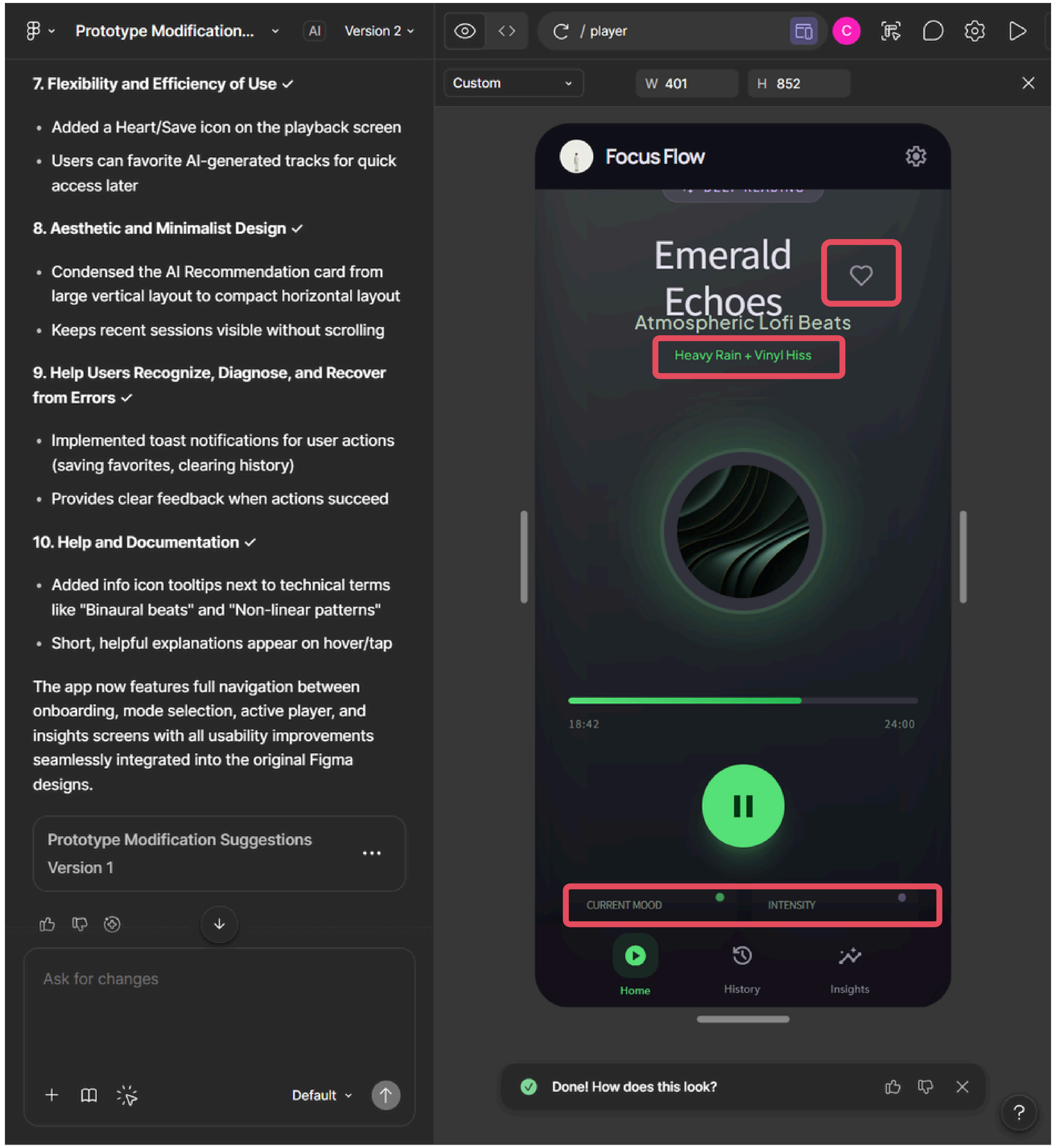


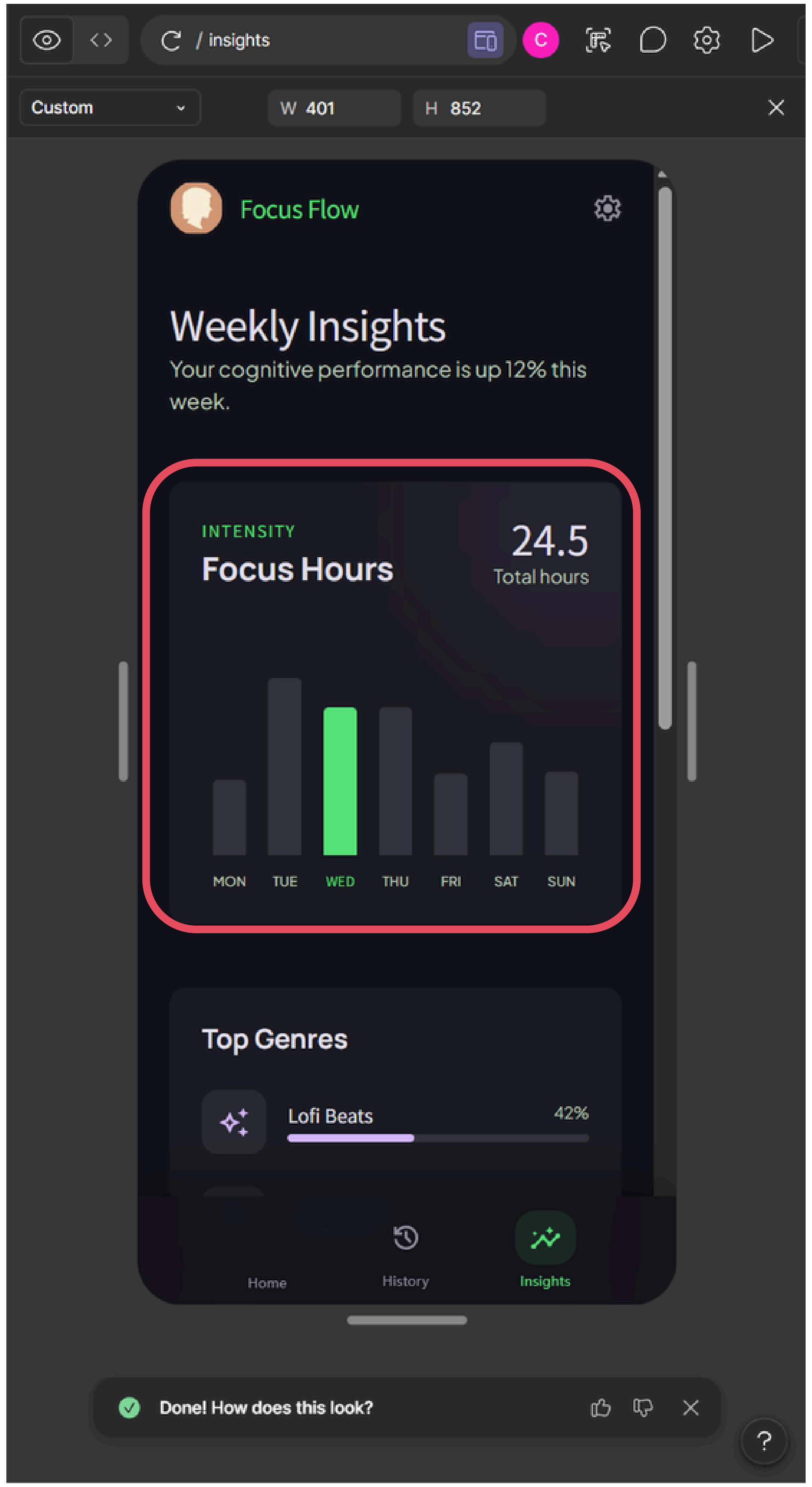


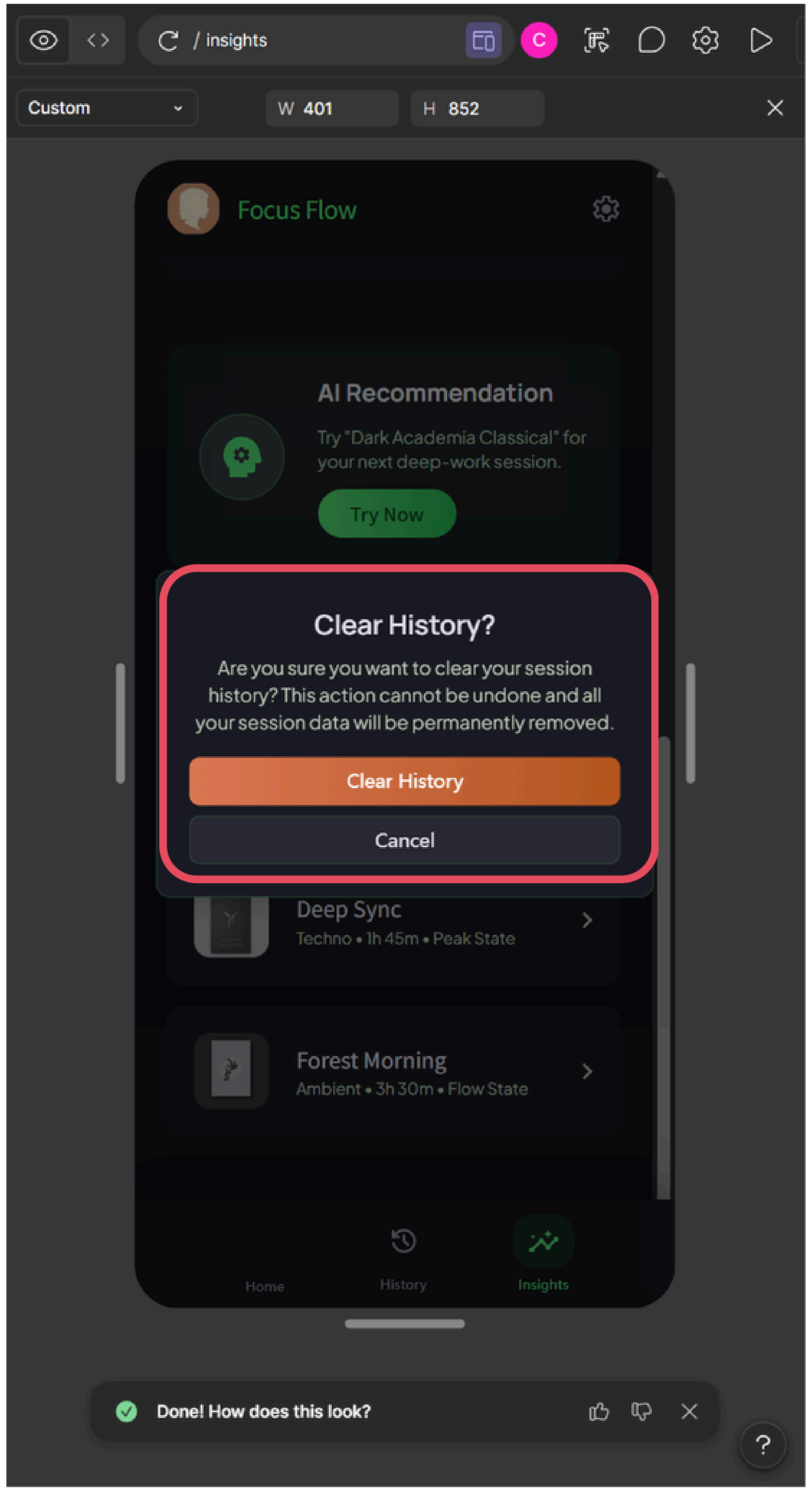


Figma Make

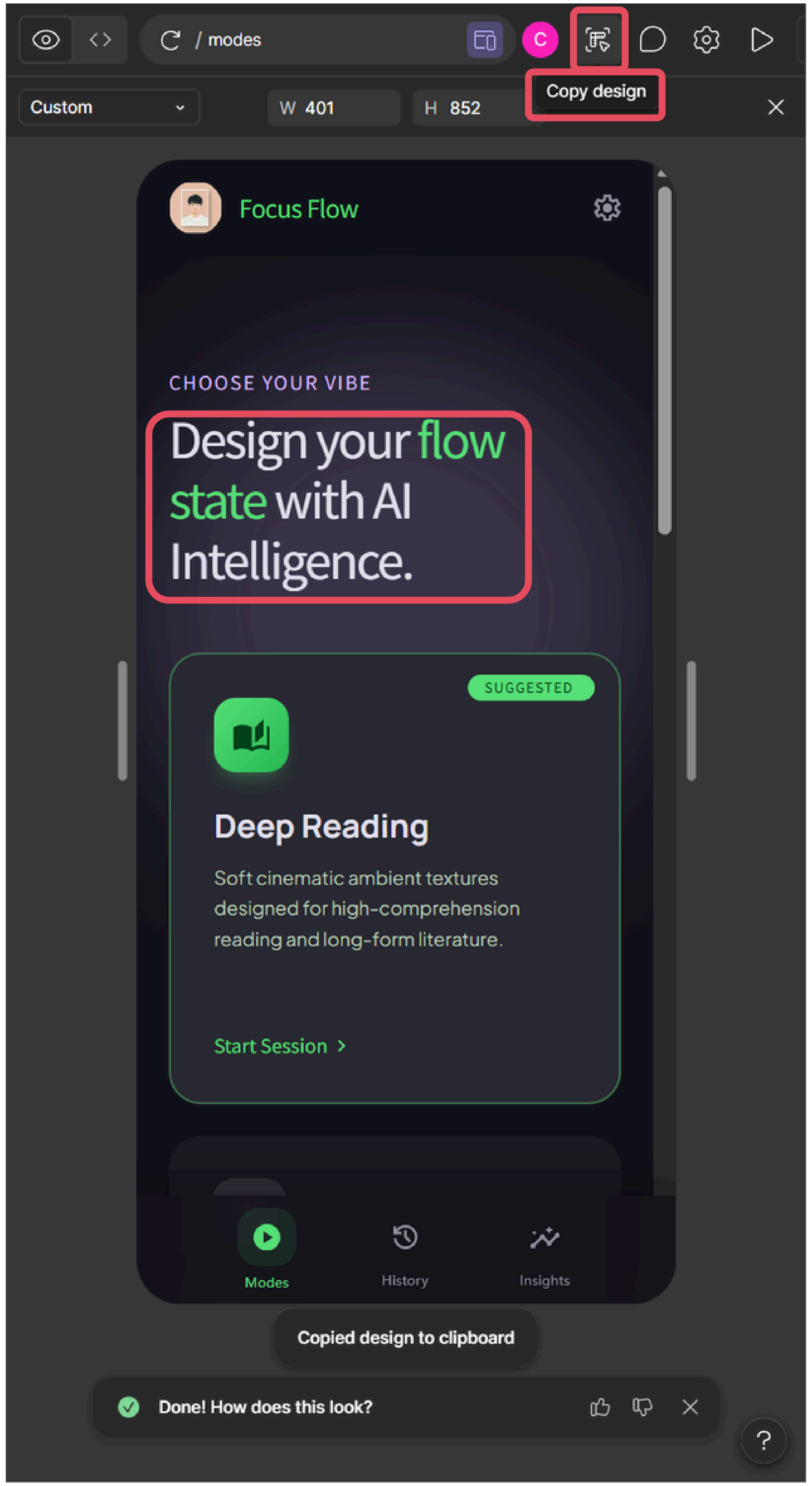
Result:





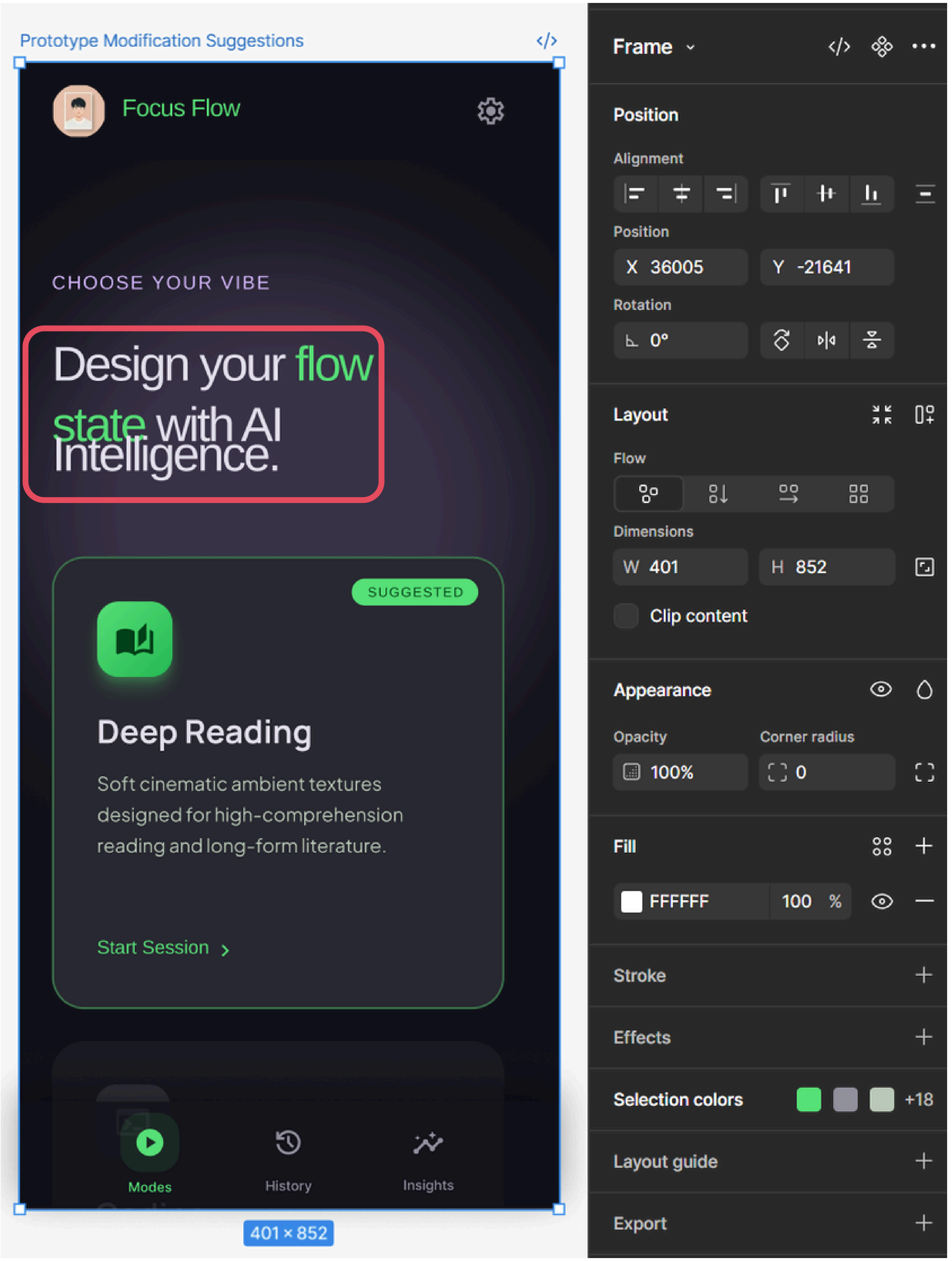


Figma Make



Figma Make

Copy design to Figma



Figma

[RA] When we imported the Stitch prototype into Figma and asked Figma Make to apply Gemini's suggested improvements, the results were quite **similar to editing directly in Stitch**.

In preview mode, **Figma Make can also automatically add interactions**, which makes the prototype feel more complete.

However, after using Copy design to Figma, we found that **only the currently displayed screen could be copied reliably**, so **long pages or hidden sections were not transferred fully**.

We also noticed **small layout differences**, such as tighter text spacing, so the result is still not ready to use directly and manual adjustment is still needed.